

Born wired: innate cortical connectivity plus local plasticity is enough to stand, walk and look

Author: Zhiwen Li.

Affiliation: Independent Researcher, No. 67 Yuanren Street, Huangjing Town, Taicang, Suzhou, Jiangsu, China.

Correspondence: rivenlee94@gmail.com . ORCID: 0009-0005-8289-6393

Version of 2026-09-29. Sections R12, R13, R14, R15, R16, R17 and Methods 5.12 and 5.13 were added on this date and describe the rebuilt engine, the nursery, the hidden-layer routes, the cost of a larger sheet, the eyes and the ears; R18, claim (P9), Figure 11 and Methods 5.14 were added the same day and describe the merge of two taught models; R19 and Methods 5.15 were added the same day and describe the opposite cell beside every sense channel, the one that fires when that channel has nothing on it; R20, claim (P10) and Methods 5.16 were added on 2026-09-29 and describe what the local rule does to a weight when it is used and when it is not, together with the three questions of the exam bank that read it. The full manuscript PDF in this directory was re-typeset from this file on 2026-09-29 and contains all of it (55 pages, 11 figures); the Nature Machine Intelligence short PDF is still the 2026-09-17 build and contains none of it, and Supplementary Figures 5 and 6 are new with this version. Every number quoted below was produced by the script named next to it; the scripts and the log of every run are in the repository, and Section 5.9 says where to find them. Experiments whose outcome contradicted our prediction are reported as such and are listed in Section 4.5.

Abstract

Brains are born with largely pre-specified cortical wiring, and the only rule locally available to a synapse is correlation between the neurons it connects; machine-learning agents instead optimise a global objective. How far does the biological combination reach on its own? We built a system of 148,032 model cortical neurons connected by 309,148 pre-specified "instinct" synapses, driving a simulated quadruped with no reward, no error signal, no gradient and no training loop. Placed prone, it stands up on 5/5 network seeds; shown red, it walks toward the stimulus on 5/5, over a median 1.73 m; with no sensory input it collapses; with the instinct table cleared the same brains can neither stand nor approach (0/5). Silencing the prefrontal population abolishes visually guided walking (0/200 ticks, 5/5) while leaving a wired reflex unchanged. The behaviour is produced by the cortex rather than by the stimulus: removing the recurrent step removes the action while sparing the reflex. Adding innate structure grows the repertoire without retraining; a 200-cell dopamine population writes one new sense-to-action link during the life of a single brain; gaze tracking emerges from static look-at rules and a motion-onset reflex rather than being written down. Acquisition becomes a recipe once the sheet is given somewhere to write: a route laid between two populations at a weight too small to do anything, twenty seconds of contact, and one cell whose firing says only that this moment counts -- after which a tone that meant nothing drives the retreat cells at 1.000, while the same lesson writes nothing if the route was not laid down or the animal was never touched. The sheet can also be grown rather than retrained: a rebuilt engine carries 750,075 cells and 1,978,219 synapses at 13.3 ms per tick on one consumer GPU, five times the cells of the reference build at about a twelfth of the cost per tick, and what caps a larger sheet is two build rules rather than the hardware. A layer that is not the last one is a signal source on the same terms as the last, and the detail a compressed layer has lost is still readable underneath it: with a plastic route from the intermediate cells, two stimuli that the compressed layer cannot tell apart drive two different motor cells, in a 17-cell case and on a 1,813-cell body alike, and over ten minutes with learning always on. The sheet is spatial, and a step costs what is lit rather than what exists -- with the external drive held fixed, eight times the cells cost 1.06 times the time -- so the only walls met on the way up are rules of our own design. Two eyes converge, focus and move to wherever the picture is changing without any of the three being written down, and two ears separate front from back and left from right from two raw waveforms, with no spectrum, delay estimate or angle computed anywhere outside the cells. Every sense channel is a pair of cells -- one carrying what came in, one carrying what did not -- so a black screen or a silent room is still a reading in the input layer rather than nothing at all. Two brains that differ only in what they were taught differ only in the numbers written on their edges: adding two such weight tables and halving them gives a third model that was never taught anything and holds both lessons, both on a 17-cell graph (both cues answered, margins +1.000 and +1.000, 5/5 seeds) and on the

1,813-cell body, where the merged animal walks 0.66 m on the cue one parent learned and braces on the cue the other learned, while a parent merged with itself reproduces its own behaviour to three decimals and two parents taught opposite answers to one cue merge into no answer at all. No rule lowers a weight in the configuration reported here (a global-decay knob exists, is off by default, and is reported in 4.6 as something that does not help), and the acquired walk does not yet stay upright; both are reported rather than hidden. The connection table also holds itself still without any decay term: over five minutes of ordinary life seven edges in ten do not move by more than 0.001, the largest change anywhere is a tenth of one per cent of a weight, and the table drifts up rather than down -- while a lesson written down the teachable route keeps 97-98% of its read-out after ten minutes of life, after ten minutes of the tone coming back, and after a second lesson on the same route, and a route that is used continuously climbs to its own ceiling and stops (1.98657, then 1.11257, then 0.02656 over three equal twenty-second stretches of teaching).

Author Summary

Most artificial intelligence today works like a machine that is fed examples and graded until it gets them right. A brain does not work that way. It is born with a great deal of wiring already in place, and the only rule its synapses can use is that cells which fire together tend to connect more strongly. I wanted to know how far that starting point alone can carry a body. To find out, I built a simulated brain of about 140,000 cells, gave it a fixed set of inborn connections, put it inside a simulated four-legged body, and let it run with no reward and no training. It got up from lying down, walked toward a red object it could see, and followed a moving object with its eyes. When I removed the inborn connections, the same brain could do nothing. When I silenced one part of it, the animal stopped using its eyes to walk but kept its reflexes. More recently I found that the animal can also be taught. If I lay a connection between two parts of its brain at a strength far too small to do anything, and then touch it while a tone plays, for twenty seconds, it learns that the tone means *back away* -- and if I never lay that connection, or never touch it, the same lesson teaches it nothing at all. The lesson only strengthens what the animal was already doing, which is why it needs the animal to already be doing something. Since then I have measured three things the first version had wrong or left out, and one thing it had not tried at all. The middle layers of a sensory network are not just a way station: they still carry the detail the last layer has thrown away, and I could make the body use it -- two things the last layer could not tell apart drove two different muscles, and that held for ten minutes with learning running the whole time. The sheet pays for what is switched on inside it and not for how many cells it has, so a bigger one is affordable, and "too small" turns out to be a measurement rather than an opinion: with ten cells per muscle, forces of 0.50, 0.51, 0.52 and 0.54 all produce the same pattern, and with a hundred they do not. The eyes now converge, focus and look at whatever is moving without any of that being written into the code, and the ears tell front from back and left from right from two raw sound waves. Every sense is also a pair of cells, one for what arrived and one for what did not, so a black picture or a silent room is something the brain is still reading rather than an absence. Two animals that each learned one different thing can also be added together: I averaged their two sets of connection strengths and got an animal that does both, which works only because each lesson had moved a different set of connections. The animal still falls over after a while, and I report that rather than hide it. A connection that is used a lot does not grow without limit, and one that is never used does not fade away: the strength of a connection climbs to a ceiling and stops, and an idle one settles back to the strength it was born with and stops there, so the whole table drifts neither up nor down without bound. The result suggests that what a brain is born with matters at least as much as what it learns.

1. Introduction

1.1 Two pictures of a layered network

The dominant picture of a deep network is a feed-forward transducer: a token or a pixel enters on the left, an answer leaves on the right (Brown et al., 2020), and the loss is back-propagated through the stack (Lillicrap et al., 2020). In that picture the intermediate layers are the computation and the output layer *is* the answer.

We work from a different picture, and the difference is not cosmetic. We state it as nine claims, (P1) to (P9), so that the results can be read as tests of specific ones.

- **(P1) A layered sensory network is a compressor.** It exists to take a very large number of peripheral channels and reduce them to a small number of cortical channels. Its output is not an answer; it is a bus. Reading the answer off the last layer is mistaking signal conditioning for thought. The anatomical version of the same idea -- a hierarchy of cortical areas in which each stage re-represents its input -- is classical (Hubel & Wiesel, 1962; Felleman & Van Essen, 1991; Mountcastle, 1997).
- **(P2) Thought is co-activation of cortical neurons** (Hebb, 1949; Hopfield, 1982; Amit, 1989; Wang, 2002). A thought is a sequence of populations, and the next population is fully determined by the current weights and the currents arriving on the somata. There is no separate inference stage. The testable consequence is sharp: remove the recurrent step and the behaviour should go with it, while anything the wiring maps directly from sense to muscle should survive untouched (R8).
- **(P3) The prefrontal population compresses again, and the path is reciprocal** (Fuster, 2001; Miller & Cohen, 2001). It takes the already-compressed signals of several other regions and reduces them further, and then thinks with exactly the same rules as everything else. Information is unavoidably lost in this second compression; that is a property of the design, not a defect. The compression is not one-way: the same pre-specified connections that carry a region *into* the prefrontal population carry its state back *out* to the regions that fed it. A prefrontal state can therefore re-enter the visual and auditory populations -- which is what imagining seeing or hearing something is in this model, and what is missing in aphantasia -- and re-enter motor memory and motor cortex, so that a state which means "walking" can start the walk with no stimulus present. One asymmetry is deliberate: the regions that report the body are read by the prefrontal population and are never written by it (2.1), because a fabricated body reading is not an idle thought -- it is an instruction.
- **(P4) Instinct is the initial state of the sheet** (Tinbergen, 1951). The innate part of the design is not a separate module and not a set of conditions in code: it is the connectivity the cortical sheet is born with, both within a region and between regions, and it includes the suppressive connections (a rule that must suppress something drives an inhibitory cell; there are no negative weights anywhere in the system). Everything the sheet does it does with that wiring, and what experience changes, it changes slowly and locally.
- **(P5) A network cannot start from silence** (Pfeifer & Bongard, 2006). A newborn cortex already has ordered wiring, because an organism that must discover standing, breathing and gaze stabilisation from an uninformative start does not survive the discovery period. The innate wiring provides an already-ordered starting repertoire; experience then writes content into it.
- **(P6) A capability can be acquired during the life of one brain, without an objective.** The innate wiring fixes the starting repertoire, but it is not the whole of what the sheet will ever do: the same local correlation rule that slowly adjusts the innate connections can also write new ones while the brain is behaving. If a reward-like population of cells is present at all, it is one more population -- a population whose firing licenses that writing -- and not an error signal computed from a target. What it buys is a link from something the brain can already sense to something it can already do (R9).
- **(P7) Every layer of a sensory hierarchy is a signal source, not only the last one.** P1 says that the last layer is a bus rather than an answer; it does not say that it is the only bus. Compression throws detail away, and detail a compressed layer has thrown away is still present in the layers underneath it, so a cell in any layer can be read by the rest of the cortex on the same terms as a cell in the last one. Two consequences are claimed with it. Depth is not a design constant: a layer with nothing to do costs building time and nothing else, so a hierarchy may carry more layers than it currently uses. And what a cell stands for is measured from what drives it rather than assigned by its place in the diagram -- intermediate cells may carry overlapping features, and no cell is a detector until a stimulus has been shown to select it.
- **(P8) The sheet is spatial, and a tick costs what is lit rather than how many cells exist.** A cortical cell has a position and sends mostly to cells near it plus a few that are not; a flat sheet puts two peripheral channels that are physically adjacent into unrelated places in the network, so the volume of a solid object never becomes a population of its own. Position is also what makes a larger sheet worth building: if the cost of a tick follows the number of cells *active* in it

rather than the number present, cells can be added until the resolution is sufficient instead of until the machine is full, and the fraction of the sheet that is afloat should *fall* as the sheet grows. R14 and R15 are the measurements of P7 and P8, and both are engineering statements that can be checked on one machine rather than scaling laws of the kind claimed for language models.

- **(P9) What one life writes can be combined with what another life wrote, without retraining.** If the instinct table and the local correlation rule are the whole of the machinery, then two brains built from the same table differ only in the numbers their own lives have written on their edges, and those numbers can be added. A third model that was never taught anything then holds both lessons. The claim is deliberately narrow: it is arithmetic on a weight vector over a shared edge list, it preserves what each parent already had, it creates no capability that neither parent had, and it needs the same cells and the same edge list on both sides. Two lessons that want the same edge in opposite directions are its boundary, and R18 measures both sides of it.
- **(P10) Stability is a property of the update rule, not of a decay term.** If the only things that move a weight are a local correlation and a pull back towards the weight the edge was born with, then an edge that is used should climb to its own ceiling and stop, an edge that is not used should return to the weight it was born with and stop there, and a table that is mostly idle should be mostly still -- with no term anywhere that lowers a weight because a network is growing too strong. R20 measures all three, and the shape it reports is the shape long-term imaging of cortical synapses reports: most of the table stable, a small minority moving, and movement bounded in both directions.

1.2 What this paper tests

If the picture above is right, then a system should be able to do real things with

- a single recurrent population of model cortical neurons,
- a fixed, pre-specified wiring diagram (which we call the *instinct table*), and
- a purely local plasticity rule that has no access to any task objective,

driving a real body. We therefore built one and asked nine questions, each of which has a falsifiable answer:

1. Does the innate wiring alone close the sensorimotor loop? **(R1)**
2. Can we attribute a behaviour to a specific population by silencing it? **(R2)**
3. Can the repertoire be *grown* by adding innate structure, without retraining and without destroying what is already there? **(R3)**
4. Does local plasticity buy anything that a fixed recogniser does not have? **(R4)**
5. Does hierarchical compression cost anything measurable, and does it cost anything behaviourally? **(R5, R6)**
6. While a behaviour is running, is it the stimulus or the cortical population that is producing it? **(R8)**
7. Can a capability be *acquired* during the life of one brain, with no objective and with nothing added to the system except a reward-like population of cells? **(R9)**
8. Does a layer that is not the last one carry information that the last layer has lost, and is the number of layers a constant? **(R14)**
9. How large does the sheet have to be before a behaviour is resolved, and what does a larger sheet cost? **(R15)**
10. Can two brains, each taught one different thing, be added into one brain that does both, and what happens when the two lessons want the same edge? **(R18)**

We also report five supporting measurements: gaze tracking that emerges from primitive reflexes with no tracking rule anywhere in the wiring **(R6)**; the tolerance of behaviour to *blurred* innate wiring **(R7)**; two eyes that converge, focus and move to wherever the picture is changing, with none of those three written down anywhere **(R16)**; ears that separate front from back and left from right without ever computing a source angle **(R17)**; and the addition of two brains that were taught different actions into one brain that performs both **(R18)**.

2. System

Figure 1 is a schematic of one tick. Senses drive two sensory hierarchies; the hierarchies drive one recurrent cortical sheet; the sheet drives the muscles. There is nothing else in the loop.

2.1 A single cortical population

The model cortex is one flat array of **148,032 binary neurons**. It is not divided into modules with separate code paths; the divisions are address ranges inside the array, and every region is updated by the same update rule on the same tick.

region	cells	role
visual cortex	23,040	output of the visual hierarchy (compressed)
visual detail	23,040	first layer of the same hierarchy (uncompressed)
visual motion	1,920	frame-difference / relay cells used by the gaze reflex
auditory cortex	30,000	output of the auditory hierarchy
prefrontal	57,436	second-stage compressor; it is fed by the other seven regions
motor cortex	160	16 muscles x 10 cells; the number of lit cells is the force
motor memory	3,576	time cells; lighting the first cell of an action plays the whole action
proprioception	480	joint-angle-derived body sense
vestibular	120	trunk tilt / tilt rate
body state	60	e.g. "is the trunk low", "is it lying down"
dopamine	200	reward signal; when lit, co-active synapses are written immediately
(only used in R9, silent otherwise)		
inhibitory pool	8,000	every suppressive instinct borrows a cell from here
total	148,032	

All 148,032 cells are updated simultaneously, once per 20 ms tick (50 Hz), by

a cell fires on this tick if the excitatory current it receives plus whatever else arrives on its soma exceeds a fixed threshold of 1.0, and it is not held below threshold by an inhibitory cell.

There is no separate "inference" pass and no read-out layer. The motor command is literally the activity of 160 cells in the same array.

One region departs from "each tick replaces the state". Inside each named block of the prefrontal population the cells are wired head-to-tail into a closed chain (6,566 synapses in seven blocks), and the main loop *adds* the newly computed prefrontal code to the code still running, so that a block outlasts the stimulus that lit it. We report R1-R8 with that persistence **switched off** -- the prefrontal code is recomputed from its input on every tick -- because that is the configuration those experiments were run in, and because it makes the causal reading of R2 and R8 unambiguous. R9, and the rows marked as persistent in R2 and R8, report both configurations side by side; Methods 5.5 gives the switch and why persistence is there at all: a code that cannot outlast its stimulus cannot be thought with (P2).

The reciprocal return line. The prefrontal population is not a sink. The seven regions are compressed into it through the network's own local connections -- three layers of 57,436 cells, 2,412,312 synapses born at fixed random weights -- and the same connection table is then used *backwards*, with the same fixed weights, to carry the prefrontal state out again (Methods 5.1). This is the model's imagination: a prefrontal state re-enters the visual and auditory populations and is seen and heard in the mind's eye -- cutting those two is the anatomical analogue of aphantasia and of the loss of inner speech -- and it re-enters motor memory and motor cortex, so that the prefrontal population can hold and steer an action and not only watch one.

Two properties of this path are decisions rather than consequences, and both are in the direction of keeping the animal from hallucinating a body that it does not have.

- **The return is thresholded once per region**, at 40% of that region's own peak, and not against a single peak taken over the whole returned vector. With one global threshold the two large regions (visual, 23,040 cells; auditory, 30,000) set the scale and a small one (vestibular, 120 cells) can never be reached at all.
- **The three body-sense regions are input-only**. The prefrontal population reads proprioception, vestibular state and body state, and never writes back into them. Those three regions are wired straight to behaviour, so a returned reading is not a thought one can have and discard: it acts. We measured it. With the return line open to all seven regions, an animal standing in front of a steady red image received 12-22 returned cells per tick into proprioception, 17 into the vestibular region and 13 into body state; that lit "tilted over -> stand up" and "stop -> keep standing" while the walk was already running, three actions were in motor memory at the same time, and it fell at tick 126. Imagining an injury does not hurt, and imagining a tilt must not straighten the body.
- **The return into motor memory sustains an action; it does not start one**. Motor memory is a timeline whose cells *are* actions: lighting one cell runs a whole 60-tick programme. The returned pattern is deliberately coarse -- a spread of "roughly what is being thought", not a code -- and a spread laid on a timeline crosses action boundaries. Measured, this is not hypothetical: a body placed prone stands up to 0.265 m and then sags back to 0.180 m, because on tick 51 the return sent back the *head cell of the next action along the timeline* ("lie down", whose first cell sits immediately after the last cell of "stand up"), and that programme started running next to the stance. It sinks the body because motor cortex encodes force by recruitment -- how many cells of a muscle's pool are lit -- so two actions lit at the same time give each muscle the **largest** of the two forces, not the average. We did not repair this by removing the path: that would leave the prefrontal population unable to reach the motor system at all, the opposite of what this section is about. The return into motor memory is instead intersected with the cells that motor memory is already running on that tick, so the route can hold an action but cannot start one. Starting an action is what the precise routes are for -- the innate 看着红 -> 前进1.2, and the visual/auditory/prefrontal synapses onto action entry cells, which are plastic. With that one change the same brain holds **0.261 m** and walks the same distance (1.85 m in the standard run). The write-up, including the variant that simply disconnects the route and why we rejected it, is in docs/修复_前额叶七路与返回线_20260918.md (section 7).

2.2 The instinct table

Behaviour-relevant wiring is written down as a table of rules of the form

```
source_block -> target_block    strength    frozen?
```

where a *block* is a named set of cells (a name is defined by a calibration procedure that lights a physical condition, e.g. "a red region is straight ahead", and records which cells responded). At load time each rule is expanded into per-cell synapses, with the stated strength divided evenly across the synapses of the rule; a strength of 2.0 therefore means "if the whole source block is lit, the target cell is driven to fire, and if only a small part of it is lit it is not".

The current table contains **585 rules**: 581 in the written table and 4 in a derived reflex table. They group as follows.

group	rules	what it is
motor repertoire	221	one rule per action the body can perform (10 static postures, 4 slow actions, 207 locomotion gaits). Writing one rule for an action wires the whole action: its time cells chain into each other so that lighting the first cell plays the action out.
balance / righting	6 + 4	trunk low or inverted -> stand up; leaning forward -> shift weight back; and 4 reflexes that hold a specific shank force above a floor when the body is tilted.
visually guided approach	6	red ahead -> forward (two rules, see R2); red on the left -> turn left; red on the right -> turn right
auditory	3	a broadband sound -> shuffle in place; a fixed narrow-band tone -> hold the standing posture and suppress the walking action
gaze	345	9 coarse direction rules, 14 per-column rules driven from the <i>uncompressed</i> visual layer, and 322 rules implementing the motion-onset reflex of R7

Expanding the table produces **309,148** excitatory synapses (plus 6,857 inhibitory ones). Everything else in the network is sparse random background connectivity generated from a fixed seed; we refer to different seeds as different *brains*.

Two of the encoding choices are old ideas in model form. A motor program that runs itself out once triggered, rather than being recomputed every tick, is the classical central-pattern- generator proposal (Grillner & Wallen, 1985); and "one cell per instant of the action" is a time cell in the sense of Eichenbaum (2014).

Three properties of this design matter for the interpretation of everything below.

1. **The innate wiring is authored, not learned.** Whoever writes the table is the "genome". For the locomotion repertoire the rules were distilled from a separately trained reinforcement-learning policy (see Methods 5.4); this is supervision that happened *before* birth, not during life.
2. **No rule is an if statement.** Action selection is competition between currents on the somata of motor-memory cells. Two rules can each be insufficient on their own and sufficient together (R2 uses exactly this), and the balance between regions is not fixed.
3. **Suppression is done by cells, never by negative weights.** Where a rule must suppress something (the tone that stops walking, the motion-onset subtraction) the table borrows an inhibitory cell and drives it, so the "negative" part of the diagram is itself a population of neurons that other rules can act on. Expanded, every synapse in the network has a positive weight; the minus sign exists only in the text file (Methods 5.1).

2.3 Plasticity

The only learning rule in the system is local and correlation-based (Hebb, 1949; Turrigiano & Nelson, 2004; Magee & Grienberger, 2020). Between consecutive ticks, pairs of cells that were co-active have their synapse strengthened, pairs that were not have it decayed, and a synapse that has been co-active often enough is *created* if it did not exist. The rule has a small fixed learning rate, saturating bounds, a co-activation requirement before a new synapse is created, and a per-tick cap on how many synapses can be created. It never sees a reward, a target, or an error. Frozen synapses in the instinct table are exempt, which is how we express "this one must not move" (for example the balance reflexes). All constants are listed in Methods 5.5.

2.4 Body and sensors

The body is a Unitree Go2 quadruped simulated in MuJoCo with joint-angle servos: the 160 motor cells decode to 16 target angles, and the physics runs ten MuJoCo sub-steps of 2 ms per cortical tick. The visual hierarchy receives a 1920x1080 first-person image, downsamples it to a 24x16 macro-pixel grid with three colour channels and 10 cell-pairs per level (one cell of a pair says its level has been reached, the other that it has not), so one visual cell subtends about 6.25 degrees horizontally. The field of view is 100 degrees horizontal by 75 degrees vertical. The auditory hierarchy receives a 4096-bin spectrum. Proprioception, vestibular and body-state regions receive hand-defined predicates over the joint angles and trunk pose. The eyes have two degrees of freedom with a 30 degree range, a strong viscous load and a muscle acceleration of 500 degrees/s², so a single tick of maximum force moves the eye about 0.1 degrees; sustained force over roughly ten ticks is needed to make a large saccade.

3. Results

R1. The innate wiring alone closes the loop (Figure 2)

We built a brain from the instinct table plus random background connectivity, attached it to the body, and gave it nothing but its own senses.

condition	measure	5 seeds
placed prone, no instruction	final trunk height	5/5 above 0.20 m; median 0.261 m (min 0.261) from a starting 0.099 m
placed upright, red region in front	forward displacement in 4 s	5/5 above 0.30 m; median +1.73 m (min +1.48, max +2.09)

condition	measure	5 seeds
placed upright, no sensory input at all	final trunk height / tilt	5/5 collapsed ; median 0.184 m, tilted 41 degrees
placed prone, instinct table cleared (background connectivity only)	final trunk height, then forward displacement with red ahead	0/5 above 0.20 m ; median 0.184 m, every seed exactly the no-sensation value; 0/5 approach (median -0.12 m)

(logs: 日志_闭环多种子_新.log, 日志_白脑_新.log; scripts 实验_闭环前提_多种子.py, 实验_白脑对照.py)

The third row is the control that matters. With the sensory regions blanked, the body does not "do nothing gracefully" — it falls over, because the motor cells are receiving no current from anywhere and the servo set-points collapse to rest. Nothing in the simulation holds the body up except the cortex. Standing up happens because the body-state and proprioceptive regions report *lying down*, and the instinct table contains the rule "lying down -> stand up".

Because the stand-up trajectory lives in the motor repertoire, the *height* reached is identical across seeds (0.261 m); what the random background changes is the walking distance, which varies between 1.48 and 2.09 m across the five seeds. We report this rather than a single run because the single-run version of this table appeared in an earlier draft of this project and was not reproducible evidence.

Standing up is not, in this body, a control problem, and R1 should not be read as if it were. Holding all twelve muscles at the constant vector 0.5 — the stance pose — brings the trunk to 0.25 m by tick 9 and leaves it at 0.261 m tilted 0.6 degrees, and the published RL policy of Section 4.4, driven on the same body, also recovers from the prone posture. What row 1 measures is therefore the *link*: the body-state and proprioceptive regions reporting "lying down" are wired, in the table, to the cells that emit that vector. Remove the table and the same body, the same muscles and the same seed stay on the floor (row 4). **The control that isolates the instinct table itself.** Row 4 is a within-seed ablation of the one thing the claims of Section 1.1 are about. Rows 1-3 leave open the possibility that the random background connectivity is doing the work and the measured table is decoration: the body, the sensors, the seed and the 100 sparse synapses per cell are all still present in row 4, and only the 585 rules are removed. On the same five seeds, with nothing else changed, not one brain stands up (every run ends at exactly 0.184 m -- the value the no-sensation control settles at, i.e. what the body does when nothing drives it) and not one approaches the red region (median displacement -0.12 m; the body drifts backwards while sagging). The behaviour in rows 1-2 is carried by the written wiring and not by the random background.

R2. The behaviour is computed by a specific population (Figure 3)

Rules in the instinct table are currents, not conditions. We exploited that to run an intervention: keep the visual input fixed (a red region straight ahead, identical image every tick), and silence the entire prefrontal population by forcing it to zero *after* it is computed and before the cortical step. Nothing else changes.

condition	ticks with the walking action lit (of 200)	first fire	path length	prefrontal cells active (mean)
red, prefrontal intact	199/200 (5/5)	tick 2	2.17-2.53 m	1460-1461
red, prefrontal silenced	0/200 (5/5)	never	0.24-1.41 m	0
black screen, prefrontal intact	0/200 (5/5)	never	0.55 m	359
blue screen, prefrontal intact	0/200 (5/5)	never	0.22-1.40 m	1554-1564
blue screen, prefrontal recomputed each tick	0/200 (5/5)	never	0.22-1.40 m	1524-1534

(logs: 日志_前额叶多种子_新A.log; script 实验_前额叶_多种子.py, five build seeds. Row 5 is the control in which the prefrontal code is *not* allowed to persist -- it is recomputed from its input on every tick.)

Two things are worth separating here. First, the ablation is *specific*: silencing a population of 57,436 cells removes a specific visually guided behaviour and leaves the sensory response itself intact — a blue screen still drives 1,524 prefrontal cells when the code is recomputed every tick, and 1,554 when it persists (rows 5 and 4), while the red screen that does drive the behaviour lights 1,460 cells in that same population — and the behaviour that needs that population's contribution is gone on 5/5 seeds.

Second, the mechanism is arithmetic rather than logical. The rule "red ahead -> walk" is written as **two** rules: the visual region contributes 0.80 and the prefrontal region contributes 0.85 to the same motor-memory cells. The firing threshold is

1.0, so neither half is sufficient alone and the sum is. Removing either half abolishes the behaviour. We deliberately do **not** call this an AND gate: the motor cell is a linear summing junction, the contributions from other regions can be present or absent depending on the situation, and the same behaviour could in another context be driven by one region alone. The ablation shows that *this* behaviour, in *this* situation, requires the prefrontal contribution; it does not show that behaviour in general is gated by the prefrontal cortex.

The path lengths in the four rows that read 0/200 are the body sagging and being dragged by whatever posture reflex is still running; the walking action itself never lights.

The same stimulus, with and without persistence. Rows 4 and 5 differ in one thing: whether the prefrontal code persists across ticks or is recomputed from its input on every tick. On the build reported here the two agree -- a blue screen delivers 0.72 to 0.74 of the 1.0 the walk needs and neither configuration crosses -- and that is a *change* from the previous build of this system, which is worth stating rather than quietly dropping. In the earlier build a blue screen with persistence reached a sum of 1.07 and started the walk at tick 7 to 18 on 5/5 seeds.

Three things about the prefrontal path changed between the two builds (2.1): the population is fed by seven regions instead of three; the return line is thresholded per region instead of against a single peak taken over the whole return vector; and the return into motor memory is now intersected with the action that region is already running. Any of the three can account for the difference, and we have not separated them, so we report the current behaviour next to the earlier number rather than attributing it. What the comparison does show is how much of the earlier effect ran through the return line: with persistence, a blue screen used to drive the 看着红 block to 0.99 of its cells and now drives it to 0.59 (the table below).

The two halves, measured separately. R2 shows that the behaviour needs both halves; it does not show what each half contributes. We measured both, tick by tick, in the same closed loop. Each row gives the mean fraction of the named block that is lit over 200 ticks, the drive that fraction contributes, the sum, and the walking tick count.

scene	视觉:中有红 lit (of 1346)	前额叶:看着红 lit (of 1440)	drive from the visual half (0.80 x)	from the prefrontal half (0.85 x)	sum	walking
<i>prefrontal recomputed each tick (the R1-R8 configuration)</i>						
red ahead	1.00	1.00	0.80	0.85	1.65	199/200 (5/5 seeds)
black	0.00	0.00	0.00	0.00	0.00	0/200 (5/5)
blue	0.29	0.57	0.24	0.48	0.72	0/200 (5/5)
green	0.32	0.60	0.26	0.51	0.77	0/200 (5/5)
<i>with the prefrontal loops (the R9 configuration)</i>						
red ahead	1.00	1.00	0.80	0.85	1.65	199/200 (5/5)
black	0.00	0.11	0.00	0.09	0.09	0/200 (5/5)
blue	0.29	0.59	0.24	0.50	0.74	0/200 (5/5)
green	0.32	0.61	0.26	0.52	0.78	0/200 (5/5)

(logs: 日志_两半_无环.log for the first block and 日志_两半_新.log for the second, same script, 5 seeds each, both re-run on the build of this paper; 实验_两半各亮多少.py; the firing threshold is 1.0)

The visual half is the same in both blocks, because it is the front end: near-binary (1.00 for red, 0.29-0.32 for blue and green, 0.00 for black). The prefrontal half is the one that moves, and on this build it barely moves with persistence: 0.57 against 0.59 for blue, 0.60 against 0.61 for green. Persistence therefore no longer drives it into saturation, the sums rise only from 0.72 to 0.74 and from 0.77 to 0.78, and neither colour crosses the threshold. The one thing persistence does change in this table is the black row: with nothing on the screen at all, the block drifts up to 0.11 of its cells over the run -- activation that no sensory input is producing -- which is the accumulation R9 reports as the unfinished part of the system.

Two things follow. First, in the configuration of R1-R8 both halves are colour-selective but not equally so: the visual half is close to binary for this colour (1.00 for red against 0.29 for blue), while the prefrontal half is a partial responder (1.00 against 0.57). Neither half reaches 1.0 by itself for any colour in this table; only the sum does, and only for the colour the instinct was calibrated on. That is the mechanism of the approach instinct in full, and it is arithmetic on the somata of the action's first time cell, not a condition in code.

Second, this is the quantitative reason the ablation above is specific rather than catastrophic. A blue screen still drives 1,524 prefrontal cells; but for *this* rule the two halves deliver 0.72 on that tick, which is short of 1.0. The system is not deciding "not red" -- it is falling short of a threshold.

R3. The repertoire grows by adding innate structure, not by retraining (Figure 4)

If behaviour is built from innate wiring, then adding wiring should add capability — and, critically, it should not cost the capabilities that are already there. We tested this by building five nested instinct tables **on the same random background connectivity** (same build seed, so the only thing that differs between brains is the table):

stage	what is added	rules in the staged brain
A	motor repertoire only	221
B	+ balance / righting premises	227
C	+ visually guided approach	233
D	+ auditory reactions	236
E	+ gaze (static, per-column, motion-onset)	581

The four derived reflex rules (Methods 5.3) are loaded from a separate file and are **not** part of the staged subsets, so the staged brains are built from the written table alone; the counts above are the counts printed by the script, not the counts in the grouping table of Section 2.2.

At every stage we re-measure **all** behaviours, not only the new one. The measures are the protocols of 5.7: standing up from prone (final trunk height), walking toward a red region (the walking action's time cells must actually fire; distance alone is not accepted, since a collapsing body also travels), reacting to a broadband sound, being stopped by the narrow-band tone *while* red is present, and holding the gaze on a small moving ball.

stage	rules	stand up	walk to red	react to sound	stopped by the tone	gaze tracking
A motor repertoire only	221	0/3	0/3	0/3	0/3	0/3
B + balance / righting	227	3/3	0/3	0/3	0/3	0/3
C + visually guided approach	233	3/3	3/3	0/3	0/3	0/3
D + auditory reactions	236	3/3	3/3	3/3	0/3	0/3
E + gaze	581	3/3	3/3	3/3	0/3	3/3

(log: 日志_分阶段_新.log; script 实验_本能分阶段.py; three build seeds per stage, and all five behaviours re-measured at every stage)

The medians behind the matrix: standing up leaves the trunk at 0.184 m at stage A and 0.261 m from stage B onwards; the walking action is lit on 0 of 200 ticks at stages A and B and on 199 of 200 from stage C; the sound response is 0 ticks at A-C and 100 ticks from D; gaze error is 15.3 degrees (unwired) at A-D and 5.6 degrees at E. The no-sensation control collapses to 0.184 m at every stage, as it should.

Three things in this table are worth stating explicitly.

The repertoire is not the behaviour. With the motor repertoire alone (stage A) the brain contains the standing-up action but never performs it: the body is placed upright and simply falls over, exactly as it does when all sensation is removed. A capability that exists in the wiring is not a capability until something lights it. This is the cleanest statement in the paper of why a brain cannot start from silence.

Each addition buys exactly the capability it encodes and nothing else. Stages C, D and E add 6, 3 and 345 rules respectively; each addition turns on its own behaviour without turning on any other.

Additions do not damage the earlier repertoire. Every capability that is present at one stage is still present at the next. Standing up, once it appears at B, passes 3/3 at C, D and E; walking, once it appears at C, passes 3/3 at D and E; the sound response, once it appears at D, passes 3/3 at E. Adding the 345 gaze rules does not disturb the two rules that drive walking, and adding the six approach rules does not disturb the rules that stand the body up. The stages are nested in behaviour, not only in the table.

One cell of the matrix fails. "Stopped by the tone" is 0/3 at every stage, including E, where all 585 rules are present. The tone *does* stop the walking: with red alone the walking action is lit on 199 of 200 ticks, and with red and the tone together it is lit on 15 of 200 (median), and the body travels 0.11 m instead of 1.83 m. Fifteen ticks is one further cycle of the fifteen-tick walk, so the action terminates within one cycle of the tone arriving. It nevertheless fails the criterion we fixed in advance (at most 10 of 200 ticks) and we report it as a failure rather than move the threshold afterwards..

R4. What sets the boundary of recognition, and how far local plasticity moves it (Figure 5)

A recogniser that is fixed -- a fixed set of weights, a threshold, a match -- has a fixed tolerance. A synapse that follows a local correlation rule has no tolerance, only a rate: it follows whatever its two cells have been doing. We tested whether that difference has behavioural consequences. The experiment produced a result we did not predict, and the result is more informative than the prediction was.

Protocol. The hue of a full-field colour starts at 0 degrees (red, the colour the approach instinct was calibrated on) and is then stepped by a fixed increment of 3, 6 or 12 degrees. After each step the walking action is **stopped** (the cortical state is cleared) and the current colour is presented afresh for 8 ticks; if the action re-lights on at least 6 of those 8 ticks the colour counts as recognised and the sweep continues, otherwise the sweep stops and we record the hue at which it stopped. Restarting the action at every step is what makes this a test of recognition: a walking action that has been lit chains its time cells into each other and runs by its own dynamics (R8), so "still walking" would measure inertia.

The boundary, before and after. The instinct that starts the walk is the sum of two cell populations (R2), so which colours start it ought to follow from those two populations. We measured both, and then measured, in the same **five** brains, whether each hue starts the walk -- once before any drift, and once after a 3-degree sweep has run to its stopping point. The "before" columns are the control for the "after" columns: the same brains, the same weights, the same stimulus schedule, with nothing but that one sweep between them.

hue	视觉:中有红 lit (of 1346)	前额叶:看着红 lit (of 1360)	visual half	prefrontal half	sum	seeds that start the walk, before the drift	after it
0 deg (red)	1346	1357	0.80	0.85	1.65	5/5	5/5
24	1192	1248	0.71	0.78	1.49	5/5	5/5
48	1149	1248	0.68	0.78	1.46	5/5	5/5
72	1038	1243	0.62	0.78	1.39	5/5	5/5
96	663	1051	0.39	0.66	1.05	4/5	5/5
120 (green)	367	844	0.22	0.53	0.75	0/5	0/5
144	439	959	0.26	0.60	0.86	0/5	0/5
168	529	1106	0.31	0.69	1.01	4/5	5/5
180 (cyan)	565	1111	0.34	0.69	1.03	4/5	5/5
204	507	1070	0.30	0.67	0.97	0/5	0/5
240 (blue)	337	794	0.20	0.50	0.70	0/5	0/5
black screen	0	3	0.00	0.00	0.00	0/5	0/5

(logs: 日志_色相红块_新.log for the two populations (identical to the pre-loops 旧日志_20260916_加环前/日志_色相红块.log), 日志_渐变边界_无环.log for the behaviour; scripts 诊断_色相_红块.py, 实验_渐变_边界.py. R4, like R1-R3 and R5-R8, was measured with the prefrontal non-persistent -- 前额叶不保留=1, Methods 5.5.)

Two things are in this table, and they came apart.

The boundary is where the sum crosses, and it is there in a brain that has never learned. Every hue whose two halves sum to 1.01 or more starts the walk on **4-5 of 5** brains; every hue whose halves sum to 0.97 or less starts it on **0 of 5**. Nothing else predicts the column, and the pattern is monotone in the sum and *not* monotone in hue: cyan at 180 degrees, the opposite of red, comes closer to starting the walk (sum 1.03) than green at 120 degrees (sum 0.75), which is only half as far around the circle. The column is also not something learning built. With the correlation rule of 2.3 switched off entirely -- no weight changes at all, before the probe or during it -- the same five brains give the same column: 96, 168 and 180 degrees recognised on four of five, every hue at 0.97 or below silent on all five (日志_色相点亮_无环不学.log, 诊断_色相点亮_前测.py with 不学=1).

The drift moves one brain, and it is the brain that was sitting on the line. The three hues that change between the two columns are exactly the three whose sums lie inside 0.05 of the threshold: 96, 168 and 180 degrees, going from 4 of 5 brains to 5 of 5. The brain that changes is seed 20260918, which sat at 0 of 8 ticks on all three of those hues before the drift and at 7 of 8 after it. Every hue at 0.97 or less stays silent in both columns, and no hue moves in the other direction. The drift does move the boundary, then -- in one brain of five, at the three hues that were within a twentieth of the line.

The sweep measures how far the rule reaches. The protocol above also gives a single number per run: how far the colour can drift before recognition stops.

step size	plasticity	hue at which the sweep stopped (one value per seed, in seed order)	median
3 deg	on	96, 96, 96, 96, 96	96
3 deg	off	96, 96, 96, 96, 72	96
6 deg	on	102, 102, 102, 102, 102	102
6 deg	off	102, 96, 102, 102, 72	102
12 deg	on	96, 96, 96, 96, 96	96
12 deg	off	96, 96, 96, 96, 72	96

(logs: 日志_渐变正式_新.log for the 3-degree rows and 日志_渐变正式_6和12.log for the rest, both on the build of this paper; script 实验_渐变_正式协议.py; the same non-persistent configuration as the table above.)

The brain that moves at every step size is seed 20260918, and it is the one whose sums at 96, 168 and 180 degrees lie within 0.05 of the threshold. With the rule off it stops at 72 degrees; with the rule on it reaches the top of the grid -- 96 at a step of 3 and of 12 degrees, 102 at 6 -- and the tail probes of that run then start the walk at 180 degrees on 7 of 8 ticks, where with the rule off they start it on 0 of 8. One further brain, seed 20260915, moves at the 6-degree step and not at the other two, and the difference between those runs is the grid rather than the rule: the sum of the two populations is not monotone in hue (it falls from 1.05 at 96 degrees to 0.75 at 120 and rises again to 1.03 at 180), so the largest grid point below the boundary is 96 degrees at a step of 3 or 12 and 102 degrees at a step of 6, and whether a brain crosses at 102 depends on which side of the line its sum at 102 degrees falls. Counting how many brains move across step sizes would be counting grid points rather than brains.

So the rule carries a marginal stimulus to the edge of its island, and no further. On the build of the main table, four brains of five already sit just above the line at those three hues, so a rule that can push a marginal stimulus across the line has almost nothing left to push, and the effect shows up in one brain of five at a step of 3 or 12 degrees and two at a step of 6. On the earlier build all five brains sat just *below* the line at the same three hues, and there the same rule carried all five out to 96 degrees and across the gap to 168 and 180. In both builds the rule moves the boundary outward as far as the edge of the visual island the colour belongs to -- 96 degrees, where the visual half of the sum falls from 0.39 to 0.22 -- and in neither build does it cross the gap at 120-144 degrees, which stays refused by every brain in every condition. The prediction we started from -- that a synapse with no tolerance, only a rate, would widen recognition wherever its cells had

been co-active -- is therefore **half right, and the half that fails is the instructive one**: the rule does widen recognition, but only over the island the sensory front end already formed, and only for stimuli whose drive is already within a few hundredths of the line, so on the build we report it is visible in one brain of five rather than in all five. Learning fills in a boundary; it does not draw one.

Why we quote two builds of the system. A hue whose two halves sum to 1.05 is 0.05 above the threshold; a hue at 1.01 is 0.01 above it. Whether a hue that marginal ignites is decided by the random background connectivity the brain happens to be born with, and by nothing we control. We know this because we measured the same protocol twice, on two builds that differ by 200 cells (the dopamine region of R9) and by the draw of that background connectivity. On the earlier build the same five brains sat *below* the line at all three of those hues before any drift -- 0 of 5 at 96, at 168 and at 180 -- while with the correlation rule off their sweeps stopped between 48 and 72 degrees, and with it on between 84 and 96: 96 degrees in 14 of the 15 runs, across step sizes of 3, 6 and 12 degrees. On the build of the main table, four brains of five sit above the line at the same three hues and one does not, and only that one moves. The measurement of the *sums* is stable across builds; which side of the line a 1.05 hue falls on is a draw, and it is that draw which decides how much work the rule has left to do. We therefore make the claim about the sums and about the reach of the rule rather than about how many brains will show it: the boundary of a behaviour assembled from two populations is the value at which those populations sum past threshold, and the local rule can carry a stimulus sitting within a few hundredths of that value across the line and as far as the edge of its island, and no further.

And the population it all rests on is not a concept. The same table shows that the block the calibration labelled "there is red in front of you" is lit on **42 percent** of its cells by a cyan screen and on 25 percent by a blue one. What was calibrated is not the category *red*: it is "these cells responded to this image and not to an empty one". This is the point of R5 measured on colour instead of direction, and it is why the behaviour's boundary is nothing more than wherever the sum of two populations crosses a threshold. Nothing in the system represents *red*.

The recorded value is the last hue the sweep still recognised, so it understates the tolerance by at most one step.

R5. Hierarchical compression has a measurable representational cost (Figure 6)

The visual hierarchy has four parameter layers, but the third and fourth are the same tensor: the value returned by the forward pass *is* the last exposed layer. There are therefore three distinguishable stages: layer 1 (uncompressed, 23,040 cells), layer 2, and layer 3 (the compressed output, 23,040 cells, which is what the rest of the cortex receives as "vision").

We measured, for a ball subtending **0.31 of a visual cell** (radius 0.05 m at 3 m), which cells in each layer newly respond relative to an empty scene.

direction	ball's column	layer 1	layer 2	layer 3 (compressed)	input layer (thermometer code)
left 3	2	7 cells, columns 2	5	7 cells, columns 1-10	11
left 2	4	5 cells, columns 3-4	5	4 cells, columns 4-12	11
left 1	6	6 cells, columns 6	3	6 cells, columns 3-15	11
centre	8	11 cells, columns 8	3	0 cells	11
right 1	9	10 cells, columns 9	1	1 cell, column 1	11
right 2	12	7 cells, columns 12	0	4 cells, columns 3-14	11
right 3	14	11 cells, columns 14	5	8 cells, columns 4-15	11

(log: 日志_前几层.log; script 实验_前几层看得见小东西吗.py)

Two things are visible. First, the uncompressed layer localises: every one of its responses lies inside the column the ball is in. Second, the compressed layer does not: its responses are scattered over as many as 13 columns, and for a ball at dead centre it produces **no new cells at all** — the object is simply not there, as far as the compressed representation is concerned. Note that the *input* layer (before any weights) sees the ball perfectly well in every case (11 cells, always) — so the loss happens inside the hierarchy, not in the image.

We then quantified the selectivity of the direction names themselves, using the same calibration procedure that the instinct table uses.

layer	cells lit by an empty scene	columns with a direction name	median cross-column sharing	worst columns
1	7293	14 / 16	0.00	all columns 0.00
2	5054	13 / 16	0.00	all columns 0.00
3 (compressed)	4030	13 / 16	0.14	0.75, 0.67, 0.50, 0.33, 0.25

"Cross-column sharing" for column *c* is the fraction of the cells in column *c*'s name that also appear in some other column's name (0 = the name is exclusive to that column, 1 = every other column also lights it). Layers 1 and 2 are perfectly exclusive. In the compressed layer the names overlap by 14% at the median and by up to 75%, and one column (the centre) has no name at all. (log: 日志_方位分辨率.log; script 实验_方位分辨率_按层.py)

We also tried to show the behavioural consequence and it did not come out as predicted. The prediction was that a gaze reflex re-wired to the compressed layer would stop working. We re-calibrated the same per-column names at each layer, wired the identical reflex to each in turn, and placed the same small ball in all 16 columns:

reflex wired to	columns where the layer has a name for that direction	columns the eye turns toward
nothing (control)	0 / 16	0 / 16
layer 1	14 / 16	10 / 16
layer 2	13 / 16	9 / 16
layer 3 (compressed)	13 / 16	10 / 16

(log: 日志_分层.log; script 实验_追踪靠的是哪一层.py)

The compressed layer still supports saccades. The reason is that a "blurred" name is still a *selective* name: the cells that respond to a ball in column *c* are scattered in space, but they still respond preferentially to column *c*, and in this reflex it is the identity of the active name that matters, not where its cells sit. So the cost of compression in this system is representational precision, not the loss of the function. We report the prediction failing rather than dropping the experiment, because it defines the boundary of the claim: *the compressed layer is not the answer, but neither is it useless*. What the measurement does establish is the thing the framing needs — that the amount of information about the outside world carried by the last layer of a hierarchy is strictly smaller than what the first layer carries, and that "read the answer off the last layer" would therefore be reading the wrong place.

R6. Gaze tracking is emergent, not written down (Figure 7)

Nothing in the instinct table refers to movement, velocity, or tracking. The gaze behaviour is assembled from two groups of rules that only refer to *where something is right now*:

- 1. 14 static rules:** "column *c* contains something -> pull the eye muscle that turns toward column *c*" (strength 2.0). These keep pulling as long as the object is off-centre, which is what makes the eye *hold* an object. The two columns straddling dead centre are deliberately left unwired, so a centred object produces no pull and the eye settles instead of oscillating.
- 2. 322 motion-onset rules:** for each column, column *c* -> trace_*c* (arrives one tick late, strength 2.0), column *c* -> change_*c* (+1.2) and trace_*c* -> change_*c* (-1.2, implemented with an inhibitory cell), so that change_*c* fires only if column *c* is active *now but was not one tick ago* — i.e. only on the onset of something. change_*c* then drives a chain of ten relay cells, and every relay cell drives the corresponding eye muscle (1.5).

The relay chain exists because the eye is heavily damped: one tick of maximum force moves the eye about 0.1 degrees, so a single-tick onset event would be invisible in the eye's motion. The relay stretches the onset into a ten-tick (0.2 s) pull. Functionally, the onset path is the "something moved — look at it" signal and the static path is the "keep looking at it" signal.

Measured behaviour:

scene	mean angular distance between the gaze centre and the object	eye at the end
0.05 m ball crossing the field from -25 deg at 0.5 m/s	5.6 deg (max 29.2 while the object is still far off)	+24.6 deg
0.05 m ball suddenly appearing at +20 deg, 2 s in	4.8 deg	+21.4 deg
empty scene for the whole trial	0.0 deg — the eye does not move at all	+0.0 deg

(logs: 日志_眼睛跟随.log, which is re-run from 实验_眼睛跟随.py every time Figure 7 is generated, so the numbers above are from the build of this paper; the same run written tick by tick is in 日志_眼睛_空场.log, 日志_眼睛_突然出现.log and 日志_模糊随机2.log)

The third row is the important control: with nothing in the field the eye is perfectly still, so rows 1 and 2 are not an artefact of a jittery or spontaneously drifting eye. Rows 1 and 2 also show the transformation that the onset path performs — a sequence of very weak, time-extended signals (the object is in column c , then column $c+1$, then $c+2$...) is converted into a single strong signal at one moment.

R7. A shifted innate map produces ordered behaviour with a systematic bias (Figure 8)

"Blurred innate wiring" is hard to test by randomising connections because randomisation destroys direction information as well as precision. We therefore used a cleaner manipulation: take every direction-bearing source name in the instinct table (细_列 c , and the trace_/change_/relay k _ names derived from them) and shift it by a whole number of columns, leaving everything else untouched. This is what "the inherited map is roughly right but not exactly right" looks like.

Ball placed straight ahead, 4 s, eye free:

shift	rules changed	walking still works	final eye angle	where the object ends up relative to gaze centre
-3 columns (-18.8 deg)	222	7/8 ticks	+16.4	-16.4
-2 columns (-12.5 deg)	246	7/8 ticks	+10.7	-10.7
-1 column (-6.2 deg)	291	7/8 ticks	+3.6	-3.6
0 (baseline)	0	7/8 ticks	+0.0	+0.0
+1	291	7/8 ticks	+0.0	+0.0
+2	246	7/8 ticks	-6.4	+6.4
+3	222	7/8 ticks	-12.9	+12.9

(log: 日志_整体偏移_正前方_新.log, re-run on the current code in the non-persistent configuration, reproducing every entry of this table; script 实验_本能表整体偏移_正前方.py)

Outside the plateau, the resting point moves by one cell for every cell of imposed shift — 6.4 degrees, within measurement noise of the 6.25-degree cell width — in both directions. The plateau at shifts of 0 and +1 is a consequence of the dead zone being two cells wide: while the object is inside the (shifted) dead zone it receives no pull, so the eye does not move. Throughout, the behaviour remains ordered: the eye turns, stops, and leaves the object somewhere in the field; what changes is *where*. With the same manipulation but the object placed 25 degrees off to the side, the same linear relationship holds for positive shifts while negative shifts drive the eye into its 30-degree mechanical stop (日志_整体偏移2.log).

The same idea, done the destructive way, gives a dose-response curve for how much of the innate wiring a behaviour can lose. We re-wired a fraction p of the 585 rules, either to a **random** other name in the same region (which destroys the direction) or to the name **one column over** (which keeps the direction but blurs it). Within one draw the substitutions are nested -- a larger p replaces everything a smaller p replaced, plus more -- and each condition was drawn three times independently, so the score is how many of three draws still work.

re-wired	walk to red	react to sound	gaze tracking
none (baseline)	1/1	1/1	1/1
random, 10%	3/3	3/3	3/3
random, 25%	3/3	3/3	0/3
random, 50%	0/3	3/3	1/3

re-wired	walk to red	react to sound	gaze tracking
random, 75%	1/3	3/3	0/3
random, 100%	0/3	3/3	0/3
one column over, 25%	3/3	3/3	1/3
one column over, 50%	0/3	3/3	1/3
one column over, 75%	1/3	3/3	1/3
one column over, 100%	0/3	3/3	0/3

(log: 日志_存活曲线_新.log; script 实验_模糊化_存活曲线.py; 3 independent draws per cell, and the baseline was run once because all draws coincide when nothing is replaced)

The pattern is not the one we expected, and it is more informative for it. The number of rules a behaviour is built from does not predict its tolerance. Walking is driven by exactly two rules that each contribute half the threshold, and it survives losing a quarter of the table; the gaze reflex is driven by 336 rules and fails on all three draws once a quarter of the table has been re-wired, because each of its rules names a specific visual column, and a destroyed column is a direction the eye no longer knows at all. What survives everything is the auditory pathway -- a single rule whose source is one named block, still working on 3/3 draws after *every* name in the auditory region has been replaced by some other name in that same region. The broadband sound is a broad stimulus; whatever the rule now points at, the stimulus lights it. Redundancy in the *stimulus* buys more robustness than redundancy in the *rules*.

R8. The stimulus starts the behaviour; the cortex runs it (Figure 9)

R2 showed that a named cortical population is necessary for a behaviour. It left open *where the behaviour is* while it is running. We therefore measured the behaviour and the two cortical codes that produce it, tick by tick, and then removed the stimulus.

A red region is present for the first half second and is then replaced by a black screen. Walking is counted by the same criterion as everywhere in this paper (the walking action's time cells are lit); the two codes are counted as the number of active cells inside the named blocks.

condition, after t = 1.0 s	walking action lit	视觉: 中有红 (1346 cells)	前额叶: 看着红 (1360 cells)
<i>prefrontal recomputed each tick (the R1-R8 configuration)</i>			
red throughout	495 / 495	1346	1360
red removed at 0.5 s	495 / 495	0	3
red removed at 0.5 s, action's time cells cleared once at 1.0 s	0 / 495	0	1
never red, same clearing at 1.0 s	0 / 495	0	2
never red (control, no clearing)	0 / 495	0	3
<i>with the prefrontal loops (the R9 configuration)</i>			
red throughout	495 / 495	1346	1360
red removed at 0.5 s	495 / 495	0	1360
red removed at 0.5 s, action's time cells cleared once at 1.0 s	0 / 495	0	1360
never red, same clearing at 1.0 s	0 / 495	0	148
never red (control, no clearing)	0 / 495	0	148

(5 seeds each; logs 日志_想还在吗_无环.log and 日志_想还在吗_有环.log, per-tick traces 日志_想还在吗_轨迹_无环.csv and 日志_想还在吗_轨迹_有环.csv; script 实验_想还在吗.py)

Row 2 makes the point in both configurations, and the difference between them is what R9 is about. The stimulus is gone, the visual code is at zero in both -- and the walking action is still lit on **every one of the 495 ticks**. What differs is the prefrontal code: 3 cells of 1360 when it is recomputed every tick, and all **1360** when it persists. In the persistent configuration there is literally a code for *red is there* running in the cortex while there is no red; in the non-persistent one there is not, and the walking continues anyway, because what is running the walk is the action's own trajectory.

Row 3 is where the two part company, and it is worth being careful about it. Clearing the action's fifteen time cells once stops the walk for good in both configurations (0 of 495 ticks) -- but for different reasons. Without persistence there is simply no drive left: the prefrontal code is at one cell. With persistence the thought is still fully lit, all 1360 cells, and the walk still does not come back -- because the drive that starts the walk is the sum of two halves, 0.80 from the visual region and 0.85 from the prefrontal (R2), and with the visual half at zero the surviving 0.85 is below the threshold of 1.0. The persistent code holds the thought; it cannot on its own re-start the action.

The last two rows of the second block also show what persistence costs. With a black screen and no red anywhere in the run, the block drifts up to 148 of its 1360 cells over three seconds -- activation that no sensory input is producing -- and it does not start the walk (0 of 495 ticks). This is the same accumulation that R9 reports as the unfinished part of the system.

The reading we take is that after the first half second the movement is not being produced by the stimulus at all. The stimulus *selected* an action; the action is a trajectory through the cortical population, and once selected it is the trajectory that produces the movement. This is also why the recognition protocol of R4 stops the action and clears the cortex between probes: "still doing it" measures the trajectory, and a trajectory runs itself.

The converse intervention. If behaviour is a trajectory through a recurrent population, then removing the recurrence should remove the trajectories -- and leave alone anything the instinct table wires directly from sense to muscle. We kept the instinct table exactly as it was and cleared the cortex at the start of every tick, so that only the current sensory signals were injected: a purely feed-forward arc, sense to cortex to muscle, built from the same wiring.

behaviour	intact cortex (3 seeds)	recurrent step removed
stand up from prone, final trunk height	0.261 m (3/3)	0.184 m (0/3)
red ahead: ticks with the walking action lit, of 200	199 / 199 / 199	0 / 0 / 0
small moving object: mean gaze error	5.6 deg (3/3)	5.6 deg (3/3)

(log: 日志_切断回响_有环.log; the same numbers are obtained in either configuration -- the control clears the whole sheet at the start of every tick, so persistence has nothing to act on; script 实验_切断回响.py)

The manipulation is not a blunt instrument, and the third row is how we know. The gaze reflex is wired as a direct map from a visual column to an eye muscle; it is unchanged **to one decimal place**, 5.6 degrees in every seed and in both conditions. Everything that consists of *doing an action* disappears: the final height of the no-recurrence body, 0.184 m, is the same value that R1's no-sensation control reaches, because the standing-up action never runs and the body does what it does when nothing drives it. (Its forward displacement is likewise not a measure: the no-recurrence bodies "travel" between -0.25 m and +1.08 m by sagging and being dragged.)

The dissociation is the point, and it is also the answer to the falsifier we wrote down in 4.5: a same-wiring feed-forward controller does not reproduce the behaviours, but it does reproduce the reflexes, which localises what the recurrent step is for. In this system, reflexes are wired and actions are trajectories.

R9. A reward-like signal acquires a new link between a sense and an innate action (Figure 10)

R1-R8 measure a repertoire that is present at birth. The claim this project is built on -- that complex behaviour is assembled on top of innate behaviour rather than trained from nothing -- is a claim about *development*, and the experiment that bears on it is one in which a capability appears during the life of one brain. R9 is that experiment. It adds two things, and it measures what each of them buys.

(a) *A 200-cell dopamine region.* It is silent in every other experiment in this paper. When it is active, a connection is written *on the spot* between cells that were active on the previous tick and cells that *became* active on this tick -- the same "who brought whom in" relation the slow correlation rule of 2.3 uses, cashed immediately instead of after four co-activations. Three constraints were each forced by a measurement. (i) Cells active on both ticks are excluded: with them in, 60 teaching ticks wrote 1.15 million connections and the auditory block grew from 609 active cells to 2202 of 2749 -- the population collapsed onto itself. (ii) Each target receives at most eight sources per reward and the drive is divided equally among them, exactly as the instinct table divides a block's drive among its targets: without this, a target received 300 units where the walk needs 0.85, and any cell that happened to light up once was written full and stayed lit. (iii) The

fast path is multiplied by the same per-region plasticity factor as the slow path (0.02 into the motor regions, see 2.3): without it, 200 teaching ticks wrote 260,000 connections into the motor cortex itself and the gait came apart -- three of five seeds fell after about a metre.

(b) *Loops inside the prefrontal population.* Inside each named block of the prefrontal population the cells are wired head-to-tail into a closed chain (6,566 synapses in seven blocks) with the same current as any other instinct, and the main loop *adds* the newly computed prefrontal code to the code already running instead of replacing it, so that a named block stays lit after the stimulus that lit it has stopped being present. The loops are written only in the prefrontal population: a loop in the motor-memory chain would block the sequence. The reason for adding this is the framing of 1.1 -- a code that cannot outlast its stimulus cannot be thought with. R9 measures separately whether the loops are needed for *acquiring* an association (they are not) and for *acting* on one (for standing up, they are).

Protocol: four stages, one brain, nothing rebuilt in between.

stage	body and senses	dopamine	walking action lit
1 birth	prone, black screen, broadband sound only, 30 ticks	off	0 / 30 (5/5 seeds)
2 innate	standing, red region ahead, no sound, 30 ticks	off	30 / 30 (5/5)
3 teaching	standing, red ahead <i>and</i> sound, 200 ticks	on	running
4 test	cortical sheet cleared, prone again, black screen, sound only , 200 ticks	off	200 / 200 (5/5)

(logs: 日志_发育多种子_A.log, 日志_发育多种子_B.log; script 实验_发育_声音叫走路_多种子.py. The dopamine-dark control is 实验_发育_无奖励对照.py (日志_无奖励对照.log); the no-loops control is 实验_发育_无环对照.py (日志_无环对照.log, 日志_无环对照B.log); the gait control is 诊断_天生步态稳不稳.py (日志_天生步态.log).)

Stages 1 and 4 are the *same stimulus on the same brain*, before and after. In stage 1 the sound lights its own block -- 18 to 34 of the 1097 cells of the named block 听觉:低频响 -- and nothing else happens: the walking action is lit on none of the 30 ticks, and the body only settles onto the floor (0.030-0.049 m). In stage 4 the walking action is lit on essentially every tick, and the body, which has been placed on its belly, first stands up (trunk height 0.118-0.156 m to 0.300-0.396 m, 5/5 seeds) and then walks 0.55-1.23 m.

Teaching writes **242,685-396,607** new excitatory connections on top of the 309,148 innate ones, and they land where the teaching happened: auditory 121,510-147,104, proprioceptive 103,491-120,985, visual 39,233-40,548, vestibular 6,634-11,172, motor 3,882-3,935, body state 1,635, motor memory 138-197, visual motion 168, prefrontal 40. Nothing here is written by hand and there is no teacher signal; the only difference between stage 1 and stage 3 is that 200 cells were lit while the behaviour was running.

The control the claim needs. The same four stages, with the dopamine region left dark: teaching writes **0 new connections**, and stage 4 stays at **0/200 ticks** of walking on 3/3 seeds -- the body never stands up (final height 0.103-0.205 m, which is the prone height). It is the reward-like signal, not the passage of time and not the sound, that does the writing.

What each addition buys. Repeating the same four stages with the prefrontal replaced every tick -- no loops, no persistence -- still acquires the behaviour: the walking action is lit on 199-200 of 200 test ticks on 5/5 seeds and the body travels 1.13-2.43 m, with 640,698-716,731 connections written. So **the association does not need persistence**. What persistence buys is the *standing up*: with the loops, 5/5 brains stand up in stage 4; without them, 1/5 does (final heights of 0.155, 0.060, 0.061 and 0.155 m against 0.352 m). The reading we take is that the sound reaches the walking entry either way, but acting on a thought about standing up requires the thought about standing up to last longer than the tick that produced it.

What persistence costs, and what we did not get right. With the loops the prefrontal population is an accumulator: each tick's code is added to the last and nothing removes it, so a named block stays lit after the stimulus that lit it has gone (R8). That is what the loops are for -- a code that cannot outlast its stimulus cannot be thought with -- and it is what lets the sound of stage 4 keep the walk running. What the loops do *not* do on this build is stay colour-selective. A blue screen drives the block to 0.59 of its cells against 0.57 without the loops, and the sums are 0.74 and 0.72 (R2), so on this build a sub-threshold colour is not carried across the line; on the earlier build of the system it was (blue reached 1.07 and the walk

started on 5/5 seeds), and R2 reports the change next to the numbers rather than hiding it. What the loops do cost is visible in the same table: with a black screen, and no stimulus anywhere in the run, the block drifts up to 0.11 of its cells, activation that the image is not producing.

The accumulator also over-drives the innate locomotor loop: the innate walk toward red, with no learning and no sound at all, topples after 66 and 68 ticks on seeds 20260915 and 20260916 with the loops, while the same two brains walk 2.26 m and 2.51 m without falling when the prefrontal is replaced every tick. A recurrent population needs a way to forget, and the loops we wrote have none: that, and not the reward rule, is the unfinished part of this system.

A knob for the accumulation, and what it buys. The accumulation is not intrinsic to the loops: what is missing is anything that subtracts. One global quantity supplies it -- let every prefrontal cell receive an extra inhibitory current equal to a constant k times the number of prefrontal cells active on this tick, delivered uniformly to that population and to nothing else. It is a knob, not a written rule, and it adds no connection. Three seeds, 120 ticks, standing:

k	prefrontal cells active, red, ticks 20 to 120 (three seeds)	blue	green	black screen	walk lit, red	walk lit, blue
0 (no knob)	2747 - 2851	2748 - 2849	2738 - 2839	47 - 441	119/120	119/120
0.0005	546, 540, 548	566, 509, 554	561, 567, 535	47 - 441	119/120	119/120, 0, 0
0.001	275, 271, 278	267, 258, 270	283, 275, 287	47 - 441	119/120	119/120, 0, 0
0.0015	30, 29, 30	24, 25, 24	24, 24, 24	47 - 441	119/120	119/120, 0, 0
0.004	0, 0, 0	0, 0, 0	0, 0, 0	47 - 441	119/120	119/120

(logs: 日志_前额叶自压_窗口.log, 前额叶自压=0.001; script 诊断_前额叶自压_窗口.py.)

Three things follow. (i) The knob does what it is meant to do: at $k = 0.001$ the red code stops growing -- 2747 rising to 2851 becomes 271 holding at 275 -- and the population is sparse in the sense 1.1 asks for. (ii) It removes the cost that mattered. On the two seeds that topple under the loops, E0 -- never taught, red *and* sound, no learning -- now travels 1.66 m and 2.65 m tilted 4 degrees, against topples at ticks 76 and 68 with the loops and no knob, and those are the same values the no-loop configuration gives. (iii) It is not free and the window is narrow. At $k = 0.004$ the prefrontal is dark for every stimulus, yet the walk is lit on 119 of 120 ticks: the ignition happens before the population is suppressed, and afterwards the action runs itself (R8), so what is left is a one-tick prefrontal event rather than a code that persists. The black screen is untouched at every k , because with no stimulus in the image the population is small and a constant times a small number is a small number. We report the knob as a bounded result, not as a solution: it trades unbounded growth for a population that is either sparse-hundreds or dark, with little room between.

What fails in R9. On 4 of 5 seeds the acquired behaviour does not stay upright: the body topples 67-105 ticks into the test, after 1.03-1.23 m (the fifth seed walks the full 200 ticks, 0.55 m, tilted 11 degrees). Two controls on the same taught brains rule out the reward-written connections and locate the rest.

seed	never taught: red <i>and</i> sound, no learning	taught: test, learning on	taught: test, learning off
20260915	topples at tick 76	topples at tick 61	topples at tick 61
20260918	walks 1.69 m, 2 degrees	topples at tick 42	walks 2.13 m, 2 degrees

(log 日志_摔是谁干的2.log; script 诊断_摔是谁干的2.py. The instability is present one stage earlier and is not confined to the test: with the loops, all five seeds also topple during the 200 *teaching* ticks -- at ticks 78, 154, 174, 154 and 175, log 日志_打折R9.log -- so from the middle of the teaching onwards the body is on the floor, while the association is still written. That the acquisition survives the body falling over is itself consistent with (a): what the reward path writes is the co-activation of the auditory code and the action's cells, and those are still running while the body is down.)

The two seeds fail for different reasons, and neither is the reward-written connections: on 20260915 the body topples under the *innate* repertoire, in a brain that has not been taught anything; on 20260918 the innate repertoire is fine and what breaks the walk is the slow correlation rule of 2.3 continuing to write while the behaviour runs -- the same brain, with the

same connections, walks 2.13 m without falling if learning is switched off at test time. What the two have in common is measured in the next paragraph but one: in both cases the walk is perturbed while it runs, and in neither case is the perturbation written into the weights that end on the muscles.

Which of the two additions owns which failure. The same three conditions, on the same two brains, with the prefrontal code recomputed every tick -- the configuration in which R1-R8 were run -- separate them.

seed	never taught: red <i>and</i> sound, no learning	taught: test, learning on	taught: test, learning off
20260915	1.66 m, 4 degrees	1.47 m, 6 degrees	1.63 m, 3 degrees
20260918	1.69 m, 2 degrees	topples at tick 40	2.09 m, 2 degrees

(log 日志_摔是谁干的3_无环.log; script 诊断_摔是谁干的2.py --无环. With the knob of the previous paragraph set to $k = 0.001$ the same two seeds give 1.66 m and 2.65 m and do not topple, log 日志_摔是谁干的4_自压001.log.)

Seed 20260915 topples only when the prefrontal accumulates: it topples at tick 76 with the loops in a brain that has never been taught, and it walks 1.66 m without them. Seed 20260918 topples in both configurations as soon as learning is switched on during the test -- 42 ticks with the loops, 40 without -- and walks 2.09-2.13 m when it is switched off. The two failures therefore have two different owners, the accumulating prefrontal code in one case and the slow correlation rule continuing to write while the behaviour is expressed in the other, and neither is the reward-written connection.

Where the perturbation comes from. The two controls above locate the failures but stop short of the mechanism: what they establish is *when* the walk breaks, not what breaks it, and one of the two owners they name turns out not to be supported. We measured the weight of everything that ends on the motor cortex -- the pathway that literally pulls the muscles -- across a whole 200-tick test. It does not change: 265,436 excitatory synapses onto the 160 motor cells at the start of the test and 265,436 at the end, carrying 494,436 units of weight both times, a difference of zero, with no new synapse formed. Freezing that pathway outright -- the same brain, the same test, with the plasticity factor onto the motor cells set to 0 instead of 0.02 -- moves the topple from tick 63 to tick 64. The motor weights are therefore not what breaks the walk. That does not exonerate the slow correlation rule -- on 20260918, switching learning off at test time still saves the walk, so something the rule writes is implicated -- but it is not written here, into the pathway that ends on the muscles, and the sentence in an earlier draft of this section which attributed the failure to the motor loop being rewritten while the behaviour ran is not supported by these measurements.

The measurement that does explain it is smaller and much more specific. The walk is played by a 15-cell time chain in the motor-memory region, and on *every tick* of the test the excitatory current arriving on that chain **from the prefrontal population is 0.850** -- and 0.850 is exactly the current that chain needs to fire, the same number that had to be divided among sources when the reward path was calibrated in (a) above. That is not a coincidence: the instinct table was calibrated so that the compressed sensory code for red and the prefrontal code for red *sum* past the chain's threshold, and each of them contributes about half. The calibration is right for a code that is recomputed from its input every tick -- the configuration of R1 to R8 -- because the contribution is present only while the stimulus is. It is wrong for a code that persists. A rule written to *trigger* an action becomes a *standing drive at the action's own threshold*; the chain is no longer free-running, and the gait drifts until the body goes over. Two measurements make this concrete. Without the loops the same brain, the same picture and the same sound produce an exactly periodic walk: the twelve muscle forces at ticks 30, 60, 90 and 120 are identical to two decimals, with the gait resetting on a 30-tick cycle. With the loops the forces acquire a slowly growing offset on four of the twelve muscles -- 0.2 to 0.5 of full force by tick 60 -- and the difference in the current reaching the motor cortex, on those ticks, comes entirely from the time chain (+138 to +140 units) rather than from the prefrontal population, whose direct contribution to the motor cortex is 0.00. The drive also does not come back through the return line: cutting the return line's projection into the motor strip leaves the topple exactly where it was (tick 76 against 76), and cutting the whole return line moves it to tick 63.

Scaling those 5,175 synapses at build time, with nothing else changed, tests that reading, and the answer is that the effect is real and localised but the number is not a repair. At 0.75 and at 0.5 the seed that toppled under the innate repertoire is repaired: 20260915 holds the stand for 191 of 200 teaching ticks (against a topple at tick 78) and in the test it stands up and walks 187 of 200 ticks at 3 degrees (against a topple at 63, after which the body lay on the floor). Nothing else is

repaired. At both strengths three of the five seeds lose the walk altogether -- 0 of 200 test ticks (at 0.5 one of them, 20260918, gives 0 of 30 innate ticks as well: the red region no longer starts the action at all) -- and a fourth, 20260916, topples at tick 101 in the test in all three configurations, so its topple was never this pathway. Five seeds, two strengths, one seed repaired: moving the same half of the sum moves the failure around rather than removing it. We therefore do not adopt the change; the table's strengths belong to the build on which every other number in this paper was measured. We report the measurement because it localises the defect. What is missing is not a weight rule that forgets; it is that a trigger written as a constant current stops being a trigger the moment the code emitting it stops going away. Fixing that needs a rule that decays while the code persists, which is a property of the rule and not of the learning.

(logs: 日志_冻死进运动.log, 日志_诊断步态.log, 日志_前额叶打走路链.log, 日志_返回线截断.log, 日志_打折R9.log; scripts 诊断_冻死进运动.py, 诊断_前额叶打走路链.py, 诊断_返回线截断.py, 试_打折R9_参数.py.)

R10. The reward can be delivered by a sensory afferent rather than by a switch (no figure)

R9 delivers its reward with a flag: the script sets 多巴胺亮 = True. The population that does the work is inside the system, but the *occasion* for reward is outside it. The framing of 1.1 says a drive is one more innate circuit, so we wired one and asked whether the acquisition of R9 survives it.

A 210-cell touch region and one innate rule per site. Seven skin sites -- head, back, belly and the four legs -- get 30 cells each, appended after the inhibitory pool. The region sits in all three "non-participating" lists of Section 2.1: it neither sends nor receives random connections and has no internal recurrence, so its only route to the rest of the cortex is innate. The instinct table gains seven fixed rules of the form 触觉:<site> -> 多巴胺:全部 at weight 2.0, 42,000 synapses in total. Touching one site at full pressure lights all 30 of its cells, and the dopamine region is lit on the following tick -- 200 of its 200 cells -- and stays lit while the contact is maintained; in the *resting* arm below it is lit on **200 of 200** teaching ticks. A brush along the same site (10 of its 30 cells) lights the dopamine region on 0 of 200 cells, and no touch gives 0 of 200. Reward requires contact, and contact of a kind that engages a whole site.

Protocol. The four stages of R9, unchanged, on three brains, in three arms that differ only in when -- if ever -- the hand is on the body: *resting* (contact on every teaching tick), *contingent* (contact only on ticks on which the walking action was active, the way an animal is trained), and *no touch*. The flag of R9 is permanently off in all three arms; there is no "give reward" statement left in the script. Adding a region re-draws the whole background connectivity, so these are not R9's brains, and the comparison of interest is within this run. Seeds are 20260914, 20260915, 20260916 in that order.

arm	hand on body, teaching ticks	new excitatory connections	test: walking ticks	max trunk height at test
resting	200 / 200 / 200	480,928 / 524,379 / 514,138	0/200, 200/200, 200/200	0.115 / 0.155 / 0.058 m
contingent	0 / 199 / 198	0 / 520,017 / 511,099	0/200, 200/200, 200/200	0.115 / 0.155 / 0.061 m
no touch	0 / 0 / 0	0 / 0 / 0	0/200, 0/200, 0/200	0.115 / 0.095 / 0.116 m

(logs: 日志_仿生奖励_一直摸.log, 日志_仿生奖励_声音叫走路.log; scripts 实验_发育_仿生奖励_三臂.py, 实验_发育_仿生奖励_声音叫走路.py, 触觉区_touch.py.)

Birth is the same in every arm (0 of 90 ticks across the three brains). The no-touch arm writes nothing and the sound still does nothing (0 of 600 test ticks), which reproduces R9's dopamine-dark control in this build. The two arms that deliver contact acquire the link, and the connection counts are in the same range as R9's 490,659-552,186 despite the different brains: the reward arrives as a sensory afferent here, and the acquisition does not depend on how the reward is delivered.

The instructive row is the first one. Seed 20260914 in the *resting* arm was rewarded on every one of the 200 teaching ticks and wrote 480,928 connections -- it is not that nothing happened -- and it still fails at test (0/200). Its teaching stage differs from the other two in one respect: the walking action was lit on **0 of 200** teaching ticks, where the red region alone lit it on 30 of 30: in this brain adding the sound to the red suppresses the walk it is supposed to be paired with. The reward therefore reinforced co-activity that did not include the behaviour, and it bought nothing. A reward is not a teacher: it can only strengthen what the brain actually did. The same brain in the *contingent* arm was never touched, for the same reason - a protocol that pays only for the behaviour cannot pay a brain that does not perform it.

What fails in R10. Both arms that acquire the link acquire it, and in both the body does not get up at test: the walking action is lit on 200 of 200 ticks while the trunk stays between 0.058 and 0.155 m and the body topples at tick 41 to 95. Two of the three brains stand up in R9's build (0.388 and 0.391 m for seeds 20260914 and 20260915). Three measurements, all in this build, locate the difference, and none of them is the reward. (i) The touch pathway is silent at test: the hand is not on the body, and nothing else enters the region. (ii) The region is not the cause: with learning off and no teaching at all, the same three brains placed prone on a black screen stand up and hold the posture -- maximum trunk height 0.261 m, the standing cells lit on 149 of 150 ticks -- which is exactly what the unmodified build gives (诊断_触觉版_天生站起来.py). (iii) What differs is the learned state, and the way it differs is that the acquired walk now wins against the righting chain. That is the class of failure R9 already reports -- acquisition that does not stay upright -- and we do not have a measurement that separates the two builds' failures from each other. We report the endogenous reward as established, and the post-acquisition stability as open, in both builds.

R11. What a wider cortex costs, and what widening does to behaviour (no figure)

Nothing in the architecture refers to the number of cells, but the number of cells here is set by the eye: the visual field is divided into a mosaic, each mosaic cell carries three channels of ten threshold pairs, and the prefrontal region is sized from the result. Widening the mosaic is therefore the natural way to scale the system, and we measured what it costs. Two things had to be true first.

The eye and the mosaic have to be the same number. The world is rendered analytically -- objects are projected onto the mosaic grid by geometry, with no rasterisation -- so the mosaic *is* the eye's resolution. Widening the front end while the world keeps drawing 24x16 would have produced interpolated cells carrying no new information. Both now read the mosaic size from one place, so a wider mosaic is finer vision: 6.25 x 3.1 degrees per cell at 24x16, and 0.52 x 1.6 degrees at 24x192.

A software ceiling had to be removed rather than worked around. The previous peak memory of the entire system was 7,005 MB against a steady state under 1 GB that the same script now reports as 543 MB -- about 50 MB of that is the index the plasticity fix below adds -- and it was one line: the sensory front ends built the "one strength per threshold pair" table densely, because the ring-wrapped pair offset spans the whole ring. At the auditory width of 30,000 cells that is 30,000 x 29,999 doubles, or 7.2 GB -- larger than the cortex itself by an order of magnitude, and the only thing standing in the way of a wider brain. The table is now filled in row blocks; because `rng.uniform(size=(n, m))` fills row by row, the block version draws the same numbers in the same order, and the resulting weights are bit-identical to the dense version for both front ends (checked by 验证_感觉网分块.py, which compares both front ends against the dense version, and confirmed by R9 reproducing digit for digit on three seeds in the unmodified configuration). The fix takes that peak at 24x16 from 7,005 MB to 1,434 MB and changes no number in this paper.

mosaic	degrees per cell (h x v)	cortical cells	excitatory / inhibitory synapses	build	tick*, no plasticity	tick*, with plasticity	resident	peak
24x16 (this paper)	6.25 x 3.13	148,032	11.6 M / 4.9 M	10.7 s	1 ms (970 Hz)	2 ms	543 MB	1,434 MB
24x32	3.13 x 3.13	214,836	16.2 M / 6.1 M	14.0 s	2 ms (653 Hz)	3 ms	700 MB	1,890 MB
24x48	2.08 x 3.13	285,876	20.9 M / 7.4 M	19.3 s	2 ms (458 Hz)	5 ms	855 MB	2,599 MB
24x64	1.56 x 3.13	356,916	25.5 M / 9.2 M	24.1 s	3 ms (363 Hz)	5 ms	1,030 MB	3,054 MB
24x96	1.04 x 3.13	498,996	34.7 M / 13.8 M	37.2 s	4 ms (268 Hz)	9 ms	1,411 MB	4,474 MB
24x128	0.78 x 3.13	641,076	43.9 M / 18.4 M	47.9 s	5 ms (187 Hz)	11 ms	1,791 MB	5,383 MB
24x192	0.52 x 3.13	925,236	62.4 M / 27.6 M	68.3 s	8 ms (126 Hz)	16 ms	2,555 MB	7,200 MB

(logs 规模试验_列16.log ... 规模试验_列192.log; script 规模试验_测量.py, which builds a network at the requested width and times it without running any behaviour, and which is run for all seven widths by 量规模_七档.py. One host, 16 GB, 16 logical cores. The pre-fix logs are kept beside them as 规模试验_列*.log.bak_优化前.)

* These two columns are a *step benchmark*, not a live tick: the network is driven with a uniform 2 per cent of cells active and no behaviour is run, so what they time is the cost of one pass over the synapse tables. The live cost of a tick, with the world rendering the mosaic and the body being driven, is 58 ms with the rule disabled and 158 ms with it enabled at 24x16 -- see (ii) below and Methods 5.5. The two benchmarks are not comparable to each other, and the live one is the one that matters.

Three things follow, and the third of them is a warning about the first two. (i) Cells are cheap in memory; plasticity is what costs, and what it costs depends on how much of the cortex is active rather than on how many synapses exist. The previous implementation of the rule gathered over all 27.6 million inhibitory synapses on every tick to decide which of them to touch, when only the synapses incident on a cell that is active can be touched at all. Indexing them by source and by target once, at build time, makes the per-tick work proportional to the active population, and against the benchmark above it takes the update on every tick from 554 ms to 16 ms at 925,236 cells. The rule is unchanged: the new update is bit-identical to the old one, checked tick by tick over 100 ticks on all 4,890,945 inhibitory synapses at 24x16 (验证_学抑制_逐拍对照.py, maximum difference 0.0). (ii) That benchmark flatters the result, and the live number is the one that matters. It drives the network with a uniform 2 percent of cells active, and after the first step that pattern does not sustain itself, so what it times is the cost of the scan; a brain that is actually looking at something holds about 6,000 of its 148,032 cells active on every tick, where the scan was not the dominant cost in the first place. In the closed loop at 24x16, with the world rendering the mosaic and the body being driven, a tick costs 58 ms with the rule disabled and **158 ms** with it enabled (诊断_整拍耗时.py; the same live tick cost 215 ms before the fix, at identical activity, log 日志_对照_学抑制_改前改后.log). (iii) The ordering is therefore unchanged -- the body is faster than the brain when the brain learns -- but the gap is a factor of eight rather than thirty, the remaining cost is spread over the current calculation and both halves of the rule rather than concentrated in one avoidable scan, and the honest statement of the ceiling is the live one: 158 ms against the 20 ms the body allows, with the four-orders-of-magnitude gap to a human cortex on top of it. The ceiling on this machine is still memory rather than time: the peak is 7,200 MB at 925,236 cells and the next doubling exceeds the host. This is a cost curve, not a scaling law, and the same asymmetry is what R9 reports from the behavioural side, where the correlation rule changing while a behaviour is expressed is what breaks it.

The cheap end of that programme, run. Nothing had to be re-derived by hand: the calibration and the table are generated from the mosaic size, so widening to 24x32 produced 34 direction names instead of 16 and 965 rules instead of 581, expanded to 357,274 innate excitatory synapses where 24x16 has 309,148. Calibration runs in 20 s and a brain builds in 51 s. We then re-ran the behaviours that go through the eye, and the righting reflex, which does not.

behaviour	24x16	24x32
righting: prone, black screen, no learning	3/3 brains stand, 0.261 m, 149/150 ticks	3/3, 0.261 m, 149/150
walk to a full-field red	5/5 seeds	5/5 seeds
gaze, 0.30 m ball crossing the field	5.5 deg mean	4.3 deg
gaze, 0.05 m ball appearing	4.8 deg	3.5 deg
gaze, empty scene	0.0 deg, eye does not move	0.0 deg, eye does not move
hues that start the walk	0-72: 5/5; 96, 168, 180: 4/5; 120, 144, 204, 240: 0/5	0: 5/5; 24: 4/5; 48, 72: 3/5; 96: 2/5; 168: 1/5; 180, 204, 240: 0/5

(logs 日志_天生站起来_列16.log, 日志_天生站起来_列32.log, 日志_放大走路边界_列16.log, 日志_放大走路边界_列32.log, 日志_跟随_列16_*.log, 日志_跟随_列32_*.log, 日志_放大_色相红块_列32.log, 日志_走路电流_列16.log, 日志_走路电流_列32.log; scripts 实验_放大_走路边界.py, 实验_放大_色相红块.py, 实验_眼睛跟随.py, 诊断_触觉版_天生站起来.py, 诊断_走路点火_多种子.py. The wider build is kept apart from the narrow one with AGI数据后缀, which names its calibration file and its table; without it the wider brain would overwrite the tables R1-R10 rest on.)

Four things follow.

(i) What does not go through the eye is unchanged, to the digit. Righting is driven by proprioception, the vestibular names and body state, and none of those region widths move with the mosaic. All three seeds stand in 0.261 m with the righting chain lit on 149 of 150 ticks -- the same numbers R1 and R10 report -- so widening the eye did not perturb the rest of the brain. The random background connectivity is redrawn, so this is not the same brain; it is the same *anatomy* doing the same thing.

(ii) A finer eye is a better eye. Both gaze measures improve by about a quarter, and the eye comes to rest closer to the object: 2.1 and 0.2 degrees off-axis at 24x32 against 3.2 and 2.3 at 24x16, with the empty-scene control still exactly 0.0. This is the first measurement in which a wider cortex buys behaviour rather than only costing memory.

(iii) The instinct strengths are calibrated to the build, and widening does not re-calibrate them. What a rule writes is the total current a source block delivers, spread over as many synapses as the block has cells (Methods 2.1), and that design survives the widening intact: the current reaching the walking head cell from its visual and prefrontal halves is 0.80 and 0.85, the same two numbers R2 reports, and the total excitation onto that cell is 1.750 at 24x16 and 1.750 at 24x32, to three decimals, because the source blocks light fully at both widths. The inhibition onto that same cell is a different matter: half of the random inhibitory wiring is scattered over the whole brain (Methods 2.1), so the inhibitory current a cell receives grows with how much cortex is active, and the eye is what changed. It rises from 2.775 on four of five seeds to a mean of 3.07, and its spread across seeds widens. Every threshold in the table therefore loses margin, and the first place it shows is the boundary of recognition. At 24x16 that boundary is exactly where the two halves sum: everything at 1.01 or more starts the walk on 4-5 of 5 brains, everything at 0.97 or less on 0 of 5. At 24x32 the static sum stops predicting: 168, 180 and 204 degrees sum to 1.05, 1.10 and 1.05 and start the walk on 1, 0 and 0 of 5. What the finer eye buys in precision it takes back in tolerance, and it gives back the false positive as well: cyan at 180 degrees, the opposite of red, reached the line at 24x16 and does not at 24x32. The consequence for scaling is that the instinct table is not width-free in effect even though it is in form -- a wider brain needs its strengths re-tuned, or a rule that normalises against total activity -- and that is a cost of the same kind as the plasticity bottleneck, not a defect of the widening.

(iv) The compressed layer still does not localise. The hierarchy's loss is a property of the compression, not of the mosaic. At 24x16 a ball one third of a cell wide is followed perfectly by layer 1 and missed entirely by the compressed layer for one direction of seven (R5). At 24x32 the same physical ball is two thirds of a cell wide, and the uncompressed layer is still perfect -- all seven directions, every newly responding cell inside the ball's own column -- while the compressed layer still scatters its few responses over up to 30 of 32 columns and still returns nothing at all for two directions of seven. The front end's receptive-field sizes are written in ring positions, not in degrees, so they shrink in angular terms as the mosaic widens. Whatever else scale buys, it does not buy the compressed layer the ability to localise.

What we did not measure. No behaviour was run above 24x32, and the rows of the table above from 24x48 to 24x192 remain a cost curve. Of the experiments, the static boundary and the gaze reflex were re-run at 24x32 and the drift of R4, the shifted map of R7, the stimulus-removal and recurrence-removal of R8, and the acquisition of R9 and R10 were not. The re-tuning that (iii) calls for has not been done either: we report the mis-calibration rather than a fix.

R12. A route that is born empty, and the four arms that say what a lesson writes on it (no figure)

R9 and R10 show that a capability can appear during the life of one brain. They also show why that is not yet a recipe. The fast path writes a synapse from *every* cell that was active on the previous tick to *every* cell that became active on this one, so what it writes is not the lesson: it is everything that happened to be co-active while the reward was on. R9 installs one association with 242,685-396,607 new connections, and R10's most instructive arm is a brain that was rewarded on all 200 teaching ticks, wrote 480,928 connections, and acquired nothing. A recipe needs somewhere for the lesson to land, and R12 measures the smallest one we could build.

Three additions, all off by default. (i) A *backup route*, named at build time as a pair of populations (backup_routes, born_wired/reflex_controller.py): every edge from the first population to the second is laid down at birth at a weight of 0.0001 -- too small to move any cell -- with a lower bound of zero, a ceiling of 1.5, and no tether to a birth weight. The graph is fixed once it is built, so an edge that does not exist can never appear later; a backup route is the other way round:

it exists, it conducts nothing, and it is the only thing a lesson is allowed to write. (ii) The dopamine population is a single ordinary cell (dopamine, tau 0.05). Its rate says "this moment counts" and nothing about what was right; it is read by `learning_modulation`, which scales the local rule instead of replacing it -- while dopamine fires the rule runs at its own rate, and while it is quiet the rule is slowed by `dopamine_gain` (0.9 here). Which edges change is still decided only by which cells fired together. (iii) One innate afferent, `touch` -> dopamine at gain 4.0, so that the keeper's hand reaches the gate through the same kind of connection as everything else. The hand is a teacher here for a reason that has nothing to do with the lesson: touching the body already drives the retreat cells, by one of the instinct rules of R10. The keeper supplies the *pairing*; the innate wiring supplies the *meaning*.

All three sit at their factory value, and the control is exact: with `dopamine_gain` and `touch_dopamine_gain` at zero and no routes named, the model is the one it was, bit for bit -- checked by comparing the whole activity array rather than a summary (`tests/test_nursery.py`).

The lesson. A 262 Hz tone, which nothing in the innate graph carries to the retreat cells; the animal walking throughout (the constant locomotion drive of R2's walking protocol); the keeper's hand on the body for one second and off for one second, twenty times. Read at the end: the weights on the route, then six seconds of the tone alone with no hand, of silence, and the tone again.

arm	hand	dopamine gate	route at birth	route after the lesson	retreat cells, tone alone	leg-drive cells	retreat cells, in silence
taught	1 s on / 1 s off	on	0.0001 (six edges)	0.95, 0.99	1.000	0.504	0.020
<i>ignored</i>	never	on	0.0001 (six edges)	0.000	0.000	0.000	0.000
<i>ungated</i>	1 s on / 1 s off	off	0.0001 (six edges)	1.45, 1.50 (the ceiling)	1.000	0.509	0.022
<i>no route</i>	1 s on / 1 s off	on	the edge does not exist	--	0.057, gone by the next phase	0.003	0.000
all arms, before the lesson	--	--	0.0001 (six edges)	--	0.000	0.000	--

(logs `nursery_20s_20260928.log`, `nursery_20s_20260928.json`, `nursery_noroute_20260928.log`; script `tools/nursery.py`. The route named is `cochlea` -> retreat and holds six edges at the shipped widths; the four arms are one seed each.) A 6 s lesson is in the same log: the route reaches 0.30 instead of 0.95, and the tone alone drives the retreat cells to 0.519. Lesson length is a knob, and 20 s is not a threshold.

Four things the arms say.

1. *A reflex is acquired.* Twenty seconds of contact took a sound that meant nothing -- the retreat cells are at 0.000 with the tone at birth -- to a sound that commands retreat, at 1.000, sustained across two test phases. The leg-drive cells, which are what actually moves the thighs, follow at 0.504 under the tone against 0.009 in silence, and the thigh angles move as well (0.85, 0.83, 0.85, 0.80 under the tone against 0.81, 0.82, 0.85, 0.88 in silence; two of the four separate clearly and two do not). The acquisition survives the removal of the teacher.
2. *Nothing is written where nothing happened.* The route in the *ignored* arm falls from 0.0001 to exactly zero while the tone plays just as often: the tone fires, the retreat cells do not follow, and the decay term of Methods 5.5 -- the one every synapse in the system already has -- takes the weight to its floor. This is the property the whole framing rests on. A correlation rule knows that two cells fired together; it has no representation of having been right, and it does not need one.
3. *The reflex is written onto the empty route.* In the *no route* arm the same lesson, the same hand and the same gate write nothing that outlasts the lesson: the tone lifts the retreat cells to 0.057 at the end of the lesson -- a trace -- and to 0.000 on the next phase. The causal claim is therefore not "dopamine wrote a reflex"; it is "dopamine wrote a reflex *onto the route that was laid down*".

4. *Co-activation decides whether a weight is written, and the gate decides how hard.* The ungated arm acquires the reflex too, and writes to the ceiling (1.50) instead of stopping just below it (0.99). That is the honest reading of the gate: what it buys is not the ability to learn -- the local rule learns without it -- but that the rule is slow when nothing is being taught. Demonstrating the second half needs a different experiment from this one, in which many cells are co-active outside the lesson. We have not run it.

Within the *taught* route, four of the six edges stay near zero (0.031, 0.000, 0.040, 0.001) while two reach 0.95 and 0.99: only the cells that actually fired together under that tone were written. The keeper names two populations; the co-activity picks the cells. That is the division of labour the framing predicts, and it is the reason the recipe can be stated without a teacher who knows anything about the inside of the brain.

What this does not show. The animal of this build shuffles rather than walks -- across these phases the body covers 0.001 m to 0.523 m in six seconds -- so the metres column of the log is not a measure of anything, and the read-outs used here are the retreat and leg-drive cells and the thigh angles. A lesson taught on this body is not a gait, and reaching one is the problem R9 and R10 also report. The arms are one seed each. On a re-run of all four arms the route weights, the cell rates and the 0.057 trace reproduce exactly and the mean thigh angles move in the third decimal, which is why they are quoted to two. And what is written is one link between one sense and one action, not a representation: before this can be called a training method, the same machinery has to be shown to write a *structure* rather than a reflex.

R13. What the rebuilt engine costs, and the two rules that stop the sheet from growing (no figure)

R11 measures what the reference engine costs and finds the tick dominated by plasticity at 24x16. Two questions follow, and both were answered by rebuilding the engine rather than by adding structure to it: what does a tick cost once the sheet is bigger, and what -- as opposed to what hardware -- stops the sheet from being bigger still.

The rebuilt engine. The same architecture, re-implemented around the same update rule and the same "instinct as initial condition" design, with raw senses only: two eyes that deliver two RGB images with no matching, no object recognition and no distance, two ears that deliver a sound pressure waveform with no source localisation, and a body that reports its own state. It runs on the graphics card, and its shipped configuration is 192x144 pixels per eye with 1,600 motor, 512 proprioceptive and 2,048 association units -- **750,075 cells, 1,978,219 synapses**. One step of the whole loop costs **13.29 ms (75.2 steps/s)**, of which the brain is 9.24 ms, the eyes 2.86 ms and the physics 0.82 ms (artifacts/profile_after_opposite.log). Against the reference build's live tick -- 148,032 cells, 158 ms per tick with plasticity enabled (R11) -- the rebuilt engine carries **five times the cells at about a twelfth of the cost**, and the whole 199-test suite of the rebuild passes in 130 s (artifacts/tests_after_opposite.log).

Three changes account for it, and all three are required to be bit-identical to the code they replace, checked by a fingerprint tool that compares cell count, edge count, the initial weight digest, and the weight and activity digests after 20 and 200 steps (tools/check_model_fingerprint.py): the synaptic update is put back on the 50 ms cadence the rule always specified instead of running on every 10 ms step; the structural quantities the rule needs are measured once at build time instead of re-derived every tick; and the per-step array arithmetic writes in place, which also removes the 3 MB-per-eye-per-step allocation that produced occasional 71 ms render spikes (now 4.8 ms worst case).

What a bigger sheet costs is connections, not cells. With the region widths held at 160x120/800-256-1024 the whole step is 10.54 ms (94.9 steps/s) over 523,579 cells; at the shipped 192x144/1600-512-2048 it is 13.29 ms (75.2 steps/s) over 750,075; at 256x192/1600-512-2048 it is 20.67 ms (48.4 steps/s) over 1,286,267. Cell count rises by 43 percent across the first two while the step cost rises by 26 percent, and most of that is the tick's fixed traffic between host and device rather than the neurons. This is the same relationship R11 reports on the reference engine -- cells are cheap in memory, and what costs is per-tick work proportional to activity -- and it is why "add cells when a capability is missing" is an engineering proposition rather than a wall.

Two build rules stop it, and neither is the hardware.

- *An incoming budget.* A region's own size is checked against the excitation its inputs can deliver. At 192x144 with the shipped widths, proprioceptive sizes of 512, 640, 768 and 896 all build, and 1,024 is refused: `ValueError:`

structural lower bounds exceed an incoming budget. The same probe widens the association region to 3,072 and 4,096 and the motor region to 3,200 without a refusal.

- *The offscreen buffer the eyes are drawn into.* The arena declares `offwidth="1280"`, and the renderer draws three images of the eye's own width, so the widest permitted eye is $1280/3$, a little under 427 pixels. 256x192, 320x240 and 384x288 render; 448x336 is refused with `Image width 1344 > framebuffer width 1280`.

Neither refusal is a cost and neither is a hardware limit: the first is a structural rule of our own design, and the second is one number in the arena file. Together they cap the sheet at 320x240 with 2,400/768/3,072 units, which builds at 1,988,843 cells and 4,312,947 synapses. Both are reported because a reader who intends to grow this system will meet them before meeting the graphics card. (`tools/probe_scale_walls.py`, `log artifacts/scale_walls_after_opposite.log`.)

A route we built, cannot read, and are reporting as zero. The eye population contains one cell (`eye_looming`) whose resting level is -0.5, which is driven at weight 1.0 by the conjunction of outward and inward motion in the two eyes, and which drives the brake. It is the beginning of a "something is coming at me" reflex. Its read-out is **0.000 in every scene we can build**: a striped wall at 0.35, 0.80, 1.60 and 3.00 m; the two eyes swept by hand at 0.6 and 3.0 rad/s; a ball walked in from 3.0 m to 0.35 m at 0.5, 1.5 and 3.0 m/s; and the animal walking by itself. It stays at 0.000 when the thresholds that feed it are lowered by a factor of five, and per-frame logs show the reason: the conjunction never leaves zero, while its two components reach 0.08 against a wall closing at 1.6 m/s and 0.11 for the single-eye rise and fall signals. A cell that needs 0.5 of drive is not reached by 0.08. We report this as a route that does nothing rather than repairing it silently, in the same spirit as R5's compressed layer and R4's boundary: the eye reflex that *does* work is the gaze and motion-onset path of R6, and this one is not evidence for anything (`tools/measure_loom.py`, `logs artifacts/loom.log`, `artifacts/loom_ball.log`, `artifacts/loom_wall.log`).

R14. The last layer is not the only place a signal can come from (no figure)

P1 says that a compressed layer is a bus and not an answer. R14 asks the question that follows from it and that the rest of the paper had not asked: if a hierarchy throws detail away, is the detail still readable from the layers underneath it, and can a behaviour use it?

A synthetic case where the two answers differ. Four position channels feed eight intermediate cells tuned to overlapping ranges of those positions, and the eight feed one heavily compressed output cell. Two positions are chosen that drive that output cell to the same value: with a read-out attached to the compressed cell alone, the two positions are one and the same. Two motor cells are then given a plastic route to the eight intermediate cells, and after training each motor cell responds to its own position and not to the other. With learning off the routes stay indifferent, and with the training pairs reversed the two motor cells reverse as well, which excludes the read-out code choosing the action (`tools/evaluate_hidden_routes.py`; the run is stored as `artifacts/hidden_routes.json`).

The same thing on a body. On the rebuilt engine, thirteen cells and their plastic edges are added between the joint-position intermediate cells and the flexion motor units that already exist (`born_wired/feature_routed.py`), and five seeds are run in three conditions: the reference route alone, the reference route plus hidden routes with learning on, and the same with learning off. With the hidden routes and learning on, presenting stimulus B, then A, then B again moves the trunk to about 0.250, 0.147 and 0.250 m: the second B recovers the first B's activity exactly (return error 0) and its height to within 6.8 micrometres. Neither control produces that separation. Fifteen cases, 750 s of simulated time, body and neural state continuous throughout (`tools/evaluate_routed_robot.py`, `artifacts/routed_robot.json`).

And over ten minutes of lived time. A separate instance (seed 11) runs for 600 s with learning *always* on, ten consecutive pairings of the two stimuli and a test of each after every pairing. The flexion activity for A stays at 1.0 in every cycle and for B stays near zero, the body and the brain are never reset, and the animal does not fall (`tools/evaluate_online_features.py`, `artifacts/online_features_10min.json`). This is the closest thing in the paper to a capability that is *maintained* rather than demonstrated once.

Depth is not a design constant. Nothing in the rebuilt engine requires a hierarchy to have a given number of layers: a cell is a cell, "layer" and "region" are labels for organisation, and a layer that is never driven costs building time and nothing else. The reference build's ladder of four parameter layers is a choice and not an invariant, and the general claim is the negative one -- no result in this paper depends on how many layers there are, only on which cells can reach which.

There is no clock to set. The rebuilt engine has no global time base and no run-mode switch. Cells are integrated with their own membrane and adaptation time constants and the loop simply runs as often as the machine allows; a step is a step rather than a division of a fixed quantum, and there is no "full-speed" mode that the same model could be put into or taken out of. What decides a tick is the state of the sheet and the number of cells active in it (R15).

What R14 does not show. The synthetic case has 17 cells, four input channels and no camera, and the body case has 1,813 cells; they are two experiments and the first must not be read as an account of the second. The body experiment declares two structural priors rather than learning them: the shared compressed route is capped at 0.10 and the intermediate bridges at 0.55, and the two routes are allowed different plastic speeds. Testing reads the learned weights with learning off, which measures an association that has formed rather than continued learning, and the ten-minute run is the evidence that the association survives with learning on. Nothing here shows that routing every intermediate layer to every motor unit is optimal, and the added routes cost connections and activity.

R15. How large the sheet has to be, and what "too small" looks like (Supplementary Figures 5 and 6)

Two engineering questions decide whether a system of this kind can be grown rather than argued about: does the shape of the sheet matter, and does the cost of a tick follow the size of the sheet or the activity in it?

A flat sheet and a solid one. Three arms of 1,000 cells each, six local and two random outgoing connections per cell, the same input volume and the same seed; the only differences are whether the coordinates are two- or three-dimensional and whether the input is wired to nearby cells at all. Each arm is shown twenty objects one at a time and the overlap between the sets of cells that end up active is measured (Jaccard; lower is better for two unrelated objects, higher is better for one object moved a little).

measure	random wiring	flat coordinates (2D)	solid coordinates (3D)
two unrelated objects (lower better)	0.306	0.041	0.081
the same object nudged (higher better)	0.792	0.852	0.869
the same object at a different depth (lower better)	0.343	0.290	0.084

Depth is therefore carried by the solid sheet and by nothing else, and it is bought at the price of some separation between different objects (0.041 to 0.081). That trade is the honest result rather than a shape that is better in every way (tools/probe_spatial_dimension.py, artifacts/probe_spatial_dimension.log, Supplementary Figure 5).

What a tick actually costs. Two engines are measured side by side at 100,000 cells. The original one takes 5.64, 5.17, 5.44, 5.18, 5.47 and 5.71 ms per step for external drives of 0, 0.01, 0.1, 1, 10 and 50 per cent -- while the number of cells alight goes from 0 to 86,064. Its cost does not depend on activity at all, which is to say that it is not a sparse implementation and its headroom is already spent. The rebuilt engine's cost does follow the activity: 0 cells alight costs 0.04 ms and 11,818 cells alight costs 13.71 ms, and the two engines agree on the network state to 3.3e-16 while the activity is partial, so the speed is not bought with a different model (tools/probe_sparse_cost.py, tools/check_sparse_engine.py; logs artifacts/probe_sparse_cost.log, artifacts/check_sparse_engine.log). When only the sheet grows -- 100,000, 200,000, 400,000 and 800,000 cells with the external drive held at 1,000 cells -- the cost per step is 1.28, 1.33, 1.30 and 1.35 ms: eight times the cells for 1.06 times the time (tools/probe_sparse_scaling.py, artifacts/probe_sparse_scaling.log).

What is alight in the build we ship. In the 160x120 build, one retina holds 367,712 of 408,328 cells, 90.1 per cent of the sheet. Cells above 0.5 activity are 6.54 per cent of the whole sheet -- but 4.15 per cent on the retinal side and 28.17 per cent everywhere else -- and 69.00 per cent of connections have an active source. One brain step is 60.18 ms with learning on and 39.47 ms with it off, and rendering the two eyes is a further 23.51 ms (artifacts/execution_limits.log). The

cost of a step in the shipped build is therefore dominated by the sensory sheet, which is exactly the part that is not yet sparse; the association sheet, the part a general system would grow, is the cheap part.

A negative result about the fraction. P8 predicts that the fraction of the sheet alight should fall as the sheet grows, and at fixed drive it does: the same 1,000 driven cells light about 1,180 of 100,000 (1.2 per cent) and about 1,180 of 800,000 (0.15 per cent). But the cost per active cell is roughly flat across the same runs -- every run that lights more than a few hundred cells lands between 1.03 and 1.73 milliseconds per thousand active cells -- so this is a statement about what the machine must do per active cell and not yet a demonstration that a larger sheet resolves the same input with fewer cells. That demonstration needs a task that improves with resolution, and it is not in this paper.

"The model is too small" is a measurable statement. If capacity is the concern, its effect should be visible as a loss of resolution. With ten motor cells per channel the smallest distinguishable force step is 0.10, and inputs of 0.50, 0.51, 0.52 and 0.54 all produce the identical pattern of active cells; at twenty cells the step is 0.05, at fifty 0.02 and at a hundred 0.01, with the round-trip error of the encoding falling from 0.050 to 0.005 (`tools/probe_motor_resolution.py`, `artifacts/motor_resolution_probe.json`). The same statement holds on the reference build's prefrontal population: the cells named "thinking about standing" number 12 and those for "thinking about sitting" number 10, and 10 of them are the same cells -- a Jaccard overlap of 0.83, with sitting a subset of standing, which is what one expects if the two are one population separated by a threshold rather than two representations (`artifacts/old_representation_probe.json`). Resolution is therefore not a matter of opinion in either direction: it can be measured, and where it is too coarse the measurement says so.

R16. Two eyes that converge, focus and look where the picture changes (no figure)

R6 reports gaze that emerges from a motion-onset reflex. R16 asks three narrower questions about the eyes themselves, because each of them turned out to have a wrong answer in our own build.

The picture is sampled, not warped. The visual front end originally remapped position and then read a single pixel at the remapped point, which magnified the centre of the field without bound and squeezed the periphery: a ball at 0.60 m was drawn as a horizontal bar. It now samples the unwarped image, dividing the field into equal shares and pulling the boundaries towards the centre so that the middle of the field carries 2.5 times as many cells per degree as the edge; each cell reports the mean of its own patch and nothing is interpolated. The consequence is honest and uncomfortable: the old front end's ability to read distance from 2.50 m was bought by that distortion. With the unwarped sampling, a target at 2.50 m lights 1 cell with a drive of 0.000 -- the same as parallel -- where the warped version lit 2 cells at 0.175; at 0.25 m both light all seven. The reading now begins where the geometry says it should, and the results of R6 are unaffected because gaze is not a distance reading (`artifacts/retina_modes.png`, `artifacts/bank_honest_vs_warped.log`; Methods 5.6).

Convergence and the distance bank. With a ball straight ahead and 600 steps to settle, the two eyes do converge, and the angle they reach agrees with the geometry over the near and middle range and falls short beyond it: 0.496 rad measured against 0.537 required at 0.20 m, 0.436 against 0.433 at 0.25 m, 0.368 against 0.363 at 0.30 m, 0.210 against 0.273 at 0.40 m, 0.173 against 0.219 at 0.50 m, 0.096 against 0.110 at 1.00 m, 0.034 against 0.061 at 1.80 m and 0.001 against 0.037 at 3.00 m. The bank of seven distance cells turns on monotonically: one cell at 3 m and beyond, two between 1.2 and 2.5 m, three between 0.7 and 1.0 m, and seven from 0.35 m inwards. Two defects are visible in the same table and are reported rather than smoothed. The shortfall beyond about 1 m has a known cause in the design: the stereo offset is specified as a number of retinal *columns* rather than as an angle, so the angle it stands for shrinks as the retina is made finer -- the wrong direction, and the reason finer eyes made convergence worse rather than better. And the left eye consistently turns further than the right (at 0.30 m, -0.234 against +0.134), an asymmetry in the muscle wiring whose root we have not found (`docs/眼睛肌肉与自动聚焦_实现_20260923.md`, `artifacts/eye_focus_bands.log`).

Looking at whatever changes. The eyes move to the place in the picture where activity is changing most, and the route that does it is made of cells rather than of a tracking rule: the picture is compared with itself one step earlier and the difference drives the eye muscles. Measured as the mean absolute angle between the gaze centre and a ball at 0.90 m over the last 230 of 250 steps, in four scenes and three configurations of the route:

scene	only the picture	the paper's route, no competition	plus one competing cell
clean, fast sweep	0.0765	0.0410	0.0449
clean, still ball	0.0080	0.0096	0.0111
clean, slow sweep	0.1571	0.0859	0.1002
furnished arena	0.0949	0.1488	0.0969

One competitive inhibitory cell between the two eyes improves the clean moving scenes by about 40 per cent and does nothing for the arena, and a clean still ball is not chased, which is the wanted behaviour. A systematic offset of about 0.1 rad remains -- the eyes come to rest about 0.1 rad to one side of the ball -- and we have not removed it (tools/measure_eye_gaze_routes.py, artifacts/eye_gaze_routes.log).

A wall that can be seen. A wall of plain colour at 0.30 m renders as five to seven grey levels and gives the contrast cells no input at all, so "seeing the wall" had never happened and not hitting it rested entirely on touch. With a striped texture the summed distance cells read the wall monotonically from 0.30 m to 1.80 m (5.941, 2.716, 1.447, 0.414, 0.146) and the near and brake populations fall from 0.938 and 0.963 to 0.017 and 0.033. This is the distance of the wall and not of a stripe on it: no cell is asked for metres (tools/probe_wall_distance.py, artifacts/wall_distance.log). The price of the stripes is that the whole picture now changes: the mean change signal rises from 0.25 to 0.95, which presses the convergence bank down -- at 0.30 m the seven cells go from all lit to none -- and the per-column "this column is busier than it was" route reads zero in nine threshold settings across five motions, because a striped wall fills the field and a ball at 3 m is one or two pixels. What carries "something is near, slow down" is the change-magnitude route together with disparity and not the looming route, which is the same finding as R13's route that reads zero, reached from the other direction (docs/墙条纹与逼近判断_20260923.md).

R17. Ears that separate front from back without solving for the source (no figure)

Two ears deliver two raw waveforms and nothing else: no spectrum, no arrival times, no distance, no direction. The front end that turns them into cell activity is 40 cells and 84 fixed synapses in seven named groups, and it performs no FFT, no cross-correlation, no peak search and no angle computation; its input is the two waveforms and its output is cell activity in [0, 1]. Two cues are therefore available to the cortex, and only two.

Left from right by a fixed coincidence. A left-preferring cell is excited by the left ear's activity delayed by three samples -- 187.5 microseconds -- together with the right ear's activity now, and a mirror pair gives the other side. Three samples is a constant written into the wiring before the brain runs, not an estimate of the arrival time taken from the input. The module is covered by ten tests, including exchange of the two channels (which exchanges the two orientation cells exactly), silence, decay with distance, and independence from the block length the audio is delivered in (docs/auditory_neurons_handoff.md).

Front from back by the shape of the spectrum, not by loudness. The outer ear and the head are modelled where they belong, in how the sound reaches the ears: each frequency band is tilted by the direction of the source -- high frequencies brighter from in front, duller from behind -- and the far ear loses high frequencies by an amount that grows with frequency, with the two ears equal for a source dead ahead or dead behind. The cells that read this compare the *shape* of the spectrum between the two ears, so a distant loud sound and a near quiet one read the same. A pure tone has no shape to compare and gives no front/back reading at all, which is what a real ear does. Measured: a source dead ahead at 0.55, 0.85 and 1.70 m gives front cells 0.190, 0.239 and 0.188 with back cells at 0.000; dead behind gives front cells 0.082, 0.057 and 0.000 with back cells 0.121, 0.126 and 0.140 (docs/auditory_neurons_handoff.md).

A way of doing it that was wrong, and what it cost. The pinna cue was first implemented as a short delay added back onto the same ear. That delay fell inside the three-sample coincidence window that the left/right circuit reads, so one ear's own echo was read as the other ear, and a source on the left turned the eyes to the right. Replacing the delay with a frequency tilt removed the fault, because a change of loudness cannot move a phase (docs/binaural_senses_handoff.md).

The cells reach the muscles, and turning is graded rather than a switch. Turning the front/back route on adds 29 per cent to the eye turn and 46 per cent to the orienting population for a source behind (0.198 to 0.256 and 0.254 to 0.370), against

11 and 18 per cent for the same source in front: the cue contributes most where it is needed, since a source in front is already located by the left/right cue. The direction is correct on both sides in every case. With a source swept in angle -- 0, 9.5, 22.6, 36.9, 49.4 and more than 59 degrees off centre -- the eye reaches 0.000, 0.011, 0.043, 0.190, 0.538 and 0.600 (the last is the mechanical limit of the eye), so the response is graded over the range rather than a threshold (tools/measure_sound_turning.py, artifacts/sound_turning.log, docs/眼睛肌肉与自动聚焦_实现_20260923.md).

It does not disturb walking. Fifteen seconds in the furnished arena with the eyes and the ears both live, three seeds: 0.922 m walked with the sound-to-orienting route off and 0.916 m with it on, with the minimum body height and the minimum vertical axis identical in all three pairs (tools/measure_sound_walking.py, artifacts/sound_walking.log).

One asymmetry we have not fixed. A source in front still drives the eyes more than twice as strongly as the same source behind (0.532 against 0.198 with the new route at zero gain). That is the older left/right route, not this one: the two ears' time difference is the same for a source dead ahead and dead behind, the difference is in the frequency content, and the coincidence circuit is more sensitive at high frequencies (three samples is 60 degrees of 880 Hz against 18 degrees of 262 Hz), so a source in front is read more strongly. Both sides are read with the correct sign; the response behind is simply weaker, and we have not equalised it.

What R17 does not show. The pinna and head-shadow cues are properties of how the sound arrives at the ears and are computed in the acoustic front end from the source geometry; the cortex receives two waveforms and computes no angle. The two "front" cells have no outgoing edge at all -- setting the route's gain to zero leaves nothing -- so what is measured is the back-cell route, and the front cells are a sense that nothing yet reads. The front end's frequency resolution is finite (512-sample frames leak between bands, a property of the frame and the window rather than of the source geometry), and nothing here is tested with echoes, with several simultaneous sources or with a moving source.

R18. Two brains, each taught one different thing, added into one brain that does both (Figure 11)

Every measurement above is a measurement of one brain. If the instinct table and the local rule are the whole of the machinery, then two brains built from the same table differ only in the numbers their own lives have written on their edges, and those numbers can be added. The claim is stated as arithmetic on a shared edge list rather than as a property of brains: a third model, which was never taught anything, takes the sum of the two weight vectors, and it should hold both lessons. We tested it on the smallest graph we have and then on the body.

One graph of 17 cells. The parents are the graph used in R14 and R15 (four sensory cells, eight tuned hidden cells, one compressed cell, two motor cells and two inhibitory cells; 62 edges, of which the eight hidden cells and the compressed cell can reach both motor cells). A parent is taught one pairing only: thirty pairings of 0.4 s of a four-channel stimulus with the teacher on one motor cell and 0.2 s of rest, with learning on throughout. Parent A is taught that cue A means motor cell 13; parent B that cue B means motor cell 14. Reading is 0.6 s per cue with the teacher absent and learning off, and the number we report for a cue is the margin by which the motor cell that cue is meant to drive beats the other one. A single lesson does not answer the other cue: parent A reads cue A at +1.000 and cue B at -0.729, parent B reads cue A at -0.726 and cue B at +1.000, and an untrained graph reads 0.000 on both. Adding the two weight tables and halving them gives +1.000 on both cues, on 5 of 5 seeds. Two other ways of combining the tables give the same result: keeping, per edge, whichever parent moved it further from its birth weight, and adding both parents' changes to the birth value.

Why that works is a fact about the two lessons, not about the arithmetic. Each lesson changes 24 of the 62 edges at all, and 5 of them by more than 0.05. Parent A's five all end on motor cell 13 and none on 14; parent B's five all end on 14 and none on 13; and no edge of the graph is moved by more than 0.05 in opposite directions by the two lessons (0 of 62). The two lessons write to disjoint edges, so halving each of them does not involve a decision about which one to keep.

Controls. A parent merged with itself reproduces that parent exactly (+1.000 and -0.729), which is what the arithmetic requires, and two untrained parents give 0.000 and 0.000. To check that what survives is the content of each lesson and not merely "both cues now answer", the same two lessons were taught with the motor cells exchanged -- cue A to motor 14 and cue B to motor 13 -- and the merge of those two parents answers cue A with motor 14 and cue B with motor 13, both at +1.000. The boundary is the conflict case: two parents taught opposite answers to the same cue (cue A to motor 13, cue A to motor 14) merge into a model with no answer at all, a margin of 0.000 to 0.001.

The same operation on the body. The 1,813-cell controller of R14 is used with the empty route extended from one innate action to two (`empty_routes=("flexion", "locomotion")`), so that the eight feature cells and the compressed cell have a route laid toward the gait cell and toward the flexion cell; both parents are built this way and the birth weights are identical. One parent is taught that cue A means walk: six pairings of 2 s of cue A with 0.6 of external drive on the gait cell and 0.5 s of rest. The other is taught that cue B means brace, in the same way, on the flexion cell. Each lesson moves 5 of the 17,745 edges by more than 0.05, and all five go to its own action cell (walk: 5 to the gait cell and 0 to flexion; brace: 0 and 5; 5 of 5 seeds). Above that threshold no edge is touched by both lessons and none is moved in opposite directions. At any size at all 9,382 to 9,497 edges change a little, because the innate edges are plastic at 0.005; 0.05 is the size at which a change starts to matter. Measured over 8 s per cue, on the same starting pose and the same cell state, so that the only difference between arms is the weight vector: an untrained brain walks 0.001 m on cue A and reads 0.004 on the flexion cell for cue B; the walk parent walks 0.787 m and reads 0.003; the brace parent walks 0.004 m and reads 1.000; the merged brain walks 0.664 m *and* reads 0.787, on 5 of 5 seeds. Combining the tables by keeping the larger move per edge gives 0.921 m and 1.000 -- better than the average, and for a stated reason: the average writes each lesson at half the strength its parent had given it, and keeping the larger move keeps each lesson at full strength. A parent merged with itself reproduces its own behaviour to three decimals (0.7868 m and 0.0027, both times), which is the check that installing a weight vector on the graph is faithful and not a re-run of the training.

What R18 does not show. The parents of a pair are identical apart from the lesson, so this is not evidence that two brains with different innate wiring, or of different sizes, can be merged; the cells and the edge list have to match. Merging preserves and does not create: a merged model cannot do anything neither parent could do, and it inherits everything the local rule cannot do, including the toppling of R9. Where the two lessons touch the same edges in opposite directions the result is no answer at all, which is measured rather than assumed. The 17-cell case reads two synthetic channels and not a retina, and each of its actions is one motor cell; the body case is one gait command and one flexion command, not an animal that gets up after falling -- a merge of walking and righting needs the righting behaviour itself to exist first (R1, R3), and we have not built that pair. Nothing here tests the larger version of the idea, in which many candidate models are written, put through a large question set and selected, and no comparison is made with further training of one parent, which this system does not have.

R19. Every sense is a pair, including the cell that fires on nothing (no figure)

Every sense channel in this system carries the same channel twice: one cell saying that a level has been reached, the cell beside it saying that it has not, so that a channel with nothing on it still has a reading. The reference build coded every channel this way at its input layer (Section 2.4), and where a channel is recruited rather than cut into levels the same two readings are the lit prefix and the unlit remainder of one population. The rebuilt engine of Section 5.12 carries a companion cell beside each of the four reflex senses -- touch, foot obstacle, foot load and foot slip, four cells each -- beside the three auditory sheets (six, six and four cells), and beside every photoreceptor; in every case the companion is driven by one minus the same number, in the same step, so the second is lit whenever the first is not. The visual pair also gets one thing no other pair has: the unlit half of the picture converges on a sheet of its own (`retina_opposite`), with the same gain and the same sector rule the lit half uses to reach retina (`born_wired/embodied.py`).

The point of the pair is that a channel with nothing on it is still a reading. Measured on the birth animal of the rebuilt engine, three seeds each, in the exam bank (`engine_v2/tools/taskbank.py`): on a black screen the unlit half of the visual input reads 0.96985 and the lit half reads 0.00000, and the unlit half's own sheet reads 0.97143; on a white screen the two swap. In a silent room the auditory channel reads 0.00000 and its opposite cell reads 0.98983. Standing with nothing touching the body, the touch, obstacle and slip channels each read 0.00000 and their opposite cells read 1.00000. The four feet of that pose carry 0.42 to 0.54 of the load channel, and each foot's load cell plus its opposite cell adds up to 1.000 (worst error 0.000 over the four feet). Four questions of the bank ask exactly this -- `the_dark_room_is_not_silence`, `a_quiet_room_is_not_a_sound`, `nothing_touching_it_is_not_nothing` and `the_weight_pair_still_carries_the_weight` -- and the birth animal passes all four on all three seeds (`engine_v2/artifacts/exam_birth_full.json`).

What the pair costs. The pair adds 165,938 cells and 165,888 synapses to the shipped configuration of R13: one companion beside every photoreceptor (165,888), eighteen cells on the unlit half's own summary sheet, and four companions on each of the four reflex senses (32). The cell and synapse counts of that configuration therefore read 750,075 and 1,978,219 where R13 reads 584,137 and 1,812,331, and a fresh profile of the same configuration reads 13.29 ms per step where the profile R13 quotes reads 11.19 ms (artifacts/profile_after_opposite.log); what part of the step is the pair itself is not separated here. The reflex companions carry no synapse.

What R19 does not show. Nothing downstream reads the second cell of any pair yet: no edge leaves an opposite sheet toward a reflex, association or motor cell, and the dark half of the picture stops at its own convergence sheet. A black picture therefore still reaches no reflex cell, and this is not a claim that the animal notices that the light has gone out. The earlier single-layer build had the same gap for a different reason and reported it: with the pair summed into one number at the last layer, a black frame left the whole output sheet inhibited and dark, because every negative cell of every pair was lit and nothing above them preferred the dark. Closing that gap needs cells that prefer the dark the way a retina has both ON and OFF cells, so that an unlit picture drives a layer above the input; that is a structural step with its own measurements, and it is not taken here.

R20. A memory that is used holds its place, and a memory that is not does not keep sinking (no figure)

Every edge of the rebuilt engine carries two numbers that decide how far it can move: the weight it was born with, and its own ceiling. Two readings of the shipped configuration, taken while the animal simply lives -- walking with the local rule running and nothing else happening, few minutes apart on three seeds (tools/measure_weight_drift.py; artifacts/weight_drift_60s_seed0.log, artifacts/weight_drift_300s_seed0.log, artifacts/weight_drift_300s_seed1.log) -- say what those two numbers do to a table of 1,978,225 movable edges.

reading	60 s of life	300 s of life
edges that rose / fell / did not move	624,843 / 728,710 / 624,672	652,178 / 710,091 / 615,956
edges that moved by more than 0.001	67,936	202,527
edges that moved by more than 0.01	10,700	18,939
largest rise	+0.0784	+0.0960
largest fall	-0.0019	-0.0253
mean change per edge	+0.000107	+0.000348
edges above their birth weight	840,281	857,483

Five times as much time buys three times as many moved edges, and a largest rise that is barely changed (0.0784 to 0.0960). The table is not sliding; it is settling. The second seed reproduces both extremes to four decimals (largest rise +0.0960, largest fall -0.0253, 203,100 edges moved by more than 0.001). The direction of the drift is up, not down: edges sitting above their birth weight rise from 840,281 to 857,483 while those below it fall from 685,557 to 677,072.

Where weights do fall, they fall on the posture drive, and they fall by almost nothing. After 300 s the largest fall anywhere in the table is 6.0000 to 5.9937 on tonic -> motor, one tenth of one per cent of a weight whose structural floor is three per cent below birth; and between those same two populations the rises outnumber the falls, 11,399 against 7,587. The other falls are inside the eye's own first layer (retinal_interneurons -> retinal_opponent, 14,571 edges of the 40,163 that moved by more than 0.001) and on the two steering populations. Nothing anywhere in the table loses a meaningful fraction of its weight in five minutes of life.

A lesson is kept. The teachable route is the one table the animal is meant to keep, and it is kept outright. Measured over one minute of life with the rule running (the_memory_survives_a_long_life), over ten minutes of walking in silence (artifacts/forget_600_quiet.log), over ten minutes with the tone coming and going (artifacts/forget_300_life.log), and with a second lesson written down the same route (a_second_tone_does_not_wipe_the_first, artifacts/forget_300_second.log):

what happened after the lesson	retreat on the tone right after	at the end	kept
sixty seconds of ordinary life	0.970	0.945	97.5%
ten minutes of walking, silent	1.000	0.975	97.5%
ten minutes, the tone coming back	1.000	0.984	98.4%
a second tone learned on the same route	0.970	0.942	97.2%

The route's own weights do not fall even where its behavioural read-out dips a little: over ten silent minutes they move from 0.9739 to 1.0038 and from 1.0285 to 1.0569, i.e. they rise. A lesson that nobody reinforces is not eroded by the rest of the animal's life at this time scale.

A route that is used stops at its ceiling. Held on to continuously, with the keeper's hand on the body the whole time (the_route_stops_growing_at_its_ceiling, artifacts/exam_new_tasks_3seeds.json), the route's six edges rose by 1.98657 over the first 20 s, 1.11257 over the next 20 s and 0.02656 over the third, ending at exactly 1.500 on its two largest edges -- their backup_cap, hit and held. So the route climbs and then stops climbing, and what stops it is its own hard ceiling rather than any weariness in the rule.

The three questions of this section are in the bank (tools/taskbank.py) and the birth animal passes all three on all three seeds, which is what makes them measurements of the shipped animal rather than of a special configuration: the bank now asks 126 questions and the birth animal passes 99, 94 and 94 of them (artifacts/exam_birth_full.json).

What the same quantities do in a brain. The shape of this rule -- most of a table stable, a small minority moving, movement bounded in both directions -- is the shape that long-term imaging of cortical synapses reports:

measurement	result	source
adult mouse visual cortex, one month	about 96% of dendritic spines stable, half-life greater than 13 months; 73% in the critical period	Grutzendler, Kasthuri & Gan (2002)
adult mouse barrel cortex, weeks	about 50% of spines persist at least a month, the rest days or less; experience-dependent plasticity increases turnover	Trachtenberg et al. (2002)
mouse cortex, whole life	a small fraction of new spines, and most spines formed early that survive elimination, are preserved for life; lifelong memories sit in largely stable networks	Yang, Pan & Gan (2009)
adult neocortex, review	overall morphology and a subpopulation of small synaptic structures remarkably stable; cell-type-specific turnover in specific circuits	Holtmaat & Svoboda (2009)
wake and sleep	wakefulness a net increase in synaptic strength, sleep a downscaling; cerebral metabolic rate falls 15-20% after about 2.5 h of sleep	Tononi & Cirelli (2006); Vyazovskiy et al. (2008)

Two of the three numbers are close: roughly seven edges in ten do not move by more than 0.001 in five minutes, and the minority that moves is bounded in both directions, which is the same shape as "a small fraction of new spines persists and the rest is removed". What the brain has and this rule does not is a *reason* for an edge to be one or the other: in the cortex a spine's stability depends on its age and on its history, while here the tether is one constant on every edge of the shipped configuration once the route is taken out.

What R20 does not show. There is no sleep phase here and no consolidation window: nothing in this system rescales the table between waking periods, and the tether is a global constant (0.2) rather than a value that reflects how old or how used an edge is. The claim is about the shape of the rule, not about a brain-like allocation of stability. It is also measured over minutes. Ten minutes of life is not a lifetime, and nothing here says what the table looks like after hours of it. The interference question is asked once, for one pair of tones down one route; that is not a claim that arbitrary pairs of lessons cannot overwrite one another. And the falling edges of the first table are only described, not attributed: which of them changes behaviour, and by how much, is not measured here.

4. Discussion

4.1 What the results support

Taken together, the measurements support a specific and unglamorous claim: **a recurrent cortical population with a fixed, pre-specified wiring diagram and a purely local correlation rule is already enough to close a sensorimotor loop and to produce behaviour that can be attributed cell population by cell population.** Nothing in the results requires a global objective, a critic, a reward, or a backward pass.

The nine claims of Section 1.1 are not equally supported, and the difference is the useful part. Table 1 states, for each of them, the experiment that bears on it and what came out of that experiment; the last row is not a claim but the question that R4 was built to answer.

Table 1. What each claim rests on.

claim	test	outcome
(P1) a sensory hierarchy is a compressor whose output is a bus	R5: localisation and name overlap, measured layer by layer	supported. Layer 1 places every newly active cell inside the column a 0.31-cell ball occupies; layer 3 scatters the same ball over up to 13 columns and produces nothing at all for a ball at dead centre. Direction names are perfectly exclusive in layers 1-2 and 14 percent shared (worst column 75 percent) in layer 3, where one column is lost.
(P2) thought is co-activation of a cortical population	R8: remove the stimulus, then remove the recurrence	supported. Stimulus removed: both codes read zero (0 and 3 cells of 1346 and 1360) while the walking action stays lit on 495 of 495 ticks. Recurrence removed: the one wired reflex is unchanged to a decimal place and every action disappears.
(P3) the prefrontal population compresses a second time	R2: silence it; measure both halves of the drive. R9: give it loops and watch it over 200 ticks	partly supported. It supplies 0.85 of the 1.65 that crosses the threshold, and removing it abolishes the behaviour. Whether it accumulates its own state over time <i>is</i> now tested (R9): given loops it does, and with no mechanism for decay the accumulation is unbounded (1581 to 2916 active cells over 200 ticks), and it over-drives the motor loop.
(P4) instinct is the initial state of the sheet	R1 row 4: clear the instinct table, leave everything else untouched	supported. The same five brains then stand on 0/5 seeds, approach on 0/5, and settle at exactly the height of the no-sensation control.
(P5) a network cannot start from silence	R1 row 4, R3 (staged additions), R7 (blurred map)	supported. Random background connectivity with no written wiring produces no ordered behaviour, and each group of rules that is added buys exactly its own capability without disturbing the earlier ones.
(P6) acquisition during life needs a reward-like signal and no objective	R9: four stages, one brain, dopamine region on or off	supported, with a boundary. With the dopamine region lit during stage 3 the sound acquires the walk on 5/5 seeds (0/30 ticks before, 199-200/200 after, standing up included); with it dark, teaching writes 0 connections and the sound still does nothing (0/200, 3/3). The behaviour does not stay upright (4/5 seeds topple 42-101 ticks in), and controls on the same brains show that is the weight-growth rule, not the reward rule.
(P7) any layer of a hierarchy, not only the last, is a signal source	R14: a synthetic case where two stimuli reach the compressed cell identically; the same route on a body, five seeds, three conditions; ten minutes with learning always on	supported, within a declared scope. With a read-out on the compressed cell alone the two stimuli are one; with a plastic route from the intermediate cells the two motor cells separate, and with learning off they do not. On the body, stimulus B then A then B gives trunk heights of about 0.250, 0.147 and 0.250 m, the second B recovering the first to within 6.8 micrometres. The scope is small (17 cells synthetic, 1,813 cells on the body) and two route caps are declared priors rather than learned.
(P8) the sheet is spatial, and a tick costs what is lit rather than what exists	R15: three arms of 1,000 cells differing only in the coordinates of the sheet; two engines measured side by side at 100,000 cells; 100,000 to 800,000 cells at fixed drive	partly supported. Depth is told apart by the solid sheet and by nothing else (overlap 0.084 against 0.290 for flat coordinates and 0.343 for random wiring), bought with some loss of separation between different objects (0.041 to 0.081). Cost follows activity in the rebuilt engine (0 cells 0.04 ms, 11,818 cells 13.71 ms) and not in the original one (5.2 to 5.7 ms whatever is alight); eight times the cells cost 1.06 times the time at fixed drive. The prediction that the alight <i>fraction</i> falls is confirmed, but the cost per active cell is flat, so a larger sheet resolving the same input with fewer cells is not yet demonstrated.

claim	test	outcome
(P9) what one life writes can be combined with what another life wrote	R18: two parents of the 17-cell graph, one lesson each; the same two lessons with the motor cells swapped; two parents taught opposite answers to one cue; two parents of the 1,813-cell controller, one taught to walk and one to brace	supported, within a declared scope. Each lesson moves 5 of 62 edges (17-cell) and 5 of 17,745 edges (body) by more than 0.05, and the two lessons of a pair write to disjoint edges, so adding the two weight tables and halving them answers both cues on 5/5 seeds (+1.000 and +1.000) and the merged body walks 0.66 m on one cue and braces on the other. The swapped-lesson control answers both cues with the swapped motor cells, so the merge carries content and not merely "both cues answer". Two parents taught opposite answers to the same cue merge into no answer at all (margin 0.000 to 0.001). The scope: one graph per experiment, parents identical apart from the lesson, nothing retrained after the merge, and no new capability created.
(P10) stability comes from the shape of the rule -- a ceiling and a spring back to the birth weight -- and not from a decay term	R20: 1,978,225 movable edges during 60 s and 300 s of ordinary life, three seeds; the teachable route measured after one minute of life, after ten minutes of walking, after a second lesson on the same route, and after three equal 20 s stretches of continuous use	supported. Five times the time buys three times as many moved edges (67,936 to 202,527 of 1,978,225) and a largest rise that barely changes (+0.0784 to +0.0960), so the table settles rather than slides; it drifts up, not down (edges above their birth weight rise from 840,281 to 857,483). The largest fall anywhere in 300 s is 6.0000 to 5.9937 on tonic -> motor , a tenth of one per cent of a weight whose structural floor is three per cent below birth, and even between those two populations the rises outnumber the falls. The teachable route keeps 97.5% of its read-out after a minute of life, 97.5% after ten minutes of walking in silence and 98.4% with the tone coming back, and 97.2% after a second lesson is written down the same route; its own weights rise rather than fall over those ten minutes. Held on to continuously it rises 1.98657, then 1.11257, then 0.02656 over three equal 20 s stretches, ending pinned at exactly 1.500 -- its declared cap. The scope: the tether is one constant on every edge of the shipped configuration rather than a value that reflects an edge's age, there is no sleep or consolidation phase, and the interference question is asked once, for one pair of tones down one route.
(Q4) is local plasticity doing anything a fixed recogniser is not? (question 4 of 1.2, not one of P1-P8)	R4: which hues start the walk, in the same brains, before and after a drift, with the rule on and off, and at three step sizes; the same protocol on two builds	supported, with a bounded reach and a size that is not reproducible. The boundary is where the sum of two cell populations crosses: every hue summing to 1.39 or more is accepted on 5/5 and every hue at 0.97 or less is rejected on 0/5. The three hues whose sums fall within 0.05 of the line (96, 168, 180) are decided by the random background connectivity each brain is born with, and the rule carries exactly those hues across the line: on the build of the main table that shows up in one brain of five at a 3- and a 12-degree step and two at 6 degrees, and in 5 of 5 on the earlier build, where every brain sat below the line. In neither build does any brain cross the gap at 120-144 degrees. What is reproducible is the reach of the rule -- to the edge of the island and no further -- not the number of brains in which it has anything to do.

Four sub-claims are worth separating, because they have different evidential status -- one of them, the last, we now report as a negative result.

Ordered behaviour without an error signal. This is the best-supported claim. It rests on R1 (five seeds, plus a no-sensation control that collapses), R2 (a population-specific ablation that removes exactly one behaviour), and R3 (a nested sequence of stages in which each addition buys a capability).

A repertoire that is assembled rather than trained. R3 supports the weaker version: adding innate structure adds capability without destroying earlier capability, and without any retraining. It does not show a repertoire that *develops*. We return to this in 4.3.

Where a behaviour lives. R8 is the result that most directly tests the framing the paper starts from. It shows both directions: after the stimulus is removed the two codes that carry it read zero while the action is still lit on every tick, and after the cortical state is removed the same wiring still supports the one reflex but no action at all. The claim that thought is co-activation of a cortical population is therefore not just a description of the diagram; removing the co-activation removes exactly the behaviours that the diagram predicts it should, and leaves the rest.

Local plasticity as the reason that recognition can drift. R4 supports a bounded version of this, and the boundary on the claim is as informative as the claim. The drift moves the recognition of one brain of five on the build of the main table, and all five on the earlier build, and the difference between the two builds is entirely in the draw of random background connectivity -- so the part that is reproducible is not how many brains move but how far: the rule carries a marginal hue across the line and out to the edge of the visual island it belongs to, and never across the gap at 120-144 degrees. What it cannot do is create a boundary: with the correlation rule switched off entirely, in a brain that has never drifted, the same five brains accept and refuse exactly the same hues, because the boundary is the value at which two populations sum past

threshold, and that value does not depend on the rule. This is the result we would most like to see replicated, because it is the one that speaks against reading a fixed recogniser and a correlation rule as interchangeable designs: here they are distinguishable, and what the rule buys is the toleration of drift that is already under way rather than a wider category to begin with.

4.2 What "the last layer" is for

The framing we started from says that a layered sensory network is a compressor: many peripheral channels in, few cortical channels out, and the output is a bus rather than an answer. R5 gives that framing a number. A ball a third of a cell across displaces the uncompressed layer exactly where it should and the compressed layer almost nowhere — at dead centre, nowhere at all. The compressed layer's direction names are 14% shared at the median and up to 75% shared in the worst columns, where the first two layers are perfectly exclusive.

It is worth being explicit about the implication, because it cuts against a habit. If one attaches a read-out to the last layer of such a hierarchy and trains it, one is training a classifier on a *deliberately impoverished* signal. That is a reasonable engineering choice when the read-out is the point. It is a poor model of a brain, in which the compressed signal is an input to further recurrent processing rather than a terminal answer. R5's failure of our own prediction — the reflex still works when wired to the compressed layer — makes the point sharper rather than weaker: the compressed layer is not *useless*, it is *less informative*, and the difference only becomes visible when one asks a question that needs the missing detail (where exactly, and is it there at all).

The measurement of R14 makes the same point from the other side, and it is the reason P7 was added to the framing. Reading the compressed layer is not merely a lossy way of reading a hierarchy; it is a way of reading a *different* signal, one in which two stimuli that the layers underneath do keep apart have already been merged. A system that reads only the last layer has thrown away that distinction before any read-out sees it, which is what the synthetic case shows directly (two positions, one compressed response) and what the body case then shows can be recovered: with a plastic route from the intermediate cells, the two stimuli end on two different motor cells, and with the route present but learning off they do not. What that means for the framing of P1 is not that the compressed layer is useless -- R5 already shows the reflex still works through it -- but that "which layer" is an empirical question about a signal rather than a design hierarchy with an answer at the top.

4.3 What this does not show

Development. The headline claim of this project is that complex behaviour is built on top of innate behaviour, in the way that an infant's reflexes precede its skills. R3 measures the *dependence* -- stage by stage, in a fixed brain -- and R9 measures the *acquisition*: one brain, four stages, one new link between a sound and an innate action, with a reward-like signal as the only thing added to the system. What R9 does **not** show is a repertoire that develops. It is one link, learned in one session, from a sense to an action the brain could already perform, with the innate wiring and the body unchanged, and with us deciding when the reward is on. R10 takes one part of the experimenter out of that loop: the reward is now delivered by contact with the body, through a sensory region, rather than by a flag in the script. Development in the strong sense -- a repertoire that grows because of what the brain has already done -- remains a design commitment, and we say so rather than dressing up R9 and R10 as more than they are.

Scale. Everything here runs on 148,032 neurons on one CPU core, and R11 measures what a wider cortex costs on the same machine: up to 925,236 neurons and 62.4 million excitatory synapses, with 7,200 MB of peak memory as the ceiling. The cheap end of that programme -- 24x32 -- we have since run with behaviour, and it is instructive in both directions: the righting reflex is unchanged to the digit, gaze improves by about a quarter, and the colour boundary narrows, because the current a rule delivers is fixed by construction while the inhibition a cell receives grows with the amount of active cortex. Widening is therefore not mechanical. The table's strengths are calibrated to the activity level of the build they were measured in, and a wider brain needs them re-tuned. Nothing above 24x32 is anything but a cost curve, what happens at 10^8 or 10^{10} neurons is not measured, and the reader should not assume that the observed qualitative behaviour is scale-invariant.

Authoring the innate wiring. The instinct table is written by us. For the walking repertoire it is distilled from a separately trained reinforcement-learning policy: we ran that policy over a grid of velocity commands, recorded the joint trajectories, and wrote one rule per trajectory. This is supervision, and it happens before the brain exists rather than during its life. A model that had to *acquire* its motor repertoire from its own reflexes would be a strictly stronger result, and it is not what we have. The repertoire is innate either way -- the motor library is written into the instinct table before the brain runs, and hand-writing every weight would be equivalent in kind; distillation is a convenience that saves writing 221 action trajectories by hand, not a training procedure that the rest of the system depends on. We have since measured that source directly (Section 4.4): its own cost is 99,483,648 physics steps, and its input contains no exteroceptive channel at all — what it supplies is a gait, not the decision to use one.

One body, one sensory layout. The body is a quadruped, the eyes have one degree of freedom each, and the visual field is 100 by 75 degrees. Nothing in the architecture is specific to quadrupeds, but we have not demonstrated that.

4.4 Relation to other work

Reinforcement learning (Mnih et al., 2015; Hwangbo et al., 2019). Every behaviour here would be straightforwardly learnable by RL. The point is not that RL cannot do it; the point is that in this system the behaviour *is not learned at all* — it is a property of the initial wiring, and local plasticity, on its own, did not change the behaviour we measured (R4). The relevant scientific question is not "can RL match this" but "how much of animal behaviour is of this kind".

We did not train a baseline of our own; we measured a published one, because a published artefact states its own cost. `diasAiMaster/unitree-go2-velocity-flat` is a PPO policy for this same Go2 model (flat ground, velocity-command task); its model card gives 8,192 parallel environments x 24 steps x 506 iterations = **99,483,648 physics steps**, 10x RTX A4000 for 18 minutes. We drove it on **our** body at our muscle parameters (`对照_现成RL_驱动.py`, `log 日志_对照_现成RL.log`): told "1.2 m/s forward" it covers 3.71 m in 4 s, and 4.98 m told 1.6; told zero it covers 0.07 m; told -1.6 it covers -2.90 m. It stays upright in every case, and it also gets up from the prone posture. Its input layer is 45 numbers: body angular velocity (3), gravity direction (3), the three velocity commands (3), joint offsets (12), joint velocities (12), and its own previous action (12). **Not one of those 45 comes from an eye or an ear.** The artefact is therefore a very good translator from three externally supplied numbers into a gait: 10^8 samples buy the lower half of the problem, and the upper half — whether to move at all — is not in it. Supplying that half is itself a learning problem, and not a small one: our own attempt (three seeds, PPO, R1's criterion) first crossed the 0.30 m line at $1.2\text{--}1.8 \times 10^5$ steps, and no seed ever reached 1.0 m, where the instinct table covers the same thing with one row and no samples at all. That asymmetry is what "innate" means here.

Predictive-coding and free-energy accounts (Rao & Ballard, 1999; Friston, 2010). Those frameworks also avoid an explicit loss, but they replace it with a generative model and an inference procedure. Here there is no generative model and no inference: there is a fixed wiring diagram, a threshold, and a correlation rule.

Developmental robotics and intrinsic motivation (Oudeyer & Kaplan, 2007). That literature builds repertoires by staged curriculum learning. R3 is an unusually literal version of the idea — the curriculum *is* the wiring — but we add nothing to the theory of curricula.

Language models (Brown et al., 2020; Kaplan et al., 2020). A separate build in this project runs the same mechanism on a completely different substrate, and we report it here for what it says about *this* system rather than about language. In that build the cells are single Unicode characters, the text is 344,787,049 characters of modern Chinese, the only rule is the same local correlation rule, and there is still no objective, no back-propagation and no language model anywhere in the loop.

What that build shows, verified step by step, is how a population of this kind holds a state and moves it. Characters are active in **groups**, not one at a time; after the prompt ends the group does not fall silent, and it does not freeze either -- membership turns over from step to step while the population keeps its characteristic size, and every prompt eventually settles into an exact periodic orbit (periods 1 to 6 in the observed window). That is a dynamical observation and we do not read it as a measure of thought.

That is the mechanism R8 and R9 need, and it is also why they need it. The prefrontal code for red outlives the red for the same reason the character group outlives the prompt: a recurrent population with a local correlation rule holds a state. But holding one is not free at any size. In the character build the same protocol with a small activity budget left 45 of 46 prompts completely dark -- the state died instead of circulating. The cortical sheet at 148,032 cells is in that regime: it does not spontaneously grow the loops that would let a thought persist, which is why R9 has to give the prefrontal population its loops explicitly. What separates the two builds is scale and connection density, not the rule.

It is **not** evidence about language and we do not claim that it is: it behaves as a bounded associative memory, not a generative model -- lighting some characters makes other characters light, but it does not learn to answer a question or to follow an instruction -- and the paper that would be needed to make a language claim is not this one. The reports and the scripts of that build are in the repository under [语言规模实验_language_scaling_20260913/](#).

Attractors, reservoirs and large-scale models. The behaviour of R8 is a trajectory through a recurrent population, which places this system in a long tradition in which the computation is the settling of a population rather than a feed-forward pass (Hopfield, 1982; Amit, 1989; Maass et al., 2002). It shares the ambition of the large-scale reconstructions of cortex but none of their method: those build a detailed model of a known circuit, whereas here the circuit is trivially simple and the question is what the *initial condition* buys (Eliasmith et al., 2012; Markram et al., 2015; Gewaltig & Diesmann, 2007).

How a system of this kind is trained. "An agent that builds its own repertoire over a lifetime, rather than one that optimises a fixed objective", is a named open problem and not a solved recipe. The 2024 position paper on open-endedness research states that a general algorithm "remains elusive" (arXiv:2406.04268), and the doctoral work that comes closest to the framing here -- self-sufficient agents that must discover not only a policy but the reward that defines it -- concludes that reinforcement learning alone cannot be sufficient, because the reward itself has to be found (Colas, 2022). Six pieces of a recipe exist in the literature and none of them is the whole: the co-generation of behaviour and its environment (Wang et al., 2019), the reference implementation of a thousand-model sensorimotor agent (Neural Computation, 2026, doi:10.1162/neco.a.1508), the only large-scale demonstration that a route from single neurons to behaviour can be composed by hand rather than trained (Eliasmith et al., 2012), the finding that a developmental prior can carry as much information as the experience that follows it (Krotov et al., 2023, doi:10.1038/s41467-023-37980-1), the classical division of labour between a fast learner and a slow substrate (McClelland et al., 1995) with its reinforcement-learning form (eLife, 2014, doi:10.7554/eLife.04811), and the intrinsically motivated exploration that an animal uses before it can be taught anything (Oudeyer et al., 2007). We take the position that these are pieces and are not the method: R12 is a candidate for one of them -- where a lesson lands -- and not for the rest.

4.5 What would falsify the framing

We list these because a framing that cannot fail is not doing work.

- If a system with the same innate wiring but *no* recurrent cortical step (i.e. the instincts driving the muscles directly) reproduced every behaviour, the claim that thought is cortical co-activation would be empty. We ran that experiment (R8): the feed-forward controller built from the same wiring reproduces the one wired reflex to a decimal place and reproduces no action at all.
- If the gradual recognition of R4 survived with plasticity switched off in every brain, the result would be attributable to the sensory front end and not to the cortical rule. We ran that control. In four of the five brains of the main build it did survive, and in the fifth it did not: that brain stops at 72 degrees with the rule off and at 96 with it on, and reaches the far island at 168-180 only with the rule on. On the earlier build of the same system it did not survive in any of the five. The claim is therefore restricted to what both builds agree on -- the reach of the rule -- and not to how many brains show it.
- If a network with the same body, the same sensors and the same random background connectivity but *no* innate wiring produced ordered behaviour, the premise that a cortex cannot start from silence would be wrong. We ran that experiment (R1, row 4): on five seeds those brains stand on 0/5, approach on 0/5, and settle at exactly the height the no-sensation control reaches.

- If the repertoire of an animal were fully described by its reflex arc -- sensor to muscle without a recurrent population in between -- then writing innate structure would be the same thing as writing a lookup table. R8 dissociates the two: with the recurrent cortical state removed, the wired sensor-to-muscle reflex (gaze) survives untouched while every behaviour that consists of *doing an action* disappears.

4.6 Outlook: what would be needed for a general architecture

We are asked, reasonably, whether a system of this kind could grow into something general. The honest answer is that the results here demonstrate one ingredient and are silent about the others. It is worth being precise about which ingredient, and about what the others would cost, because the framing makes sharp predictions in both directions.

What is demonstrated. A born-wired recurrent cortical sheet, with no objective function and with a plasticity rule that cannot see a target, already closes a sensorimotor loop, produces an ordered repertoire, and lets a behaviour be attributed to a named population. The repertoire is extended by *writing more innate structure*, and each extension leaves the earlier ones intact.

The learning rule was never the bottleneck for the first problem. The tempting reading of these results is "learning is overrated". That is not our reading. Our reading is narrower and, we think, more useful: the first problem an organism has is not to *learn* — it is to *do something ordered*, and a random network with a correlation rule has nothing ordered to correlate. The scarcity is in initial structure, not in optimisation.

The falsifiable prediction for AI. If that is right, then an agent given a large innate behavioural repertoire before training should need dramatically less experience than an identical agent that must discover its first ordered behaviour. Every current agent is of the second kind. R3 is a one-step version of the claim inside this system; the cross-system version has not been run.

The falsifiable prediction for neuroscience. If behaviour is produced by co-activation of a cortical population rather than by a sensor-to-muscle map, then removing the recurrence should remove behaviours whose expression is an action, while leaving wired reflexes intact. R8 is that experiment, and it comes out as predicted.

The bet, stated so that it can be lost. The position this paper takes compresses into one sentence: in an architecture of this kind, capability is bought with *structure*, not with optimisation, and structure can be written down as fast as it can be understood. If that is right, then the distance between the 585 rules used here and a general agent is not a missing algorithm. It is more rules, more regions, a richer body and a much larger sheet -- expensive, and mostly engineering. Every step of that programme has a worked example in this paper: add a group of rules, get the capability (R3); blur the written map, get a biased but ordered behaviour rather than a broken one (R7); clear the written map, get nothing (R1, row 4).

If the position is wrong, it is wrong in a way that is visible rather than vague. At some point, adding structure must stop buying capability, and what is missing will be something the local correlation rule cannot supply. The experiments that would expose that are ordinary ones: keep adding groups of rules until a group that ought to work does not, or run a development experiment in which a brain with an innate repertoire fails to acquire a capability that its structure appears to allow. This paper fixes the starting point of that argument and shows that the first three steps are real; it does not settle the outcome, and the reader should not let it.

What is missing, concretely. We list the ingredients in the order in which we think they should be attacked, with the honest cost of each.

1. *A trigger that stays a trigger while the code that emits it persists.* This is now the first problem, and R9 is where it shows, but the first two explanations we offered for it were both wrong and the measured one is narrower. It is not the reward-written connections (R9 excludes them). It is not a weight rule that fails to weaken connections it does not need: the weight of everything ending on the motor cortex is unchanged across a 200-tick test (265,436 synapses, 494,436 units, difference 0), and freezing that pathway entirely moves the topple by one tick (R9). It is one calibrated quantity. The innate rules from the prefrontal population to the 15-cell walking chain deliver 0.850 on every tick, and 0.850 is exactly what the chain needs to fire, because the table was calibrated so that the sensory code and the

prefrontal code *sum* past the threshold. While the prefrontal code is recomputed from its input every tick, that is a trigger; once the loops make the code persist it is a standing drive at the action's own threshold and the gait is no longer free-running. Scaling those 5,175 synapses by 0.75 and by 0.5 repairs the seed that toppled under the innate repertoire, costs the walk on three of five seeds and leaves a fourth toppling at the same tick as before, so the repair is not a coefficient: a rule that triggers an action has to be allowed to decay while the code that emits it does not. A homeostatic weight decay -- the standard fix for unbounded connection growth -- was the obvious candidate and it does not help here; it hurts. A per-tick multiplicative decay on every plastic excitatory synapse is a knob in `皮层连接_cortex_links.py` (全局衰减率, off by default, so that no other number in this paper moves). With it on at 0.004, the seed that survives the test without it topples at tick 101; at 0.0005, an eighth of that dose, it topples at the same tick 101 and the second seed topples *earlier* than it does without the decay (tick 45 against 75). The reason is in the design rather than in the dose: only 30% of the innate synapses are frozen (固化比例), so a flat decay eats the 70% that carry the calibrated current the instinct table exists to deliver -- and that calibration, not an unbounded weight, is what this failure is made of. A homeostatic term is therefore not a bolt-on: the table would have to be re-tuned to the equilibrium of growth and decay, exactly as R11 reports for a wider mosaic. (Logs 日志_全局衰减000.log, 日志_全局衰减0005.log, 日志_全局衰减004.log; script 试_全局衰减_修R9.py.) The rebuilt engine of Section 5.12 answers the same worry with a term of a different shape rather than with a flat decay, and R20 measures it. There every edge carries the weight it was born with, and the tether is a spring back to that weight: proportional to how far the edge has already moved, so it is strongest on the edges that have changed least, and removed altogether on the one table the animal is meant to keep. A flat decay eats the calibrated edges that carry the instinct table's current; a spring to the birth weight leaves those edges where the table put them. That is why the rebuild does not meet the failure this paragraph reports, and the reason is the shape of the rule rather than a better constant. The other half of the same failure is R9's prefrontal accumulator, in which every tick's code is added to the last with nothing subtracting it; that one is answerable with a single global inhibitory current (R9's knob), and it is the one place in this paper where a structure we added is clearly not finished. The two ways a recurrent state can fail -- it dies, or it settles into a cycle it cannot leave -- are both visible in the character build of 4.4, on the same rule and at a much larger scale, which is why we treat them as properties of the rule rather than accidents of this build.

2. *Development over lived time.* R9 acquires one link in one session, with us deciding when the reward is on. The header claim needs a repertoire that grows over lived time because of what the brain has already done, with no experimenter in the loop. Nothing in the framing says this is different in kind from R9, and R3 says the repertoire is additive; what is missing is a drive that is not satisfied by the first action it releases (item 4) together with the stable rule of item 1. R10 removes one part of the experimenter from the loop -- the reward arrives as an afferent rather than as a flag -- but the experimenter still decides when the contact happens.
3. *More sensory channels and modality-appropriate front ends.* Vision here is 24x16 macro-pixels with a thermometer code; audition is a spectrum. The architecture does not care what an input cell represents — a pixel level is an arbitrary choice of the designer — but a general system would need many more channels and would need the hierarchy to be deeper. The measurements of R15 and R16 say what the choice of coordinates and of sampling costs. Coordinates decide what an input volume can become: a solid sheet tells two objects apart by depth where a flat one cannot (R15), and a front end that samples an unwarped image at a density that falls towards the periphery keeps the distance reading that the earlier, centre- magnifying sampling had been buying by distortion (R16). Both are the kind of engineering that a general system needs to get right rather than to inherit by accident.
4. *A body with more than one thing to do.* The Go2 has 16 muscles and the repertoire is 221 actions. Behaviour of the kind that is interesting — tool use, social interaction — needs a body with many more degrees of freedom and with hands. This is engineering, not theory, but it is not small.
5. *Drives.* Nothing here wants anything. In our framing a drive is not an objective function but one more innate circuit — a salient object lights a population that lights locomotion. The system already does exactly this for a red region; what it does not have is a drive that is *not* satisfied by the first action it releases.
6. *Scale.* R11 measures the cost of a wider cortex on one machine: 925,236 neurons and 62.4 million excitatory synapses cost 68 s to build, 16 ms per tick with plasticity on in the step benchmark and 2,555 MB resident with a 7,200 MB

peak, which is the ceiling of a 16 GB host. At 24x32, the one wider mosaic on which behaviour was run, the eye becomes measurably finer (gaze error 5.5 to 4.3 degrees) while the innate strengths, calibrated to the 24x16 activity level, lose margin and the boundary of recognition narrows. That is four orders of magnitude below a human cortex, and it is a cost curve rather than a scaling law: nothing above 24x32 was run. Nothing here is a scaling claim of the kind made for language models (Kaplan et al., 2020).

The rebuilt engine changes the shape of that curve rather than its direction. Cells are cheap and connections are not, and a tick costs what is active in it: with the sparse engine, eight times the cells cost 1.06 times the time at fixed drive, and 0 cells alight cost 0.04 ms where 11,818 cost 13.71 ms (R15). What that buys is the honest version of the scaling claim: the first wall we meet going bigger is two build rules of our own (R13), the second is that the sensory sheet is not yet sparse in the shipped configuration (90.1 per cent of the cells are retina, 28.17 per cent of the non-retinal cells are above 0.5 activity), and the third is that a bigger sheet has to *resolve* the same input with fewer cells, which needs a task that improves with resolution and is not in this paper. "The model is too small" is measurable rather than an opinion: with ten motor cells per channel the force step is 0.10 and 0.50, 0.51, 0.52 and 0.54 are the same pattern, while a hundred cells give a step of 0.01 (R15).

7. *A posture that is held, not a posture that is assumed.* The instinct table carries the body from prone to standing, and it has balance reflexes that push against a tilt once one has developed, but it contains no route that engages the standing pattern when the body is already upright and nothing else is happening. Placed directly into the standing pose, the same brain, the same seed and the same wiring emit no muscle current for the first ~30 ticks (0.6 s), tip past the point the reflexes can recover, and end on their side; having stood up by themselves, they hold 0.260 m for six seconds without moving a step. We tried the two obvious repairs -- driving the standing motor pattern, and driving the standing memory entry, from the trunk-height population that is already lit at 0.261 m. The first leaves the behaviour unchanged; the second disrupts walking (median displacement over 4 s falls from 2.13 m to 0.70 m). The posture that survives a stand-up is therefore held through the action sequence rather than by a posture rule. We record the negative result because it runs against the intuition one has while writing the table: a posture here is not a state the body is put into, it is something the cortex has to keep doing, and it is not yet a single missing line.
8. *Somewhere for the lesson to land, and a teacher who is cheaper than the answer.* R12 is the first complete instance in this project of a recipe rather than a mechanism, and it is short enough to state: name two populations as worth connecting, lay the whole route down empty, and let one cell decide when a moment counts. The teacher has to be present, has to touch the animal, and has to know that those two populations ought to talk to each other -- it does not know which cell, how strongly, in which direction, or whether what it wrote was right, and it never sees the inside of the brain. What R12 also shows is the price of the honesty that the *ignored* arm reports: a weight whose two ends are never together decays to its floor, and a weight that is written stops decaying, so the total weight of the sheet is bounded by experience rather than by a global decay term -- which is the repair this project tried and abandoned in item 1. What is missing is stated in R12: this writes one reflex between one sense and one action, and the same machinery would have to be shown to write a *structure* before the recipe can be called a training method. That is the experiment we would run next, and it is the one the position papers cited in 4.4 say does not exist yet.
9. *Combining what several lives wrote, which is now measured rather than hoped for.* R18 shows that two models whose cells and edges are the same and whose lessons differ can be added: the average of the two weight tables answers both cues, on the small graph and on the body, and it answers the cue with the motor cell that cue's own parent had taught it. Two things follow for a programme of this kind. The first is that a model need not be trained from scratch for every new capability; a lesson written in one life can be moved into another as numbers, which is the cheapest form of development this project has and the only one that needs no teacher at all. The second is what R18 does *not* carry: merging preserves and never creates, so a search over candidates can only find combinations of what the writing already put there, and the capability itself still has to arrive through item 8 or through R3. Three things are then needed before the idea becomes a method rather than an observation. A question set: every model has to be scored on *all* the behaviours, not on the one it was built for, or a model that happens to be good at something nobody asked about is thrown away. A way to write many candidates cheaply: a candidate is a weight vector over a fixed edge list, so it can

be stored as a seed plus the lesson, and the cost of scoring is simulation rather than memory. And a limit we have not got past: models whose *cells and edges differ* cannot be merged this way at all, so a search of this kind needs the graph fixed while the numbers move, which is the opposite of the developmental picture in item 2 where the sheet grows. We have the merge and we do not yet have the search.

What we would not claim. We do not claim that this architecture will reach general intelligence, and we would distrust any paper that claimed it on this evidence. What we do claim is that the *first* step — an ordered, attributable, extensible behavioural repertoire with no objective function anywhere in the system — is not the step that is blocking, and that current agents have been paying for it with an optimisation loop they may not need for that step.

5. Methods

5.1 Cortex

The cortex is a flat boolean array of 148,032 cells (皮层连接_cortex_links.py). One tick is 20 ms. The unit is deliberately cruder than a conductance-based or spiking model (Izhikevich, 2003) and the tick is a coarse stand-in for cortical dynamics (Buzsaki & Draguhn, 2004); both simplifications are choices, and R8 is the experiment that asks how much they cost. The update is a single vectorised operation over the whole array:

incoming excitatory current is the sum of the weights of all synapses whose source fired on the previous tick, restricted to that cell's incoming synapses; the cell fires if this current is at least the firing threshold (1.0) and it is not being held down by an inhibitory cell.

Both the excitatory and the inhibitory part are pre-computed as edge lists (源, 目标, 权重, 固化) so that a tick is a pair of np.bincount scatter-adds. Suppression is implemented as a separate population: an inhibitory cell receives excitatory synapses from its source and projects to its targets with weight 1.5; a cell that is the target of an active inhibitory cell is held sub-threshold. There are 8,000 inhibitory cells and the instinct expansion borrows them: a rule that must suppress something carries a negative number in the *text file's* strength column, which is a notation in the table only and never a weight, and is expanded into source -> inhibitory cell -> target. No synapse in the network is negative.

Background connectivity is sparse and random: 100 outgoing excitatory synapses per cell, 50% of them to other regions, born with weights uniform in [0, 0.15], 30% of them frozen; 100 outgoing inhibitory synapses per cell, half of them within a 24-cell radius. Three regions receive no random excitation (motor, motor memory, visual motion) and one region emits none (visual detail), so that adding the detail region could not perturb the global excitation/inhibition balance. The background is generated from a seed; different seeds are different brains.

The prefrontal population is not wired by the instinct table. It sits behind its own three-layer feed-forward network (前额叶神经网络 in 前额叶区_prefrontal.py): three layers of 57,436 units, 21 fixed local contacts per unit, weights uniform in [0, 0.1], hidden-layer threshold 0.55 and output-layer threshold 1.2, so that a wide, sparse input becomes a narrow, sparse output. Only the output layer is part of the cortical array; the two hidden layers are the compression itself. The network is fed by seven regions -- visual, auditory, proprioception, vestibular, body state, motor memory and motor -- each of which is first blurred by its own half-width (10, 10, 5, 5, 5, 10, 1 cells) so that a sparse code can drive a fixed-threshold layer at all. That table (输入路) is the single source of truth: the tick and the return line both read it, so a region cannot be added to one and forgotten in the other.

The return line (天生互惠返回线) is the same network used in reverse -- the same neighbour tables and the same fixed weights, no plasticity, no learning -- with two differences. The return is thresholded against *each region's own peak* at 40%, not against a single peak taken over the whole returned vector, because with one global threshold the large regions set the scale and the vestibular region (120 cells) can never be reached. And the return into motor memory is intersected with the cells that motor memory is already running: a returned cell there starts a whole action rather than shading one, so this route sustains and does not ignite (2.1). The three body-sense regions carry no return line at all: they are read by the prefrontal population and never written by it (2.1).

5.2 Regions and calibration

Names are not hand-written. Calibration (本能工具_instincts.py: 标定) lights a physical condition — an image, a sound, a body posture — runs the relevant front end, and records which output cells respond *relative to a reference condition*. For example a visual direction name is defined by rendering a red ball in that direction minus rendering an empty scene. The eyes were calibrated with a ball of radius 0.30 m; the per-column names used by the gaze reflex were re-calibrated with a ball of radius 0.05 m, because a ball that fills a whole cell is visible to every layer and would not distinguish them.

Region widths: visual 23,040; visual detail 23,040; auditory 30,000; prefrontal 57,436; motor 160 (16 muscles x 10 cells); motor memory 3,576; proprioception 480; vestibular 120; body state 60; visual motion 1,920; inhibitory 8,000.

5.3 Instinct table

本能表.txt is a text file, one rule per line: source_region:name -> target_region:name strength frozen. A second, derived file (反射表.txt) contributes 4 further rules computed from the same trained locomotion policy: in a given tilt condition, hold a given shank above a force floor. The table currently holds 585 rules and expands to 309,148 excitatory synapses plus the inhibitory edges.

A rule whose source is a *motor memory* name (rather than a sensory name) is expanded differently: the action's time cells are chained to each other, each time cell is wired to the muscles that should be active at that instant, each time cell suppresses itself on the next tick so that exactly one instant is active at a time, and the intermediate time cells suppress the first one so that a still-present sensory drive cannot restart the action mid-execution. The effect is that lighting a single cell commits the system to a whole action, and the action then runs itself out of the cortex's own dynamics. 221 of the 585 rules are of this kind: 10 static postures, 4 slow actions, 207 locomotion gaits.

5.4 Where the motor repertoire came from

The locomotion gaits were copied from a reinforcement-learning policy trained to make the same Go2 model walk (取动作_从训练好的模型.py, 建动作库.py), following the approach of Hwangbo et al. (2019) and Rudin et al. (2022). We swept forward/lateral/turning velocity commands on a grid, recorded the resulting joint trajectories, and stored each as one action of 1-2 s. This is the only place in the system where a learned artefact is used, and it is used *before* the brain is created: the trajectories become fixed synapses in the instinct table and are never updated online. A reader who objects that this is supervision is right; our claim is about what happens *after* birth, and we would rather state the provenance than hide it. The source is the public checkpoint diasAiMaster/unitree-go2-velocity-flat (a 4.5 MB PPO actor, C:\go2_policy\vel\model_500.pt), and its behaviour on our body is measured separately by 对照_现成RL_驱动.py.

What the repertoire is, and is not, is worth two numbers. Our forward row is 15 time cells, i.e. a 0.30 s cycle at 20 ms per tick, and the motor chain advances exactly one cell per tick, both open-loop and under the whole brain (诊断_步态相位.py). Replayed from its head cell with no eye and no prefrontal it carries the body 2.58 m in 4 s; under the whole brain 2.0 m, against R1's five-seed median of 1.73 m with the rule running. The source policy covers 3.71 m in the same 4 s on the same body. The row is a faithful copy of the source's *trajectory* and not of its *feedback*: what the missing margin buys the source is a correction on every tick, which this brain applies to posture but not to a gait it is playing out.

5.5 Plasticity

Plasticity is part of the tick, not a mode of the system: in every run reported above the rule runs on every tick from birth, on the same synapse table the instinct table wrote (Section 2.3). The "learning off" condition appears only where a result needs a control — R4's hue boundary with the rule disabled, R9's dark-dopamine condition — and it disables the update; it is a knife for attribution, not a second way of running the brain. Its price at present is real and is stated in R11: in the closed loop at 24x16 a tick costs 58 ms with the rule disabled and 158 ms with it enabled, against the 20 ms the body allows, so a brain that learns on every tick currently runs slower than the body it drives. Most of that gap was one avoidable scan and is now gone -- the update used to touch all 27.6 million inhibitory synapses per tick and now touches only those incident on an active cell, a change that is bit-identical to the code it replaces (验证_学抑制_逐拍对照.py) and that takes the 24x192 step from 554 ms to 16 ms. Because the update touches the same synapses with the same numbers, it

also produces the same behaviour, and we checked that at the level of the results rather than only at the level of the rule: R1, R4 and R6 were re-run after the change and reproduce their published numbers to the digit -- 5/5 seeds standing and approaching (median 1.73 m), the hue sweep stopping at 96 degrees on seed 20260914 in a 1040-tick run with the same closing controls, the hue boundary table unchanged (0-96, 168 and 180 degrees lit on 4/5 seeds; 120-144 and 204-240 on 0/5; black on 0/5), and gaze error 5.6 and 4.8 degrees with the empty scene at 0.0 (logs 日志_行为回归_改后.log, 日志_回归_R4_不保留.log, 日志_回归_色相前测_不保留.log, 日志_回归_R6_改后.log). Making the whole live loop fit inside the tick remains an engineering target rather than an architectural question; until it is met, what the results above support is that the rule is sufficient, not that it is cheap.

皮层连接_cortex_links.py:

parameter	value
excitatory learning rate	0.005
decay per tick	0.995
weight of a newly created synapse	0.001
co-activations required before creating a synapse	4
maximum pairs examined per tick	4,000
maximum synapses created per tick	200
trace decay / threshold	0.9 / 0.05
inhibitory learning rate / cap	0.0005 / 0.6
disinhibition learning rate	0.02
plasticity multiplier for motor, motor memory, vestibular, proprioception, body state	0.02

The update on a synapse that is co-activated in the accepted order is $w \leftarrow w + \eta * m * (1 - w)$, where m is the per-region multiplier of the last row of the table; a synapse whose presynaptic cell was active but whose postsynaptic cell did not follow is decayed by $w \leftarrow w * (1 - (1 - d) * m)$ with d the per-tick decay. A new synapse is created at 0.001 once its pair has been seen in the accepted order four times, and then follows the same rule. In the configuration reported here there is no term that lowers a weight because the postsynaptic cell fired anyway, which is the failure discussed in 4.6. Frozen synapses (a flag in the instinct table) are never modified. The 0.02 multiplier is how "balance reflexes barely change in a lifetime" is expressed; it is a single number, not a special case in code.

R9 adds a second, faster path on the same principle. The dopamine region is a named block of 200 cells (本能工具_instincts.py); when any of it is active on a tick, 皮层连接_cortex_links._多巴胺强化 writes, on that tick, a synapse from every cell that was active on the previous tick to every cell that *became* active on this one. Per target cell at most eight sources are used, and the total drive delivered by one reward is fixed at 0.50 and divided equally among them (the same convention as the instinct table); the per-region factor of the slow path is applied to the fast path as well (这对学多快). Three experimental switches in 主循环_完整的一拍.py are used by R2 and R9 and are not part of the architecture: 前额叶按灭 forces the whole prefrontal population to zero; 前额叶不保留 replaces the prefrontal code every tick instead of adding to it, i.e. removes the loops described in R9; and the prefrontal self-inhibition of R9 is one more inhibitory current on that population, k times the number of prefrontal cells active on the tick, with k read from the environment (前额叶自压=0.001) and 0 by default (皮层连接_cortex_links.前额叶自压).

The two configurations of the same system. R1-R8 were run with 前额叶不保留, i.e. with the prefrontal code recomputed from its input on every tick and no persistence. R9, and the two persistence rows of the R2 table, are the only measurements in this paper that use the loops; the configuration is stated wherever the two differ. Either configuration can be selected for any script from the environment (前额叶不保留=1), so every protocol in this paper can be re-run both ways with the code in the repository.

The nursery: a route that is born empty, a gate, and a keeper. A backup route is written by backup_edge() in born_wired/reflex_controller.py for each pair of populations named in backup_routes, as the complete set of edges from the first population to the second. Every such edge is born at weight 0.0001, with its lower bound set to zero, its ceiling set to backup_cap (1.5), its plasticity set to backup_plasticity (1.0), and its tether removed; the tether is what

holds an instinct at the strength the table gave it, and removing it is what lets a lesson keep what it writes. Because the edge exists from birth, no edge has to be created later, and the rule that writes to it is the ordinary local rule, not a second code path. The gate is a single cell (dopamine, tau 0.05) whose rate is read by `learning_modulation()`, which returns `righting_modulation() * (1 - dopamine_gain * (1 - dopamine_rate))`; with `dopamine_gain` at zero this is exactly the modulation the animal had before the nursery existed, which is what makes the control bit-exact. `touch_dopamine_gain` is the strength of the only innate edge into that cell (`touch -> dopamine`), and it is zero unless a keeper is being simulated. The lesson protocol, the four arms and the read-outs are `tools/nursery.py`; its assertions are `tests/test_nursery.py`.

5.6 Body, sensors, eyes

The body is a Unitree Go2 in MuJoCo, driven by joint-angle servos (stiffness 80, damping 4, maximum torque 25 Nm), ten 2 ms physics sub-steps per cortical tick. The visual hierarchy receives 1920x1080 RGB, reduces it to 24x16 macro-pixels (three channels, 10 threshold pairs each, i.e. a thermometer code), and passes it through four parameter layers of radius 10 with weights uniform in a range of 0.1 around a threshold of 0.6; the exposed tensors are layers 1, 2 and the compressed output do. Visual field 100 x 75 degrees. The auditory hierarchy receives a 4096-bin spectrum through the same kind of network. The eyes have range 30 degrees, muscle acceleration 500 degrees/s² and damping 14 /s, so a single tick of maximum force moves the eye about 0.1 degrees. The touch afferent of R10 is a 210-cell region -- seven skin sites of 30 cells each, appended after the inhibitory pool -- with no random connections in or out and no internal recurrence, so it is entered only by contact and reaches the rest of the cortex only through its seven fixed rules into the dopamine region. It is absent from the configuration that produced every other number in this paper, and the code that adds it is switched on by an environment variable, so that the measurements above are made on the build they were made on.

The retina (R16). The image is sampled without being warped: the field is divided into as many shares as there are cells and the dividing boundaries are pulled towards the centre so that the middle of the field carries `centre_gain` times as many cells per degree as the edge (`centre_gain = 2.5` in every measurement here); each cell reports the mean of its own patch, the patches tile the whole field with no gap and no overlap, and no interpolation is performed. The render resolution is set to `ceil(centre_gain)` times the cell count, since finer rendering changes no cell's value. Thus the periphery is sampled coarsely rather than deformed, and the earlier unbounded magnification of the centre is gone.

The eyes (R16). Each eye is one rotational degree of freedom driven by opposing muscles. Three routes act on it: a disparity bank of seven cells that reads the agreement between the two images at a range of offsets, whose pattern is monotone in the distance of the target; a route driven by how much the picture differs from the picture one step earlier; and, through the cortex, the prefrontal and orienting populations. The stereo offset is declared as a number of retinal columns (`STEREO_OFFSET_MAX = 6`) rather than as an angle, which is the known cause of the shortfall of convergence beyond about a metre: as the retina is made finer, six columns stand for a smaller angle. The seven-cell bank's thresholds are a fixed list; no cell computes a distance in metres.

The ears (R17). Two raw waveforms, one per ear, are the only auditory input. `AuditoryNeurons` is 40 cells and 84 fixed synapses, with no learning: raw half-wave receptors, per-ear frequency coincidence cells that compare a band's activity with its own activity one period ago, band pools, left- and right-preferring binaural coincidence cells (a fixed three-sample delay on one ear against the other ear now), binaural energy cells and two orientation cells. Front and back are read by `pinna_rates`, four cells that compare the shape of the spectrum between the ears; the acoustic front end supplies that shape by tilting each band with the direction of the source (`_PINNA_TILT = 0.45`) and by removing high frequencies from the far ear (`_HEAD_SHADOW = 0.55`). No FFT, cross-correlation, peak search or source localisation is performed anywhere in the path.

Walls (R16). The arena's walls carry a striped texture (`models/wall_stripes.png`, 2048x16, bar widths drawn from the geometric series 3, 4, 6, 9, 14, 21, 32, 48 and 72 by seed 97, two grey levels 243 and 88). A plain-coloured wall gives the contrast cells no input at the distance at which the animal meets it, so the stripes are what makes the wall visible to the visual cells at all; the cost is that the whole picture changes at every step, which is reported in R16.

5.7 Protocols

Every experiment in this paper is a frozen protocol: the brain is built from a fixed seed, the condition is applied, and the run is recorded to a log file that is kept in the repository. Each result section names its script and its log. Multi-seed results use seeds 20260914-20260918. Where a result is a single run we say so. Results that contradicted our prediction (R4, R5) are reported with the prediction stated, and where a result is visible on one build of the system and not on another, both builds are reported (R4).

The scale measurements of R11 do not perturb any frozen protocol. The mosaic size is read from the environment (AGI行数 / AGI列数) by both the visual front end and the world, and the cost numbers are single runs on one host (16 GB, 16 logical cores) in which the network is built at the requested width and stepped, with and without plasticity, but no body and no behaviour are run. The touch-region configuration of R10 is likewise switched on by an environment variable (AGI触觉宽度, 210 in R10) rather than being part of the default build.

The two 20-second-of-simulation baselines used throughout are: *walking* is counted as present if the walking action's time cells fire on at least 40 of 200 ticks, and *gaze* is counted as tracking if the mean angular distance between the gaze centre and the object is at most 12 degrees. A reflex that is not wired at all gives 15.3 degrees on the gaze measure, so 12 degrees sits inside the gap between "wired" (5.6) and "unwired" (15.3).

5.8 Reproducing

The PDF of this manuscript is typeset from this Markdown file by `model/出一份PDF.py`, which converts it to HTML with `python-markdown` and prints it with headless Chrome, so the paper can be rebuilt on a machine that has neither pandoc nor a TeX installation (`python model/出一份PDF.py`).

```

# R1-R8 are run with the prefrontal recurrence switched off (Methods 5.5). In PowerShell:
$env:前额叶不保留 = '1' # bash: export 前额叶不保留=1
python 本能工具_instincts.py 标定 # rebuild the calibrated names (needed after changing a
front end)
python 生成本能表.py # regenerate the instinct table (585 rules)
python 实验_闭环前提_多种子.py # R1
python 实验_前额叶_多种子.py # R2, with and without persistence
python 实验_本能分阶段.py # R3
python 实验_渐变_正式协议.py 3,6,12 20260914,... # R4
python 实验_方位分辨力_按层.py # R5, representational part
python 实验_前几层看得见小东西吗.py # R5, localisation part
python 实验_追踪靠的是哪一层.py # R5, behavioural part
python 实验_眼睛跟随.py 小横着飘 # R6
python 实验_本能表整体偏移_正前方.py # R7
python 实验_模糊化_存活曲线.py # R7, destructive version
python 实验_想还在吗.py # R8, stimulus removed / action cleared
python 实验_切断回响.py # R8, recurrent step removed
python 实验_两半各亮多少.py # R2, the two halves measured separately
python 实验_发育_声音叫走路_多种子.py # R9, four stages (logs written into `logs/`)
python 实验_发育_无奖励对照.py # R9, dopamine left dark
python 实验_发育_无环对照.py # R9, without the loops
python 诊断_天生步态稳不稳.py # R9, innate gait with and without the loops
python 诊断_摔是谁干的2.py 20260915 20260918 # R9, where the topple comes from
python 诊断_摔是谁干的2.py --无环 20260915 20260918 # ... and the same without the loops
python 诊断_前额叶打走路链.py 20260915 # R9, where the topple comes from
python 诊断_冻死进运动.py 20260915 # R9, the motor weights do not change
python 诊断_返回线截断.py 20260915 # R9, the return line is not the cause
python 试_打折R9_参数.py 0.75 # R9, the one-number repair, five seeds
python 实验_发育_仿生奖励_三臂.py 一直摸 20260914 20260915 20260916 # R10, reward by contact
python 实验_发育_仿生奖励_声音叫走路.py 20260914,20260915,20260916 # R10, contact only while
walking
python 诊断_触觉版_天生站起来.py 20260914 20260915 20260916 # R10, innate righting in the
touch build
python 规模试验_测量.py # R11, cost (set AGI列数 for
other widths)
python 验证_感觉网分块.py # R11, the memory fix is bit-
identical
python 实验_放大_走路边界.py # R11, behaviour at 24x32
python 实验_放大_色相红块.py # R11, the two halves at 24x32
python 诊断_走路点火_多种子.py # R11, the current onto the
walk head cell
python 对照_现成RL_驱动.py # 4.4, the published RL checkpoint driven on our body
python 诊断_步态相位.py # 5.4, the gait cycle is played at the recorded rhythm
python 验证_学抑制_逐拍对照.py # 5.5, the fast inhibition update is bit-identical
python 诊断_整拍耗时.py # R11, what a live tick costs (rule on / off)
python 对照_学抑制_改前改后.py # R11, the same live tick before the fix
python 量规模_七档.py # R11, the cost table (seven widths, one log each)

```

R10's touch region is off unless AGI触觉宽度 is set, and R11's widths are selected with AGI列数; both are environment variables rather than edits, so that every number above stays attached to the configuration that produced it. The 24x32 build of R11 is kept apart from the 24x16 one with AGI数据后缀=_列32, which names its calibration file and its instinct table; without it the wider brain would overwrite the tables that R1-R10 rest on.

对照_现成RL_驱动.py needs the third-party checkpoint at C:\go2_policy\vel\model_500.pt (4.5 MB, downloaded from Hugging Face); it trains nothing and takes seconds to run.

R1-R8 are run with 前额叶不保留=1, which recomputes the prefrontal code from its input on every tick and so removes the loops of R9; R9 and R10 are run without it. It is a live switch and not a build option, and it changes the numbers of the vision experiments (R4, R5, R6): run without it, a script reproduces the persistent configuration, which is what the third row of R2's table and panels b and c of Figure 9 report. The two R8 scripts were run both ways, and they take the name of their output from 轨迹名 so that the two runs do not overwrite each other.

5.9 Data and code availability

The repository addressed by this manuscript is <https://github.com/leeR1ven/born-wired-cortex> . It contains: the cortex simulator (皮层连接_cortex_links.py), the instinct-table generator (生成本能表.py) and the calibration tool it depends on (本能工具_instincts.py), the full tick (主循环_完整的一拍.py), the MuJoCo body and world (身体_go2.py, 世界_world.py), one script per result section named in that section, the log file that each script wrote, and the plotting script (出一张图_论文_20260915.py) that turns those logs into the figures. Every number in this paper can be regenerated by running the named script; no number is quoted from a run that is not in the repository. The locomotion trajectories were distilled from a policy trained with a public quadruped-learning implementation; the distillation script (取动作_从训练好的模型.py) and the resulting action library (建动作库.py) are included so that the provenance of the repertoire can be checked. The third-party checkpoint itself is not redistributed: it is diasAiMaster/unitree-go2-velocity-flat on Hugging Face, and 对照_现成RL_驱动.py is the script that drives it on our body. The repository also carries a four-panel playback page (回放_大脑_四幕.html, built by 出一份四幕页面.py) which renders the four behaviours of Figure 2 and R8 as looping animations, one panel per sensory condition, with each panel recorded from a freshly built brain of the same seed, so a reader can watch the behaviour without running the simulator.

Two trees, because there are two implementations and no result mixes them. R1-R11 are produced by the reference model in model/ with the logs in logs/; R12 and R13 are produced by the rebuilt engine in engine_v2/ with the runs cited in those sections under engine_v2/artifacts/. A reader who wants to check a single claim needs one tree and the log named next to the claim.

5.10 Use of AI assistance

The code, the experimental protocols and this manuscript were produced in an interactive session with the DeepSeek large language model acting as the programmer, at the direction of the human author, who does not write code. The human author specified the architecture, set the research questions, rejected results and interpretations he judged wrong, and required the reporting of contradicted predictions. Every number in this paper is produced by a script that is in the repository, with the log of the run that produced it; the model's output was not accepted without a run. Readers should treat the *codebase* as reviewed by the human author's behaviour tests rather than by line-by-line reading, and should reproduce independently.

5.11 Statements

Author contributions. L.Z. conceived the architecture, wrote the claims of Section 1.1, authored the instinct table, directed every experiment, rejected results and interpretations he judged wrong, and required contradicted predictions to be reported rather than dropped. The code was written in an interactive session with a large language model (Section 5.10).

Competing interests. The author declares no competing interests.

Funding. This work received no external funding.

Acknowledgements. The locomotion repertoire was distilled from a reinforcement-learning policy trained with a public quadruped-learning implementation; the citation is in the reference list. The code, the experimental protocols and the manuscript were written in an interactive session with the DeepSeek large language model, at the direction of the human author (Section 5.10).

5.12 The rebuilt engine

R12 to R17 and R19 use a second implementation of the architecture, and no result before R12 does. It was written because the reference engine's tick (R11: 158 ms with plasticity enabled at 24x16) could not be run in a closed loop with a body, and because the quantity that decides a tick turned out to be connections rather than cells. The invariants are the ones of Section 1.1: one population of cells updated on one tick; no negative weights, suppression always by an inhibitory cell; no separate inference pass and no read-out layer; every input a raw sense (two RGB images, a sound pressure waveform, body state) with no distance, direction or object computed anywhere outside the cortex; and learning by a local correlation rule that cannot see a target.

parameter	value
cells, shipped configuration	750,075
synapses, shipped configuration	1,978,219
eye	192x144 pixels per eye, three channels, one cell per pixel and channel plus the companion cell of R19
motor / proprioceptive / association units	1,600 / 512 / 2,048
tick	10 ms of body time; the synaptic update runs every 50 ms of accumulated time
engine	graphics card (CUDA), with an automatically selected host fallback
whole step	13.29 ms (75.2 steps/s), brain 9.24 ms, eyes 2.86 ms, physics 0.82 ms

The correspondence with the reference build is at the level of the architecture and the rule, not of the cell count, and the two builds are therefore never mixed inside one result: every number in R1-R11 comes from the build of Methods 5.1, and every number in R12 to R17 and R19 from this one. Where the rebuild reproduces a published result it reproduces it in kind rather than to the digit (the gaze reflex, the righting sequence, the red-approach walk).

The rebuild's own test suite is run with `python -m unittest discover -s tests` (199 tests, 130 s, `artifacts/tests_after_opposite.log`), and its fingerprint tool is `tools/check_model_fingerprint.py`, which is the instrument every change was required to pass. The scripts behind R12 and R13 are `tools/nursery.py`, `tools/probe_scale_walls.py`, `tools/probe_model_scale.py`, `tools/profile_live_step.py`, `tools/check_model_fingerprint.py` and `tools/measure_loom.py`; their logs are named in those sections and are in `engine_v2/artifacts/`. The source tree of the rebuild is published beside the reference model as `engine_v2/`, unchanged except that its own artifacts directory keeps only the runs cited above.

5.13 Measurements added on 27 and 28 September

R14 to R17, R19 and the scale numbers of R15 use the second implementation of Section 5.12, whose development line is published beside it as `engine_v2/`. Every number is produced by a script whose name is given next to it and every run is kept as a log in `engine_v2/artifacts/`: the sparse-cost and sparse-scaling probes, the engine-comparison tool, the spatial-dimension probe, the eye convergence sweep, the gaze-route measurement, the sound-turning and sound-walking measurements and the wall-distance probe. Two numbers of R15 come from the reference build rather than from the rebuilt engine -- the motor resolution and the prefrontal overlap -- and are computed from that build's own source and its saved cell names (`artifacts/motor_resolution_probe.json`, `artifacts/old_representation_probe.json`).

The plotting scripts for Supplementary Figures 5 and 6 are `出一张图_补充56.py`, which reads the same `.log` and `.json` files as the text. The rebuilt engine's test suite was run again after the last change of 28 September, and reports 199 tests, all passing (`artifacts/tests_after_opposite.log`).

5.14 The merge measurements of 28 September

Both merge experiments use one definition of a model: the cells, the edge list, the structural bounds and the injection of external drive are fixed, and a model is a weight vector over that edge list. Merging is arithmetic on that vector. No model is retrained after a merge, and no gradient, reward, critic or objective appears anywhere in this section; neither does a candidate connection table, since the edges that may be written already exist in both parents at a strength too small to do anything (R12).

tools/fuse_parents.py builds the 17-cell graph of born_wired/distributed.py with hidden_routes=True and trains one pairing per parent: 30 pairings of 0.4 s of a four-channel stimulus with 1.2 of external drive on one motor cell, each followed by 0.2 s of rest with learning on. Reading is 0.6 s per cue with the teacher absent and learning off, averaged over the last 0.15 s, and both the raw motor table and the margin in favour of whatever that arm is meant to answer are reported. The arms are: two untrained parents, each parent alone, the arithmetic mean of the two weight vectors, the per-edge maximum of the two moves from birth, birth plus both changes, a parent averaged with itself, the same two lessons with the motor cells exchanged, and two parents taught opposite answers to one cue.

tools/fuse_robot_parents.py uses the controller of R14 with the empty route extended to two innate actions, so that a lesson can be written beside the gait cell as well as beside the flexion cell (empty_routes=("flexion", "locomotion")). A lesson is six pairings of 2 s of one cue with 0.6 of external drive on the taught action cell and 0.5 s of rest. Every arm restores the same starting pose, the same joint noise and the same cell state before it is read, so the only difference between arms is the weight vector; restoring the pose is not a weight edit and the pose is the protocol's, not the brain's. The check on the operation itself is the arm that installs a parent's own weights into a fresh brain, which reproduces that parent's behaviour to three decimals. Both scripts are published with their logs (artifacts/fusion_parents.json, artifacts/fusion_parents.log, artifacts/fusion_robot_parents.json, artifacts/fusion_robot_parents.log) and Figure 11 is drawn from those two files.

5.15 The opposite cell on every sense channel

Each sense is two sheets of cells. For the reflex senses the pair is made in one place: born_wired/reflex_controller.py creates body_touch_opposite, foot_obstacle_opposite, foot_load_opposite and foot_slip_opposite (four cells each), cochlea_opposite (six), auditory_spatial_opposite (six) and auditory_pinna_opposite (four), and in every step the driven sheet receives the sensory vector while its companion receives one minus that vector, taken from the same array. For vision, born_wired/embodied.py creates one cell per pixel per colour channel beside each photoreceptor, driven by one minus the pixel, and that sheet converges on retina_opposite with the same gain and the same three-sector rule the lit sheet uses to reach retina.

The lit and the unlit half are kept on separate cells on purpose. A cell that summed both halves of a pair with the same sign would lose the brightness altogether, and retina is this animal's colour reading -- red drives the aversive cells, green the appetitive ones, blue the curiosity cells -- where "not red" is not red. The unlit halves therefore have sheets of their own, and at present no innate valence and no reader: no edge leaves any opposite sheet toward a reflex, an association or a motor cell, and the only consumers are the four exam questions named in R19 and the live window's own readout (retinal_opposite_activity).

This is structure and not a parameter: a companion sheet adds no gain, no threshold and no rule of its own beyond the injection of one minus the same number, and it is born with the animal like every other innate cell.

5.16 What the local rule does to a weight, and the three questions that read it

RegulatedSynapses (born_wired/regulation.py) and the same expression on the graphics card (born_wired/torch_execution.py, SynapseTensors.update) move every movable edge by

```
change = learning_rate * modulator * (post - target_activity) * source_rate - tether * (w - anchor)
```

and one write is $w \leftarrow w + \text{plasticity} * dt * \text{change}$, projected onto the edge's own bounds and onto the incoming budget of its target cell on every write. The first term is the correlation: an edge is raised when its source fires and its target sits above target_activity (0.15) and lowered when its source fires and its target sits below it. The second term is the tether: a spring back to anchor, the weight the edge was born with. There is no decay term -- decay is zero in the constructor -- and no term anywhere lowers a weight because its target fired anyway, which is the failure discussed in Section 4.6. The ceiling, the structural floor and the per-cell budget are all hard, because the projection runs after every proposal and not at the end of a run.

The shipped configuration gives almost every edge `tether = 0.2` and `learning_rate = 0.12` (`born_wired/innate.py`); the teachable route of Section 5.5 is the exception at `tether = 0`, which is what lets a lesson keep what it writes, and the posture edges that R12 marks as learned are the exception in the other direction. The rule's own answer to "does this stay put" is therefore: an edge that is not used relaxes towards the weight it was born with and stops there, and an edge that is used climbs to its own ceiling and stops there. Both ends are what R20 measures.

The three questions of R20 are in `tools/taskbank.py`. `the_memory_survives_a_long_life` teaches for 20 s, then lets the animal walk for 60 s with the rule running and the tone coming and going in 6 s blocks, then asks about the tone alone. `a_second_tone_does_not_wipe_the_first` teaches 262 Hz, then teaches 880 Hz down the same route, then asks about both tones separately. `the_route_stops_growing_at_its_ceiling` holds the keeper's hand on the body for three equal 20 s stretches with no pause and reads the route's own weights before the hold and at the end of each stretch. All three need the same animal across phases, which `nursery.Run` does not provide (it builds a fresh animal for each phase), so they use a `_Life` helper in the same file: one body, one brain, one ear per tone. Nothing in any of them writes a weight, and the only thing that drives the dopamine cell is the keeper's hand, through the innate touch route. Their readings are `artifacts/weight_drift_60s_seed0.log`, `artifacts/weight_drift_300s_seed0.log`, `artifacts/weight_drift_300s_seed1.log`, `artifacts/forget_600_quiet.log`, `artifacts/forget_300_life.log`, `artifacts/forget_300_second.log` and `artifacts/exam_new_tasks_3seeds.json`; the two drift tools are `tools/measure_weight_drift.py` and `tools/measure_forgetting.py`, and both are run against the same nursery configuration the exam bank uses, which carries the six route edges on top of the 1,978,219 of Section 5.12. The suite was run again after the three questions were added and reports 199 tests, all passing (`artifacts/tests_after_memory.log`).

Figures

Each figure is generated from the logs cited in the corresponding result section; the plotting scripts are `出一张图_论文_20260915.py` (Figures 1-9), `出一张图_发育.py` (Figure 10), `出一张图_融合.py` (Figure 11) and `出一张图_补充56.py` (Supplementary Figures 5 and 6).

- **Figure 1 — One tick of the system.** A schematic of the whole tick: senses drive the two sensory hierarchies, which drive one recurrent cortical sheet; the prefrontal population compresses the already-compressed signals a second time and returns its output to the sensory and motor regions through the return line; the motor-memory time cells drive the 160 motor cells, which are the muscles. Generated by `出一张图_系统总览.py`.
- **Figure 2 — The innate wiring closes the loop.** Four panels, five network seeds each: (a) final trunk height for a body placed prone; (b) forward displacement for a body shown a red region; (c) final trunk height for the no-sensation control, which collapses; (d) the same five brains with the instinct table cleared — final trunk height and approach displacement, both at or below the no-sensation control. Sources: `日志_闭环多种子_新.log`, `日志_白脑_新.log`.
- **Figure 3 — Causal intervention on the prefrontal population.** Bar chart of walking ticks out of 200 for the five conditions of R2 (five seeds each), with the mean number of active prefrontal cells annotated. Sources: `日志_前额叶多种子_新A.log`, `日志_前额叶多种子_新B.log`.
- **Figure 4 — The repertoire grows stage by stage.** A five-row matrix (stages A-E) x five behaviours, cells shaded by how many seeds passed. Source: `日志_分阶段_新.log`.
- **Figure 5 — Gradual recognition.** (a) how far the colour may drift before recognition stops, as a function of step size, with plasticity on and off, one point per seed; (b) the same five brains, hue by hue, whether a hue starts the walk before any drift and after the 3-degree sweep has run to its stopping point. Sources: `日志_渐变正式_新.log`, `日志_渐变边界_无环.log`.
- **Figure 6 — The cost of compression.** (a) Cross-column sharing of direction names in layers 1, 2 and 3; (b) for a ball one third of a cell across, the columns in which each layer's newly active cells lie. Sources: `日志_方位分辨力.log`, `日志_前几层.log`.
- **Figure 7 — Emergent gaze tracking.** Eye azimuth and object azimuth over time for (a) an object crossing the field, (b) an object appearing suddenly, (c) an empty scene. Source: `日志_眼睛跟随.log` (regenerate with `实验_眼睛跟随.py`

小横着飘).

- **Figure 8 — Blurred innate wiring.** (a) Resting position of the object relative to the gaze centre as a function of the imposed map shift (R7); (b) survival of three behaviours as a function of the fraction of the instinct table that has been re-randomised or shifted by one cell. Sources: 日志_整体偏移_正前方_新.log, 日志_存活曲线_新.log.
- **Figure 9 — Where the behaviour lives.** (a) three behaviours with the intact cortex and with the recurrent step removed (dots = individual seeds); (b) walking, the visual code and the prefrontal code tick by tick, with the stimulus removed at 0.5 s; (c) the same run with the action's time cells cleared once at 1.0 s. Sources: 日志_切断回响_有环.log, 日志_想还在吗_轨迹_有环.csv (panels b and c are the persistent configuration of R8, in which the prefrontal code for red is still lit after the red is gone).
- **Figure 10 — A capability acquired during life.** (a) Walking-action ticks in each of the four stages of R9, five seeds: none before teaching, nearly all after. (b) Connections written by stage 3 with the dopamine region lit and with it dark. (c) Trunk height reached in stage 1 and stage 4, and forward displacement, for each seed. Sources: 日志_发育多种子_A.log, 日志_发育多种子_B.log, 日志_无奖励对照.log. Generated by 出一张图_发育.py.
- **Figure 11 — Two lessons, one brain.** (a) five seeds of the 17-cell graph: for each arm, the margin by which the motor cell each cue is meant to drive beats the other one, for cue A and for cue B, in favour of that cue's own answer. Ten arms: nothing taught, each parent alone, the average, the per-edge maximum, the sum of the changes, a parent with itself, two untrained parents, the same lessons with the motor cells exchanged, and two parents taught opposite answers to one cue. (b) and (c) the same body, five seeds, on the 1,813-cell controller: metres walked in 8 s on cue A, and flexion cell activity on cue B, for six arms. Sources: artifacts/fusion_parents.json, artifacts/fusion_robot_parents.json. Generated by 出一张图_融合.py.
- **Supplementary Figure 5 — Flat coordinates and solid coordinates.** Overlap (Jaccard) between the sets of cells that end up active for three comparisons — two unrelated objects, one object nudged, one object at a different depth — in three arms of 1,000 cells that differ only in the coordinates of the sheet (random wiring, flat 2D coordinates, solid 3D coordinates). Depth is separated by the solid sheet alone. Source: artifacts/probe_spatial_dimension.log. Generated by 出一张图_补充56.py.
- **Supplementary Figure 6 — What a tick costs, and what it costs when the sheet grows.** (a) The original engine at 100,000 cells: cost per step against the number of cells alight, which rises from 0 to 86,064 while the cost stays between 5.17 and 5.71 ms; (b) the rebuilt engine at 100,000 cells: the same axes, from 0.04 ms at 0 cells to 13.71 ms at 11,818 cells; (c) the rebuilt engine at 100,000, 200,000, 400,000 and 800,000 cells with the external drive held at 1,000 cells, which is 1.28 to 1.35 ms across an eight-fold change in size. Sources: artifacts/probe_sparse_cost.log, artifacts/check_sparse_engine.log, artifacts/probe_sparse_scaling.log. Generated by 出一张图_补充56.py.

References

- Amit, D. J. (1989). *Modeling Brain Function: The World of Attractor Neural Networks*. Cambridge University Press.
- Brown, T. B., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33.
- Buzsaki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical networks. *Science* 304, 1926-1929.
- Eichenbaum, H. (2014). Time cells in the hippocampus: a new dimension for mapping memories. *Nature Reviews Neuroscience* 15, 732-744.
- Eliasmith, C., et al. (2012). A large-scale model of the functioning brain. *Science* 338, 1202-1205.
- Felleman, D. J., & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex* 1, 1-47.
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience* 11, 127-138.
- Fuster, J. M. (2001). The prefrontal cortex - an update: time is of the essence. *Neuron* 30, 319-333.
- Gewaltig, M.-O., & Diesmann, M. (2007). NEST (NEural Simulation Tool). *Scholarpedia* 2, 1430.

- Grillner, S., & Wallen, P. (1985). Central pattern generators for locomotion, with special reference to vertebrates. *Annual Review of Neuroscience* 8, 233-261.
- Grutzendler, J., Kasthuri, N., & Gan, W.-B. (2002). Long-term dendritic spine stability in the adult cortex. *Nature* 420, 812-816.
- Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley.
- Holtmaat, A., & Svoboda, K. (2009). Experience-dependent structural synaptic plasticity in the mammalian brain. *Nature Reviews Neuroscience* 10, 647-658.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *PNAS* 79, 2554-2558.
- Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *The Journal of Physiology* 160, 106-154.
- Hwangbo, J., et al. (2019). Learning agile and dynamic motor skills for legged robots. *Science Robotics* 4, eaau5872.
- Izhikevich, E. M. (2003). Simple model of spiking neurons. *IEEE Transactions on Neural Networks* 14, 1569-1572.
- Kaplan, J., et al. (2020). Scaling laws for neural language models. *arXiv:2001.08361*.
- Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience* 21, 335-346.
- Maass, W., Natschlager, T., & Markram, H. (2002). Real-time computing without stable states: a new framework for neural computation based on perturbations. *Neural Computation* 14, 2531-2560.
- Magee, J. C., & Grienberger, C. (2020). Synaptic plasticity forms and functions. *Annual Review of Neuroscience* 43, 95-117.
- Markram, H., et al. (2015). Reconstruction and simulation of neocortical microcircuitry. *Cell* 163, 456-492.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience* 24, 167-202.
- Mnih, V., et al. (2015). Human-level control through deep reinforcement learning. *Nature* 518, 529-533.
- Mountcastle, V. B. (1997). The columnar organization of the neocortex. *Brain* 120, 701-722.
- Oudeyer, P.-Y., & Kaplan, F. (2007). What is intrinsic motivation? A typology of computational approaches. *Frontiers in Neurorobotics* 1, 6.
- Pfeifer, R., & Bongard, J. (2006). *How the Body Shapes the Way We Think: A New View of Intelligence*. MIT Press.
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience* 2, 79-87.
- Rudin, N., Hoeller, D., Reist, P., & Hutter, M. (2022). Learning to walk in minutes using massively parallel deep reinforcement learning. *Conference on Robot Learning (CoRL)*.
- Tinbergen, N. (1951). *The Study of Instinct*. Oxford University Press.
- Tononi, G., & Cirelli, C. (2006). Sleep function and synaptic homeostasis. *Sleep Medicine Reviews* 10, 49-62.
- Trachtenberg, J. T., Chen, B. E., Knott, G. W., Feng, G., Sanes, J. R., Welker, E., & Svoboda, K. (2002). Long-term in vivo imaging of experience-dependent synaptic plasticity in adult cortex. *Nature* 420, 788-794.
- Turrigiano, G. G., & Nelson, S. B. (2004). Homeostatic plasticity in the developing nervous system. *Nature Reviews Neuroscience* 5, 97-107.
- Vyazovskiy, V. V., Cirelli, C., Tononi, G., & Tobler, I. (2008). Cortical metabolic rates as measured by 2-deoxyglucose-uptake are increased after waking and decreased after sleep in mice. *Brain Research Bulletin* 75, 591-597.
- Wang, X.-J. (2002). Probabilistic decision making by slow reverberation in cortical circuits. *Neuron* 36, 955-968.
- Yang, G., Pan, F., & Gan, W.-B. (2009). Stably maintained dendritic spines are associated with lifelong memories. *Nature* 462, 920-924.

(Bibliographic details in this list still need a check against the original sources before submission; the citation list has not been machine-verified.)